TENNESSEE
EMERGENCY MEDICAL SERVICES

ALS/BLS BLENDED PROTOCOLS
2027-2028 REVISION

BLS/ALS Protocols

PURPOSE:
To provide guidance on ALS/BLS treatment of patients

SCOPE:
All Personnel

INDEX

Introduction
Definitions
Medical Director's Statement

Cardiac Emergency (Adult & Pediatric)
- **SOP # 101** Automatic External Defibrillator (AED)
- **SOP # 102** Symptomatic Bradycardia
- **SOP # 103** Acute Coronary Syndrome/STEMI
- **SOP # 104** Chest Pain / Non-Cardiac
- **SOP # 105** Pulseless Electrical Activity (PEA)
- **SOP # 106** Premature Ventricular Complexes
- **SOP # 107** Supraventricular Tachycardia (SVT) / Regular Narrow Complex Tachycardia
- **SOP # 108** Torsades de Pointes
- **SOP # 109** Ventricular Asystole
- **SOP # 110** Ventricular Fibrillation/Pulseless Ventricular Tachycardia
- **SOP # 111** Persistent Ventricular Fibrillation / Pulseless Ventricular Tachycardia
- **SOP # 112** Ventricular Tachycardia with a Pulse
- **SOP # 113** Post Resuscitation

Environmental Emergency (Adult & Pediatric)
- **SOP # 201** Chemical Exposure
- **SOP # 202** Drug Ingestion
- **SOP # 203** Electrocution / Lightning Injuries
- **SOP # 204** Hyperthermia
- **SOP # 205** Hypothermia
- **SOP # 206** Drowning/ Near Drowning
- **SOP # 208** Venomous Snake Bite
- **SOP # 209** Radiation / HazMat Exposure
- **SOP # 210** Carbon Monoxide Exposures
- **SOP # 211** Riot Control Agents
- **SOP # 212** Dive (SCUBA) Injury / Accidents
- **SOP # 213** Acetylcholinesterase Inhibitor Exposure (Carbamates, Nerve Agents, Organophosphates)
- **SOP # 214** Topical Chemical Burn
- **SOP # 215** Beta Blocker Poisoning / Overdose
- **SOP # 216** Calcium Channel Blocker Poisoning / Overdose

Medical Emergency (Adult & Pediatric)
- **SOP # 300** Medical Complaint Not Specified Under Other Protocols
- **SOP # 301** Abdominal/GU Pain (non-traumatic) / Nausea and Vomiting
- **SOP # 302** Acute Pulmonary Edema / CHF
- **SOP # 303** Anaphylactic Shock
- **SOP # 304** Cerebrovascular Accident (CVA) / Stroke
- **REFERENCE** C-STAT Stroke Assessment Tool
- **REFERENCE** Pre-Hospital Screen for Thrombolytic Therapy
- **SOP # 305** Croup
- **SOP # 306** Family Violence

SOP # 307 Headache
SOP # 308 HEENT (Head, Eyes, Ears, Nose, Throat) Complaints
SOP # 309 Hyperglycemia Associated with Diabetes
SOP # 310 Hypertensive Crisis
SOP # 311 Hypoglycemia
SOP # 312 Medications at Schools
SOP # 313 Non-Formulary Medications
SOP # 314 Respiratory Distress (Asthma/COPD)
SOP # 315 Seizures
SOP # 316 Sexual Assault
SOP # 317 Sickle Cell Crisis
SOP # 318 Unconscious / Unresponsive / Altered Mental Status
SOP # 319 Syncope
SOP # 320 Brief Resolved Unexplained Event (BRUE) & Acute Events in Infants
SOP # 321 Back Pain (Non-Traumatic)

Shock / Trauma (Adult & Pediatric)

SOP # 401 Air Ambulance Transport
SOP # 402 Abdominal / Pelvic Trauma
SOP # 403 Avulsed Teeth
SOP # 404 Cardiogenic Shock
SOP # 405 Eye Trauma
SOP # 406 Hypovolemic Shock
SOP # 407 Major Thermal Burns
SOP # 408 Musculoskeletal Trauma
SOP # 409 Neurogenic Shock
SOP # 410 Septic Shock
SOP # 411 Soft Tissue / Crush Injuries
SOP # 412 Spinal Cord Injury
SOP # 413 Trauma - Blunt
SOP # 414 Adult Trauma Arrest (Blunt)
SOP # 415 Trauma - Penetrating
SOP # 416 Adult Trauma Arrest (Penetrating)
SOP # 417 Adult Traumatic Cardiac Arrest
SOP # 418 Tension Pneumothorax
SOP # 419 Traumatic Amputation(s)
SOP # 420 Blast Injuries
SOP # 421 High Threat Considerations / Active Shooter Scenario

Obstetrical Emergencies

REFERENCE APGAR Scoring
SOP # 500 Obstetrical / Gynecological Complaints (Non-Delivery or Gynecological Only)
SOP # 501 Normal Delivery
SOP # 502 Abruptio Placenta
SOP # 503 Amniotic Sac Presentation
SOP # 504 Breech or Limb Presentation
SOP # 505 Meconium Stain
SOP # 506 Placenta Previa
SOP # 507 Prolapsed Umbilical Cord
SOP # 508 Pre-eclampsia and Eclampsia

Miscellaneous

SOP # 601 Discontinuation / Withholding of Life Support
SOP # 602 Field Determination of Death
SOP # 603 Mandatory EKG
SOP # 604 Patient Refusal or Declination of Care – Patient Non-Transport Situations

REFERENCE Mini Mental Status Exam
SOP # 605 Physician On-Scene
SOP # 606 Bystanders on the Scene
SOP # 607 Procedure for Deviation from Standing Orders
SOP # 608 Spinal Protection
SOP # 609 Stretcher Transport
SOP # 610 Terminally Ill Patients

Pediatric Cardiac Emergency
SOP # 613 Neonatal Resuscitation

Behavioral Health Emergency
SOP # 701 Agitated or Combative Patient
SOP # 702 BEHAVIORAL HEALTH EMERGENCIES
SOP # 703 Delirium with HyperAgitation
SOP # 704 Physical Restraint

Hazardous Materials
SOP # 801 Ammonia
SOP # 802 Chlorine
SOP # 803 Cyanide Exposure
SOP # 804 Heavy Metals
SOP # 805 Hydrogen Fluoride
SOP # 806 Hydrogen Sulfide
SOP # 807 Methyl Bromide
SOP # 808 Nitrogen Oxides
SOP # 810 Crush Syndrome
SOP # 811 Airway / Respiratory Irritants

Additional Optional Procedures and Medications
Medications
Lidocaine Substitution
Benzodiazepine Substitution
Zofran
Ibuprofen
Acetaminophen
Toradol
TXA
Cyanocobalamin
Norepinephrine (LevoPhed)
Agitated Patient management
Droperidol
Haldol
Cardiac
Double Sequential Defibrillation
Irregular Narrow Complex Tachycardia/ Afib with RVR

Patient refusal of care

Medical Director's Authorization

Procedures

- **PROCEDURE** Blood Collection Time Critical Illness
- **PROCEDURE** Capnography
- **PROCEDURE** Chest Decompression
- **PROCEDURE** Continuous Positive Airway Pressure (CPAP)
- **PROCEDURE** Delayed Off Load of Patients to the ED
- **PROCEDURE** Endotracheal Tube Introducer (Bougie)
- **PROCEDURE** External Cardiac Pacing
- **PROCEDURE** Fever / Infection Control
- **PROCEDURE** Hemorrhage Control Clamp
- **PROCEDURE** Induced Hypothermia Following ROSC
- **PROCEDURE** Indwelling IV Port Access
- **PROCEDURE** Intranasal Medication
- **PROCEDURE** Intraosseous Access
- **PROCEDURE** Lucas CPR Device
- **PROCEDURE** Radio Report- Adult
- **PROCEDURE** Radio Report- Pediatric Trauma
- **PROCEDURE** Radio Report- Trauma
- **PROCEDURE** ResQPod
- **PROCEDURE** Tourniquet
- **PROCEDURE** Vascular Access

Reference

- **REFERENCE** Consent Issues
- **REFERENCE** Civilians Riding During Transport
- **REFERENCE** Hospital Capabilities
- **REFERENCE** Life Vest
- **REFERENCE** LVAD
- **REFERENCE** MCI Plan Response Levels
- **REFERENCE** Non-Viable Patients on Public Scenes
- **REFERENCE** Patient Assessment Flow Chart
- **REFERENCE** Physician Orders for Scope of Practice (POST)
- **REFERENCE** Pulse Oximetry
- **REFERENCE** QI Documentation Criteria
- **REFERENCE** Sepsis
- **REFERENCE** S.T.A.R.T. TRIAGE
- **REFERENCE** Trauma Assessment / Destination Guidelines
- **REFERENCE** Trauma Treatment Priorities
- **REFERENCE** Trauma Score
- **REFERENCE** Glasgow Coma Scale
- **REFERENCE** Common Medical Abbreviations

Pharmacology

- **REFERENCE** Glucose Reference
- **REFERENCE** Ketamine Reference
- **REFERENCE** Opiate Reference
- **REFERENCE** Pain Management Protocol
- **REFERENCE** Post Intubation
- **REFERENCE** Medication Dosage
- **REFERENCE** Drug Infusion Admix Dosage Guidelines

Pediatric Guidelines

- **REFERENCE** Pediatric Points to Remember
- **REFERENCE** Pediatric Trauma Score
- **REFERENCE** Triage Decision Scheme
- **REFERENCE** Age, Weight, and Vitals Chart
- **REFERENCE** Age and Weight Related Equipment Guidelines

Introduction

These Standing Orders and Protocols may be used by EMS personnel licensed by the State of Tennessee Department of Health, Office of Emergency Medical Services to render appropriate care. All EMRs, EMTs, AEMTs, and Paramedics are to familiarize themselves with these SOPs. These Standing Orders and Protocols are applicable regardless of the final destination of the patient and/or the personnel's duty session.

Notes:

1. The Emergency Medical Responder (EMR) will function under the current guidelines as stated in the AHA-BLS Healthcare Provider text. They shall also be responsible for other duties as assigned, within their scope of practice, as assigned by the AEMT or Paramedic.

2. These Standing Orders and Protocols are in addition to the minimum guidelines for patient care as outlined in the DOT EMT Curriculum. The EMT and AEMT will assist ALS personnel as requested or as needed.

3. When the Emergency Ambulance is out of quarters for any reason, the Paramedic will be in charge and will be responsible for all the actions and or activities as it relates to the Emergency Ambulance. On the scene of an emergency, the Paramedic will be responsible for patient care. The Senior AEMT will oversee the BLS units when applicable. The EMT or AEMT ***will*** act within their scope of practice to any request for patient care or maintenance of the unit as directed by the Paramedic. Patient care is limited to acts within their scope of practice. The EMT or AEMT is responsible for reviewing all documentation and signing in the required manner.

4. It is the responsibility of the most qualified provider caring for the patient to ensure transmission of all aspects of the patient assessment and care to the responding Emergency Unit or Medical Control.

5. When reporting a disposition to Medical Control or the responding unit, provide the following minimum information, utilizing the Radio Template in these protocols;
    a. Patient's age and chief complaint
    b. Is the patient stable or unstable, including complete vital signs and level of consciousness
    c. Interventions performed
    d. Name of physician providing order (if orders received)
    e. Provide other information as requested

6. For every call, the first directives are scene safety and body substance isolation precautions. Always utilize appropriate PPE for any patient contact.

7. For any drug administration or procedures outside these Standing Orders and Protocols, the EMS Provider must receive authorization from Medical Control. Paramedics en-route to the scene are not authorized to issue medication orders.

8. The minimal equipment required for all patient calls:
    a. When the patient is in close proximity to the unit or fire company:
        Jump bag, cardiac monitor, and oxygen or other equipment as may be indicated by the nature of the call.
    b. When the patient is not in close proximity to the unit or fire company:
        The above equipment, stretcher and any other equipment that may be needed as dictated by the nature of the call.

9. The senior Paramedic riding on the ambulance has the ultimate responsibility to ensure that all patient records and reports are properly completed. The Senior AEMT will be responsible for all patient records on BLS Units when applicable. The patient care report should accurately reflect the clinical activities undertaken. If there is a patient refusal, declination, or dismissal of service at the scene of the incident,

the incident report should reflect the details as well as the party or parties responsible to discontinue all evaluations and treatment.

10. Although the SOPs and Protocol procedures have a numerical order, it may be necessary to change the sequence order or even omit a procedure due to patient condition, the availability of assistance, or equipment. Document your reason for any deviations from protocol.

11. EMRs, EMTs, and AEMTs are expected to perform their duties in accordance with local, state, and federal guidelines and within the State of Tennessee Statutes and Rules and Regulations of the Tennessee Department of Health, Office of Emergency Medical Services. The Paramedic will work within their scope of practice dependent on available equipment.

12. The ePCR shall be completed within 24 hours. Prior to the end of shift each Paramedic will verify (or AEMT on BLS Units will verify) that all of their electronic documents including addendums have been posted to the online documentation system. When on the 1st Responder Company, the ePCR will be completed by the end of the shift. This will ensure proper documentation of the continuity of care.

13. In potential crime scenes, any movement of the body, clothing, or immediate surroundings should be documented, and the on-scene law enforcement officer should be notified of such.

14. All patients should be transported to the most appropriate facility according to the patient or family request or to the facility that has the level of care commensurate with the patient's condition. Certain medical emergencies may require transport to a facility with specialized capability. A document with the capabilities of area facilities is available to EMS providers.

15. EMS personnel may transport the patient in a non-emergency status to the hospital. This should be based on the signs and symptoms of the patient, mechanism of injury or nature of illness.

16. The following refusal situations should be evaluated by a Paramedic:
    a. Hypoglycemic patients who have responded to treatment
    b. Any patient refusing transport who has a potentially serious illness or injury
    c. Patients age less than **4** years or greater than **70** years
    d. Chest pain, any age or cause
    e. Drug overdose / intoxicated patients
    f. Potentially head injured patients
    g. Psychiatric disorders

17. The use of a length-based assessment tape is **required** for all pediatric patients as a guide for medications and equipment sizes. The tape will be utilized on all pediatric patients under the age of 8 years and appropriate for their weight. When assessing a child 8 or older that is small in stature for their age, you should consider using the length-based tape for compiling a complete accurate assessment of the patient. This information will be passed along to the receiving facility during the radio report and documented in the PCR.

Clinical Notes:

1. EMTs may administer the following medications and procedures provided they have received appropriate training:
    a. Beta Agonists (aerosolized/nebulized) for dyspnea and wheezing;
    b. Oral Glucose for suspected hypoglycemia;
    c. Oral Aspirin for chest pain of suspected ischemic origin;
    d. Sublingual nitroglycerine from patient's own prescribed medication;
    e. Epinephrine (for anaphylactic reaction) Auto-injector;
    f. Auto-injector Antidotes for hazmat exposure;
    g. Opioid Antagonist Autoinjector;

h. Over the Counter analgesics for pain or fever;
i. EMTs are authorized to use glucometers in the assessment of patients.
j. Nontracheal airways
k. Tourniquet Application
l. 12 Lead ECG acquisition and transmission
m. CPAP

AEMTs may administer the above listed medications and procedures as well as the following, provided they have received proper training:

a. Immunizations:
b. Glucagon for Hypoglycemia;
c. Sublingual Nitroglycerin for chest pain of suspected ischemic origin;
d. Patient triggered inhaled analgesics (nitrous oxide);
e. Attend patients receiving Intravenous Antibiotics;

2. A complete patient assessment, vital signs, treatments and continued patient evaluation are to be initiated immediately upon contact with a patient and continued until patient care is transferred to a higher medical authority. Refer to the Patient Assessment Flow Chart located in these SOPs.

3. The ongoing assessment times are considered:

High Acuity	***Low Acuity***
Every 3-5 Minutes	Every 5-15 Minutes

4. If a glucometer reading is less than 80 mg/dL and the patient is asymptomatic, consider the administration of oral glucose. If a glucometer reading is less than 80 mg/dL and the patient is symptomatic, administer oral glucose or start an IV NS and administer dextrose. Reassess patient every 5 minutes, repeat PRN.

   Note: Any parenteral administration of dextrose must be given through an IV line running normal saline and **NOT VIA AN INT**. Blood glucose should be rechecked after administration of dextrose or oral glucose. Normal blood glucose values for adults are 80 – 120 mg/dL.

5. Blood Glucose and Stroke Screening will be performed on all patients with altered mental status. Glucose should be titrated slowly to restore normal levels while avoiding large rapid changes in serum glucose levels. Be aware that elevated glucose levels are detrimental in conditions such as stroke.

6. Supportive care indicates any emotional and/or physical care including oxygen therapy, repositioning patient, comfort measures, and patient family education.

7. Upon arrival at the receiving hospital, all treatment(s) and monitoring initiated in the field will be continued until hospital personnel have assumed patient care or the ambulance returns to service.

8. The initial blood pressure **MUST** be taken manually. If subsequent blood pressures taken by machine vary more than 15 points diastolic, then the machine reading will be verified by a manual blood pressure.

9. EMTs and AEMTs may obtain and transmit EKG monitoring tracings and 12 Lead EKGs. Paramedics **ONLY** may interpret, treat, and determine destination based on the 12 Lead EKG.

10. Indications for football helmet removal:
    - When a patient is wearing a helmet and not the shoulder pads
    - In the presence of head and or facial trauma
    - Patients requiring advanced airway management when removal of the facemask is not sufficient.
    - When the helmet is loose on the patient's head

- In the presence of cardiopulmonary arrest (the shoulder pads must also be removed)

When the helmet and shoulder pads are both on, the spine is kept in neutral alignment. If the patient is wearing only the helmet or shoulder pads, neutral alignment must be maintained. Either remove the other piece of equipment or pad under the missing piece. *All other helmets must be removed to maintain spinal alignment.*

Clinical Notes – Airway:

1. All EMTs have standing orders for insertion of an approved airway device for patients meeting the indications.

2. I-Gels are approved for pediatric use.

3. Airway maintenance appropriate for the patient's condition includes any airway maneuver, adjunct, or insertions of tubes that provide a patent airway.

4. Pulse Oximetry should be utilized for all patients complaining of respiratory distress or chest pain (regardless of source). Oxygen therapy should be geared to get patient oxygen saturation to >92%. Use oxygen judiciously with this goal in mind.

5. **Continuous waveform capnography** is **MANDATORY** for all intubations and non-tracheal airways. Reliability may be limited to patients less than 20 kg. Use other methods to assist in confirmation.

6. The use of head blocks or other head restraint post intubation (Blind Insertion Airway Device or ETT) is recommended to reduce the chance of accidental extubation. This is in addition to the tube securing devices currently in use. Cervical collars may cause a limitation in blood flow to the brain in patients with low flow states such as cardiac arrest/CPR and should not be too tightly placed.

Clinical Notes – Cardiovascular:

1. Cardiac Arrest Best Practices:

   All efforts should be made to incorporate the following practices in all cardiac arrests:

   a. CPR is most effective when done continuously, with minimum interruption. Maintain rate of 100-110 BPM, depth of 2", and compression fraction of >80%. Utilize metronomes or timers to ensure proper rate of compression and ventilation.
   b. Initiate compressions first, manage airway after effective compressions for two minutes.
   c. All IV/IO drugs given are to be followed by a 10 cc NS bolus.
   d. Elevate the extremity after bolus when given IV.
   e. Consider other airway maneuvers (i-Gel Airway) whenever intubation takes longer than 30 seconds.
   f. Apply nasal cannula oxygen 2 – 4 LPM during initial CPR.
   g. Tidal volumes should not exceed 500cc in Cardiac Arrest.
   h. Consider the use of Mechanical CPR device if available. Make sure that placement of the device takes no longer than 20 seconds. Longer pauses in CPR substantially decrease the likelihood of a successful resuscitation.
   i. Utilize the ITD16 (ResQPod) in all cardiac arrest situations. High quality CPR is necessary to achieve maximum benefit.
   j. If using Active Compression/Decompression CPR (ResQCPR) or Mechanical CPR Device, ensure utilization of the impedance threshold device (ITD). The ITD can be placed onto the BVM if an adequate facial seal can be maintained.
   k. Remove impedance threshold device upon ROSC.
   l. If CPR needs to be reinstated, perform 2 minutes of CPR supine prior to head elevation.

2. Treat the patient not the monitor.

3. Defibrillation and synchronized cardioversion joules are based on the use of the current biphasic monitor.

4. If a change in cardiac rhythm occurs, provide all treatment and intervention as appropriate for the new rhythm.

5. In the case of cardiac arrest where venous access is not readily available, paramedics and AEMTs may use the IO as the initial access. Humeral access site is preferred in medical conditions.

Clinical Notes – IV:

1. AEMTs and Paramedics have standing orders for precautionary IVs and INTs. AEMTs have a standing order for the insertion of an IV or INT under the following guidelines:
    a. The patient must have some indication that they are unstable (see definitions page).
    b. Limited to two attempts in one arm only. (IV cannulation of legs or neck is not allowed).
    c. Drug administration will be followed by a minimum of 10 cc of fluid to flush the catheter.
    d. Blood glucose will be obtained for all patients with altered mental status.
    e. IVs should not be attempted in an injured extremity.
    f. TKO (To Keep Open) indicates a flow rate of approximately 50 cc/hr ***(peds 5-10 mL/hr).***
    g. IVs will not be started in arms with shunts.
    h. IVs appropriate for patient's condition:
        i. If patient is hypotensive, give bolus of fluid
        ii. If patient's blood pressure is normal, run IV TKO or convert to saline lock (INT)
    i. A bolus of fluid is 20 cc/kg for all patients.
    j. Attempt to obtain blood sample tubes on all patients with time critical illnesses.

2. For external jugular IVs attempted by Paramedics, IV catheters should be 18 gauge or smaller diameter based on the patient.

Paramedics, when properly equipped and trained, may utilize indwelling access ports such as Port-A-Cath in an **EMERGENCY ONLY**. This procedure should be done with a Huber needle utilizing sterile technique.

DEFINITIONS

Medical Director: The physician who has ultimate responsibility for the patient care aspects.

Unstable (symptomatic): Indicates that one or more of the following are present:

a. Chest Pain
b. Dyspnea
c. Hypotension (systolic B/P less than 90 mmHg in a 70-kg patient or greater)
d. Signs and symptoms of congestive heart failure or pulmonary edema
e. Signs and symptoms of a myocardial infarction
f. Signs and symptoms of inadequate perfusion
g. Altered level of consciousness

Stable (asymptomatic): Indicates that the patient has no, or very mild signs and symptoms associated with the current history of illness or trauma.

Firefighter/Non-EMT: Personnel trained only in basic first aid and CPR. Responsible for immediately identifying and providing patient care and assisting other personnel upon their arrival and ensuring continuity of patient care.

Emergency Medical Responder (EMR): Personnel licensed by the Tennessee Department of Health, Office of EMS and authorized by the Medical Director to perform lifesaving interventions while awaiting additional EMS response. EMR may also assist higher level personnel at scene and during transport under medical direction and within scope of practice.

Emergency Medical Technician (EMT): Personnel licensed by the Tennessee Department of Health, Office of EMS and authorized by the Medical Director to provide basic emergency care according to the standard of care and these Standing Orders and Protocols.

Advanced Emergency Medical Technician (AEMT): Personnel licensed by the Tennessee Department of Health, Office of EMS and authorized by the Medical Director to provide limited advanced emergency care according to the standard of care and these Standing Orders and Protocols.

Paramedic: Personnel licensed by the Tennessee Department of Health, Office of EMS and authorized by the Medical Director to provide basic and advanced emergency patient care according to the Standard of Care and the EMS BLS and ALS Standing Orders and Protocols.

Transfer of Care: Properly maintaining the continuity of care through appropriate verbal and/or written communication of patient care aspects to an equal or higher appropriate medical authority.

Higher Medical Authority: Any medical personnel that possesses a current medical license or certificate recognized by the State of Tennessee with a higher level of medical training than the one possessed by Emergency Medical Services Personnel (MD, DO).

Medical Control (transport): The instructions and advice provided by a physician, and the orders by a physician that define the treatment of the patient; to access Medical Control, contact the Emergency Department physician on duty of the patient's first choice of destinations. If the patient does not have a preference, the patient's condition and/or chief complaint may influence the choice of medical treatment facilities.

All EMS Providers are expected to perform their duties in accordance with local, state, and federal guidelines.

Medical Director's Statement

I have taken great care to make certain that doses of medications and schedules of treatment are compatible with generally accepted standards at the time of publication. Much effort has gone into the development, production, and proof reading of these Standard Operating Procedures and Protocols. Unfortunately, this process may allow errors to go unnoticed or treatments may change between the creation of these protocols and their ultimate use. Please do not hesitate to contact me if you discover any errors, typos, dosage, or medication errors.

I look forward to any questions, concerns, or comments regarding these protocols. I expect all EMS personnel to follow these guidelines, but also to utilize and exercise good judgment to provide the best care for all our patients.

EMS Protocols Copyrighted. Authorized for exclusive use by EMS services under EMR Medical Direction.



Joe Holley, MD FACEP FAEMS
EMS Medical Director

CARDIAC EMERGENCY
SOP # 101 Automatic External Defibrillator (AED)

Signs and Symptoms
Patient in cardiopulmonary arrest
Basic life support in progress
AED in use

TREATMENT PATHWAY

1. If AED available, apply to patient and follow prompts.
2. Oxygen and airway maintenance appropriate to patient's condition. All CPR rates of compression are **100-110 per minute for all ages. Res-Q-Pump compression rate is 80 per minute.** Ventilation rates are 2 breaths for every 30 compressions ***(peds – 2 breaths for every 15 compressions)*** if advanced airway is not in place. If an advanced airway IS in place, give 1 breath every six seconds (10 breaths per minute) for all age groups. Utilize ITD.
3. Continue CPR according to current AHA Healthcare Provider Guidelines, specific for patient's age.
4. If AED is in use (defibrillating) prior to arrival, allow shocks to be completed, and then evaluate pulse:
    a. If no pulse, continue to provide CPR and basic life support
    b. If a pulse is present, evaluate respirations and provide supportive care appropriate for the patient's condition

EMT STOP HERE

5. IV NS bolus (20 cc/kg), then TKO.

AEMT STOP HERE

6. Monitor patient and treat per SOP specific for the arrhythmia.

PARAMEDIC STOP

Notes
7. AED is relatively **contraindicated** in the following situations:
    a. If the victim is in standing water
    b. Trauma cardiac arrest
8. Victims with implanted pacemakers, place pads 1 inch from device.
    If ICD/AICD is delivering shock to the patients allow 30 to 60 seconds (2 complete treatment cycles) before using the AED.
9. Transdermal medication patch at site of AED pads:
    If a medication patch is in the location of the AED pad, remove the medication patch and wipe the site clean before attaching the AED electrode pad.

CARDIAC EMERGENCY
SOP # 102 Symptomatic Bradycardia

Signs and Symptoms
Heart rate less than 60 beats per minute and symptomatic
Decreased / altered LOC
Chest pain / discomfort
CHF / pulmonary edema
Head trauma
Elevated ICP
Dyspnea
Hypothermia
Hypoglycemia
Drug overdose
Signs of decreased perfusion
Rhythm may be sinus bradycardia, junctional, or heart block
Heart rates < 80 /min for infant or < 60 /min for child

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
   If patient will not tolerate NRB, apply Oxygen at 6 LPM BNC ***(peds- 4 LPM. Use bag-valve-mask if no response with oxygen by nasal cannula)***
2. Supportive care.
3. Pulse Oximetry. Consider Opiate antagonist if indicated.
4. Glucometer check.
5. Cardiac monitor – 12 Lead EKG, transmit if indicated.

EMT STOP HERE

6. IV access and administer NS TKO. Attempt for large bore (18 or better) in AC, with second line as time permits.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
8. Consider Naloxone 0.4-2mg IV/IN/Auto-injector. If utilizing pre-filled delivery systems, dose per manufacturer's instructions.

AEMT STOP HERE

9. If patient is asymptomatic and heart rate is less than 60 bpm, monitor and transport.
10. If PVCs are present with bradycardia, **DO NOT** administer Amiodarone/lidocaine.
11. Adults:
12. NOTE: If respiratory failure or inadequate ventilation is contributing to bradycardia, transcutaneous pacing alone will not restore adequate perfusion. Prioritize airway management and ventilation.
    a. If patient is unconscious/unresponsive and bradycardic with hemodynamic instability: Refer to Cardiac Arrest protocol and initiate CPR
    b. Consider CPR initiation at earliest signs of hemodynamic compromise (not just BP < 90 threshold), including altered mental status, severe bradycardia with poor perfusion signs, or respiratory failure.
       If 2nd and 3rd degree blocks are present apply transcutaneous pacer pads (if available), administer Atropine 0.5mg IV

c. If systolic BP < 90 mmHg and heart rate < 60/min continues
    i. Administer Atropine 0.5 mg up to 0.04 mg/kg (3 mg for adults) ***(peds 0.02 mg/kg, repeat once in 3 to 5 minutes PRN, max single dose 0.5 mg, max total dose 1 mg)***

d. If systolic BP < 90 mmHg and heart rate < 60/min continues
    i. Notify Medical Control and begin External Pacing per protocol
    ii. CAUTION - False Electrical Capture: Ensure MECHANICAL capture is confirmed with each pacing spike. Absence of mechanical capture despite electrical activity indicates ineffective pacing.
    iii. Consider:
    EPINEPHrine 2-10 mcg/min IV slow push.

To prepare EPINEPHrine 10 mcg/ml:
1. Draw up 9 ml of normal saline into 10 ml syringe
2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) from a cardiac syringe
3. Label syringe

Adult Push Dose Pressor IV Administration:
Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.

Pediatric:
- ***a. Heart rates < 80 /min for infant or < 60 /min for child***
- ***b. Signs of poor perfusion, respiratory distress, or hypotension***

Yes – start chest compressions, IV/IO
1. ***EPINEPHrine 1:10,000 (now 0.1 mg/mL) - 0.01 mg/kg IV/IO q 3-5 min.***
2. ***Contact Medical Control***
    a. ***Consider external cardiac pacing***

Consider EPINEPHrine 0.1-1 mcg/kg/min.

13. If beta blocker ingestion is suspected, consider Glucagon 1-2 mg IM/IV if unresponsive to Atropine. ***(peds – Glucagon 0.5 mg/dose if less than 20 kg, or 1 mg/dose if 20 kg or greater)***

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 103 Acute Coronary Syndrome/STEMI

Signs and Symptoms	Notes
Determine the quality, duration, radiation of pain Substernal oppressive chest pain (crushing or squeezing) Nausea and/or vomiting Shortness of breath Cool, clammy skin Palpitations Anxiety or restlessness Abnormal pulse rate or rhythm History of Coronary Artery Disease or AMI Currently taking cardiac medications JVD Distal pulse for equality/strength to assess for aneurysm Diaphoresis, pallor, cyanosis Breath sounds – congestion, rales, wheezing Motor deficits	**P** – Provocation of pain/discomfort (anything that increases discomfort) **Q** – Quality of pain **R** – Radiation of pain **S** – Severity of pain/discomfort (scale of 1 – 10) **T** – Time of pain/discomfort onset; type of pain The elderly, women, and/or diabetic patients may complain of nausea, weakness, shortness of breath or other vague symptoms. Screen all such patients for possible silent MI.

TREATMENT PATHWAY

1. Oxygen at 2 – 6 LPM BNC and airway maintenance appropriate to patient's condition. If the patient is in severe respiratory distress, consider Oxygen 12 – 15 LPM NRB ***(peds – 4 LPM BNC. Use bag-valve-mask if no response with oxygen by nasal cannula).***
2. Supportive care.
3. Pulse oximetry – provide Oxygen sufficient to keep SATs > 92%.
4. If systolic BP is > 100 and the patient is symptomatic, may assist patient with their own Nitroglycerine tablet or spray sublingually and reassess every 5 minutes up to a maximum of three doses.
5. Administer 324 mg of Aspirin (Chewable non-enteric coated) if patient has no contraindications or has not already self-dosed.
6. Cardiac Monitor – assist with 12 lead EKG and transmission if applicable. Obtain and transmit EKG to PCI capable hospital within the first 10 minutes of patient contact. Consider repeat EKGs. Approximately 10% of STEMI patients will have at least 1 normal EKG.
7. Glucose check.

EMT STOP HERE

8. IV access and administer NS TKO. Attempt for large bore (18 or better) in AC, with second line as time permits.
9. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
10. If systolic BP > 100 and patient is currently symptomatic, administer one 0.4 mg Nitroglycerin tab or sublingual spray q5min X 3 doses if SBP remains > 100.
11. Contact Medical Control for additional doses.
    Caution: The three doses include any doses given prior to arrival or patient's own medication administration. Use with caution in patients taking erectile dysfunction medication as this may cause profound hypotension.

AEMT STOP HERE

12. Perform serial EKGs in order to document progression of EKG changes. Treat arrhythmia appropriately.

13. Patients with probable AMI should be transported to an appropriate PCI capable facility as soon as possible.
14. Systolic BP is < 100 mmHg, give 250 ml NS bolus (assess for signs of pulmonary congestion).
    a. If PVCs > 15 /min and symptomatic, administer Amiodarone 150 mg over 2 minutes
15. If chest pain/discomfort continues after adequate Nitrate therapy:
    a. Continue Nitrate therapy
    b. Complete Thrombolytic screening
    c. If chest pain is greater than **7** on scale of 1 – 10, administer pain medications per pain management protocol
    d. Contact Medical Control
    e. Transport

PARAMEDIC STOP

Note: If EMS suspects a true Acute Coronary Syndrome/STEMI in a patient less than 18 years old, immediately contact online medical control.

CARDIAC EMERGENCY
SOP # 104 Chest Pain / Non-Cardiac

Signs and Symptoms	Notes
Determine quality, duration, and radiation of pain Atypical chest pain NO nausea and/or vomiting NO shortness of breath NO cool, clammy skin History or chest injury, persistent cough NO history of coronary artery disease or AMI NOT currently taking cardiac medications Distal pulse for equality/strength to assess for aneurysm No diaphoresis, pallor, cyanosis Normal breath sounds	**P –** provocation of pain/discomfort (anything that increases discomfort) **Q –** quality of pain **R –** radiation of pain **S –** severity of pain/discomfort (scale of 1 – 10) **T –** time of pain/discomfort onset, type of pain The elderly, women, and/or diabetic patients may complain of nausea, weakness, shortness of breath or other vague symptoms. Screen all such patients for possible silent MI.

TREATMENT PATHWAY

1. Oxygen at 2 – 6 LPM BNC and airway maintenance appropriate to patient's condition. If the patient is in severe respiratory distress, consider Oxygen at 12 -15 LPM NRB ***(peds- 4 LPM. Use bag-valve-mask if no response with oxygen by nasal cannula).***
2. Supportive care.
3. Pulse oximetry.
4. If systolic BP is > 100 and the patient is symptomatic, may assist patient with their own Nitroglycerine tablet or spray sublingually and reassess every 5 minutes up to a maximum of three doses.
5. Administer 324 mg of Aspirin (Chewable non-enteric coated) if patient has no contraindications or has not already self-dosed.
6. Cardiac Monitor – assist with 12 lead EKG and transmission if applicable. Obtain and transmit EKG to PCI capable hospital within the first 10 minutes of patient contact. Consider repeat EKGs. Approximately 10% of STEMI patients will have at least 1 normal EKG.
7. Glucose Check.

EMT STOP HERE

8. IV access and administer NS TKO. Attempt for large bore (18 or better) in AC, with second line as time permits.
9. Administer 324 mg of Aspirin (Chewable non-enteric coated) if patient has no contraindications or has not already self-dosed.
10. If systolic BP > 100 and patient is currently symptomatic, administer one 0.4 mg Nitroglycerin tab or sublingual spray q5min X 3 doses if SBP remains > 100. Contact Medical Control for additional doses.
    Caution: The maximum dosage of Nitroglycerine is three sublingual administrations, whether before or after your arrival. Use with caution in patients taking erectile dysfunction medications. Profound hypotension may occur.
11. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

12. If chest pain/discomfort persists after adequate nitrate therapy and pain is greater than a **7** on a scale of 1 – 10, administer pain medications per pain management protocol.
13. Perform serial EKGs in order to document progression of EKG changes. Treat arrhythmia appropriately. Be sure to turn place pads and turn on alarms!
14. Contact Medical Control.
15. Transport.

PARAMEDIC STOP

CAUTION: Patients with true cardiac disease may have subtle, atypical symptoms. Always err on the side of the patient's safety.

Note: For pediatric patients complaining of chest pain, please contact online medical control before administering aspirin, nitroglycerine, or opiates.

CARDIAC EMERGENCY
SOP # 105 Pulseless Electrical Activity (PEA)

Signs and Symptoms
Presence of electrical cardiac rhythm without palpable pulse
Confirm rhythm with electrodes in two leads

TREATMENT PATHWAY

1. Utilize AED if available.
2. Oxygen and airway maintenance appropriate to the patient's condition.
3. CPR as indicated, consider Naloxone.
4. Glucose check.
5. EKG Monitor, 12 lead and transmission if applicable.

EMT STOP HERE

6. IV NS, bolus of fluid (20 cc/kg).
7. Check fingerstick glucose level. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. EPINEPHrine 1:10,000 (now 0.1 mg/mL) 1 mg IVP/IO ***(peds – EPINEPHrine 1:10,000 (now 0.1 mg/mL) - 0.01 mg/kg IV/IO q 3-5 min)***. Administer as early in the arrest as possible. Early administration results in better outcomes in PEA.
9. Search for underlying cause of arrest and provide the related therapy:
    a. Hypoxia – ensure adequate ventilation and oxygenation.
    b. Hypovolemia – fluid administration/fluid challenge – adult 20 cc/kg ***(peds 20 cc/kg bolus).***
    c. Cardiac tamponade – adult up to 2 liter bolus ***(peds 20 cc/kg bolus).***
    d. Tension pneumothorax – needle decompression.
    e. **KNOWN** hyperkalemia or tricyclic antidepressant overdose – Sodium Bicarbonate 8.4% 1 mEq/kg, may repeat @ 0.5 mEq/kg q 10 min ***(peds <1 month use Sodium Bicarbonate 4.2% - 1 mEq/kg), (peds >1 month use Sodium Bicarbonate 8.4% - 1 mEq/kg may repeat at 0.5 mEq/kg q 10 min)*** and $CaCl_2$ 500 mg IVP ***(peds 20 mg/kg).***
    f. **KNOWN** Acidosis in prolonged arrest: consider Sodium Bicarbonate 8.4% 1-2 mEq/kg IV
    g. Drug Overdose: Naloxone (Narcan) 0.4-2.0 mg IV/IO/IM/IN titrated to adequate ventilation ***(peds 0.1 mg/kg IV/IO/IM/IN titrated to adequate ventilation, max dose 2 mg)***. If utilizing pre-filled delivery systems, dose per manufacturer's instructions. May repeat dose. Synthetic Opiate overdoses may require much larger doses of Naloxone. Physically manage airway if no response after 8 mg Naloxone.
    h. Hypothermia: initiate patient re-warming, stop chest compressions with return of spontaneous circulation.
10. Consider External Cardiac Pacing per protocol.
11. PEA Continues: Continue CPR, transport to appropriate facility.

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 106 Premature Ventricular Complexes

Signs and Symptoms
Any PVC in AMI setting with associated chest pain
Multi-focal PVCs
Unifocal and >15 /min
Salvos/couplets/runs of V-Tach (three or more PVCs in a row) and symptomatic
PVCs occurring near the “T-wave”

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient’s condition.
2. Supportive care.
3. Pulse Oximetry.
4. Glucose check.
5. EKG Monitor. 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. INT or IV NS TKO.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. If significantly symptomatic PVCs are present with heart rate >60/min:
   Amiodarone 150-300 mg IV/IO ***(peds 5 mg/kg, may repeat up to total of 15 mg/kg)***

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 107 Supraventricular Tachycardia (SVT) / Regular Narrow Complex Tachycardia

Signs and Symptoms
Adult patients with heart rates in excess of 160 bpm ***(peds – infant rate > 220 bpm, child rate > 180 bpm)***
(QRS width < .12 sec [3 small blocks])
Pediatric SVT typically has no P waves and no beat to beat variability
Patients may exhibit symptoms of dyspnea, chest pain, radiating pain, altered mental status, hypotension (systolic BP < 90 mmHg)

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Supportive care.
3. Pulse oximetry.
4. Glucose check.
5. Monitor, 12 Lead EKG and Transmission if applicable.

EMT STOP HERE

6. INT or IV NS TKO.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. Valsalva maneuver for 15 seconds, then immediately lay patient flat, administer Adenosine, and lift legs 45 degrees for 15 seconds.
9. Adenosine 12 mg rapid IV ***(peds 0.1 mg/kg 6 mg max, may repeat at 0.2 mg/kg with 12 mg maximum dose if needed)***. May repeat dose of 12 mg once. Flush with 10 cc NS after each dose. OR Verapamil 5 mg slow IV/IO (if available) OR dilTIAZem 0.25 mg/kg slow IV/IO (if available).
    a. If rhythm does not convert to < 150 /min and patient is significantly symptomatic, or if patient is unstable and significantly symptomatic, prepare for synchronized cardioversion.
        Sedate as necessary: Diazepam (Valium) 2-5 mg IV ***(peds 0.1 mg/kg)*** or Midazolam (Versed) 2-5 mg IV ***(peds 0.1 mg/kg IV)*** or Ketamine/Pain Medication per protocol.
        Synchronized cardioversion @ 50, 100, 200 joules ***(peds 0.5 j/kg then 1 j/kg, then 2j/kg)***.
        Cardiovert until heart rate < 150 /min.
    b. If rhythm converts to rate < 150 /min: reassess for changes, maintain systolic BP > 90 mmHg, transport, and contact Medical Control.

PARAMEDIC STOP

Note: Due to the increased sensitivity to drug effects, in heart transplant patients and those on Tegretol (Carbamazepine), give ½ the normal dose of Adenosine.

10. Adenosine is administered through a large bore IV in the Antecubital Fossa.
11. Other vagal maneuvers may include asking the patient to hold their breath or Trendelenburg position.

12. Carotid sinus pressure should be applied on the right side if possible. If no effect, then try the left side. **NEVER** massage both sides at once.
13. Unstable SVT may be synchronized cardioverted immediately in frankly unstable patients prior to IV access. Assess the situation and make a good decision. Cardioversion hurts, utilize pain management protocol!
14. Significant symptoms include diaphoresis, hypotension, poor color or perfusion, mental status. changes, chest pain > 7/10.
15. Do NOT give dilTIAZem or verapamil if wide complex QRS or if SBP < 100. Calcium channel blocker medications may not be the best treatment for patients with impaired ventricular function and medical command should assist with this decision.
16. May substitute verapamil 5 mg IV/IO slowly over 3-5 minutes. May repeat verapamil once at 5-10 mg dose after 15 minutes.

CARDIAC EMERGENCY
SOP # 108 Torsades de Pointes

Signs and Symptoms
Decreased/altered LOC
Dyspnea
Chest pain/discomfort, suspected AMI
Hypotension (Systolic BP < 90 mmHg) ***(peds – 70+2x age)***
CHF/pulmonary edema
Heart rate > 160 /min with QRS > .12 sec (three small blocks [wide complex]) and twisting of points

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Supportive care.
3. Pulse Oximetry.
4. Glucose check.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. INT or IV NS TKO.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. Systolic BP
    a. If < 90 mmHg – unstable/symptomatic:
        i. Prepare for cardioversion at 50, then 100, then 200 joules, escalating as needed. ***(peds – 0.5-1 joule/kg in synchronized cardioversion).***
        ii. Sedation as necessary.
        Diazepam (Valium) 2-5 mg IV ***(peds 0.1 mg/kg)*** OR Midazolam (Versed) 2-5 mg IV ***(peds 0.1 mg/kg)*** or Ketamine and/or Pain Medications per the pain management protocol.
        iii. If rate <160/min – monitor for changes, transport, Magnesium Sulfate 1-2 g IVP over 2 minutes ***(peds – 50 mg/kg IV, max 2 g).***
        iv. If rate > 160 /min – Contact Medical Control, consider Amiodarone 150 – 300 mg IV/IO ***(peds 5 mg/kg, may repeat up to a total dose of 15 mg/kg)***.
    b. Cardiac arrest (pulseless Torsades):
        i. Defibrillate beginning at 100 J
        ii. Magnesium Sulfate 1 – 2 g IVP over 2 min
    c. If > 90 mmHg – stable/asymptomatic:
        i. Magnesium Sulfate 1 – 2 g IVP over 2 min

9. Transport.

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 109 Ventricular Asystole

Reversible Causes:

Hypovolemia	Table (Drug overdose)
Hypoxia	Tamponade (cardiac)
Hydrogen ion (acidosis)	Tension pneumothorax
Hyperkalemia/Hypokalemia	Thrombosis-heart
Hypothermia	Thrombosis-lungs

TREATMENT PATHWAY

Signs and Symptoms
No pulse or respirations
Confirm cardiac rhythm with electrodes in 2 leads on monitor
Record in two leads to confirm asystole and to rule out fine V-Fib

1. AED if available.
2. CPR appropriate for patient age.
3. Oxygen and airway maintenance appropriate for the patient's condition, consider Narcan.
4. Glucose check.
5. Cardiac Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. IV NS bolus (20 cc/kg bolus fluids)
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. EPINEPHrine 1:10,000 (now 0.1 mg/mL) – 0.5 mg IO/IVP every 3-5 minutes ***(peds EPINEPHrine 1:10,000 (now 0.1 mg/mL) - 0.01 mg/kg IV/IO q 3-5 min).*** Administer as early in the arrest as possible. Early administration results in better outcomes in Asystole.
9. For prolonged resuscitation, with known acidosis consider: Sodium Bicarbonate 8.4% 1 mEq/kg IV/IO followed by 0.5 mEq/kg q 10 min **(peds < 1 mo, use Sodium Bicarbonate 4.2% - 1 mEq/kg) (peds >1 mo, use Sodium Bicarbonate 8.4% - 1 mEq/kg may repeat at 0.5 mEq/kg q 10 min).**
10. Consider:
    a. Magnesium Sulfate 1 – 2 gm IV slow push over 2 minutes ***(no pediatric dosing).***
    b. Defibrillation for possible fine ventricular fibrillation masquerading as asystole
    c. Consider external pacing under the following circumstances:
    If cardiopulmonary arrest was witnessed by an experienced provider, and the patient is in asystole, prompt application of the transcutaneous cardiac pacemaker is appropriate prior to the administration of EPINEPHrine when a patient converts to asystole as a primary rhythm during EKG monitoring.
    d. $CaCl_2$ if arrest secondary to renal failure, or history of hemodialysis, 500 mg IV ***(peds 20 mg/kg IV/IO bolus; Non-Arrest infuse over 30-60 min).***
    e. Consider discontinuing efforts if criteria are met under Discontinuation/Withholding of Life Support Standing Order.
    f. Consider Naloxone if opiate ingestion is a concern

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 110 Ventricular Fibrillation/Pulseless Ventricular Tachycardia

Notes

- Defibrillation should not be delayed for any reason other than rescuer or bystander safety.
- Prompt defibrillation is the major determinant of survival. Time on scene should be taken to aggressively treat ventricular fibrillation.
- Consider transport of patient **after** performing 2 CPR/Defibrillation cycles, securing the airway, obtaining IV/IO access, and administering of at least two rounds of drugs. This will provide the best chance of return of a perfusing rhythm.
- EPINEPHrine doses beyond 3mg total are unlikely to be of benefit. Ensure CPR is of high quality.

TREATMENT PATHWAY

Signs and Symptoms
Ventricular Fibrillation
Ventricular Tachycardia
Pulseless
Apneic

Confirm and record cardiac rhythm with electrodes verified in 2 leads

1. AED if available.
2. CPR.
3. Oxygen and airway maintenance appropriate to the patient's condition, consider Naloxone.
4. Glucose check, treat if indicated.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. IV NS TKO.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. Defibrillate at manufacturer's recommended energies. ***(peds 1-2 j/kg)***
9. Administer:
   EPINEPHrine 1:10,000 (now 0.1 mg/mL) – 0.5 mg IVP/IO (only if no other option) q 4 mins ***(peds EPINEPHrine 1:10,000 (now 0.1 mg/mL) - 0.01 mg/kg IV/IO q 3-5 min)***. Administer as early in the arrest as possible. Early administration results in better outcomes.
10. Administer:
    a. Amiodarone 300 mg IV or IO, repeat after 5 min at 150 mg ***(peds 5 mg/kg, may repeat up to total of 15 mg/kg).***
    b. For prolonged resuscitation or known acidosis consider: Sodium Bicarbonate 8.4% 1 mEq/kg IV/IO followed by 0.5 mEq/kg q 10 min, ***(peds <1 mo, use Sodium Bicarbonate 4.2% - 1 mEq/kg), (peds >1 mo, use Sodium Bicarbonate 8.4% - 1 mEq/kg may repeat at 0.5 mEq/kg q 10 min).***
    c. Optional: Instead of Amiodarone, Lidocaine 2% ***(peds 1 mg/kg, max dose 3 mg/kg. Repeat if infusion initiated more than 15 after the initial dose).***
    d. $CaCl_2$ 500 mg IVP ***(peds 20 mg/kg)***, if arrest secondary to renal failure, or history of hemodialysis.
    e. Magnesium Sulfate 1 – 2 gm IV slow push over 2 min ***(no pediatric dosage).***

Epinephrine beyond 3 doses is less likely to be effective.

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 111 Persistent Ventricular Fibrillation / Pulseless Ventricular Tachycardia

Signs and Symptoms	Notes
Unresponsive Pulseless Persistent ventricular fibrillation/tachycardia or returned to this rhythm post ROSC/other rhythm changes	**For use after SOP Ventricular Fibrillation / Pulseless Ventricular Tachycardia Protocol has been ineffective** **Limit EPINEPHRine to 3 mg max if possible. Further dosing of EPINEPHrine may decrease chances of successful resuscitation.**

TREATMENT PATHWAY

PARAMEDIC ONLY

If there is no change in V-Fib:

1. Complete 5 cycles of CPR, check rhythm and pulse.
2. Repeat defibrillation without further pulse checks.
3. Resume CPR.

If there is a change in V-Fib:

1. Apply new defibrillation pads at new sites.
2. Complete 5 cycles of CPR, check rhythm and pulse.
3. Repeat defibrillation, pause 5 seconds maximum to check rhythm and pulse.
4. Resume CPR.

Notes:

- Recurrent ventricular fibrillation/tachycardia is successfully broken by standard defibrillation techniques, but subsequently returns. It is managed by ongoing treatment of correctible causes and use of anti-arrhythmic medication therapies.
- Refractory ventricular fibrillation/tachycardia is an arrhythmia not responsive to standard external defibrillation techniques. It is initially managed by alternate pad placement and defibrillation.
- Prolonged cardiac arrests may lead to tired providers and decreased quality. Ensure compressor rotation, summon additional resources as needed, use mechanical CPR device if available and ensure provider rest and rehab during and post event.

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 112 Ventricular Tachycardia with a Pulse

Signs and Symptoms
Confirm and record cardiac rhythm with electrodes in two leads
Check for palpable carotid pulse
Decreased / altered mental status
Dyspnea
Chest pain / discomfort, suspected AMI
Hypotension (systolic BP < 90 mmHg)
CHF / pulmonary edema
Heart rate > 150 /min ***(peds >200 /min)*** and QRS > .12 sec ***(peds QRS> .09 sec)*** (3 small blocks)

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Supportive Care.
3. Pulse Oximetry.
4. Glucose check.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. INT or IV NS TKO.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. If rhythm is stable, regular, and monomorphic, administer 12 mg Adenosine via rapid IV push.
9. If rhythm possibly Torsades de Pointes – Go to Torsades de Pointes protocol.
10. If BP < 90 mmHg prepare for synchronized cardioversion.
    a. Administer sedative as necessary – Diazepam (Valium) 2-5 mg IV ***(peds 0.1 mg/kg)*** OR Midazolam (Versed) 2-5 mg IV ***(peds 0.1 mg/kg***) or Ketamine and/or Pain Medications per pain management protocol.
    b. Synchronize cardiovert beginning at 50 joules initial energy level until heart rate < 150 /min ***(peds begin at 0.5 j/kg).***
    c. If rhythm converts, monitor for changes, transport. If rhythm does not convert, administer Amiodarone 150 mg over 10 minutes ***(peds 5 mg/kg)***. Reattempt cardioversion @ 100 joules ***(peds 0.5 j/kg).***
    d. Contact Medical Control.
11. If systolic BP > 90 mmHg – stable/asymptomatic.
    a. Have patient perform Valsalva Maneuver for 10 seconds and administer Amiodarone 150 mg ***(peds 5 mg/kg, may repeat up to a total of 15 mg/kg)*** over 10 minutes.
    b. If rhythm converts, monitor for changes, transport. If rhythm does not convert, administer Amiodarone 150 mg over 10 minutes (maximum 3 150 mg doses) ***(peds three doses of 5 mg/kg).***

PARAMEDIC STOP

CARDIAC EMERGENCY
SOP # 113 Post Resuscitation

Signs and Symptoms	Notes
Completion of arrhythmia treatment	• Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for patient's condition.
2. Supportive care.
3. Pulse oximetry.
4. Glucose check.
5. 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. IV NS TKO.
7. Check fingerstick glucose level. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
8. Assess BP – if systolic < 90 mmHg administer 250 ml NS Bolus ***(peds systolic BP 70 + 2 x age, 20 cc/kg bolus)*** repeat until BP > 90 mmHg or appropriate for pediatric age.
9. Raise Head of Bed 30°.

AEMT STOP HERE

10. Medications:
    a. If anti-arrhythmic is administered:
        i. Amiodarone – 300 mg IV ***(peds 5 mg/kg, may repeat x 2)***, if one dose given and arrhythmia persists, give a second dose 150 mg.
    b. If continued hypotension and/or bradycardia despite volume replacement:

    EPINEPHrine 2–20 mcg/min ***(peds 0.1-1 mcg/kg/min).***

    To prepare EPINEPHrine 10 mcg/ml:
    1. Draw up 9 ml of normal saline into 10 ml syringe
    2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) (1:10,000)
    3. Label syringe

    Adult Push Dose Pressor IV Administration:

    Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

    Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.
11. For adults, continue ventilator support to maintain $EtCO_2$> 20. Respirations < 12 per minute ideally. ***(Peds – infants-preschool minimum respiratory rate should be 30, School aged children, minimum respiratory rate should be 20).***
12. Initiate Induced Hypothermia protocol if appropriate.
13. Ensure Head of Bed elevated 30°.

Treatment – Protocol

If patient does not tolerate ET tube or for purposeful movement during CPR:
Utilize patient sedation. **Note:** Use soft restraints if necessary for patient safety (to prevent extubation).

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY

SOP # 201 Chemical Exposure

Signs and Symptoms	Notes
History of exposure to chemical Identify substance and verify with documentation, if possible. Material Safety Data Sheets (M.S.D.S.), if available Stay within the appropriate zone for protection	Personnel safety is the highest priority. Do not handle the patient unless they have been decontaminated. All EMS treatment should occur in the Support Zone after decontamination of the patient. Appropriate PPE will be utilized.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Supportive care.
3. IV NS TKO or INT PRN.
4. Treatment – Standing Order

If Internal Exposure and Conscious:

a. Treat as Drug Ingestion
b. Contact Medical Control

If External Exposure:

a. Remove victims clothing, jewelry, glasses, and contacts
b. Decontaminate – EMS Personnel must be wearing proper protective clothing prior to helping with the decontamination process

Powder or like substance:

a. Brush off patient
b. Flush with copious amounts of water for at least 20 minutes; assess for hypothermia q 5 minutes
c. Transport and continue flushing if necessary and possible

Liquid substance:

a. Flush with copious amounts of water for at least 20 minutes; assess for hypothermia q 5 minutes
b. Transport and continue flushing if necessary and possible

If Inhalation:

a. Reconsider Self-Contained Breathing Apparatus
b. Remove victim from source ensuring there is no danger to personnel
c. Oxygen and airway maintenance appropriate to patient's condition

If Ocular:

a. Immediately flush eye with tap water or normal saline for 15 minutes
b. Contact Medical Control

Note: Coordinate through the HazMat officer prior to transport

EMT AEMT PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 202 Drug Ingestion

Signs and Symptoms	Notes
History of drug ingestion Level of consciousness (**A**lert, **V**erbal, **P**ain, or **U**nresponsive) Neurologic status (LOC, pupils) General appearance (sweating, dry or flushed skin, signs of trauma)	• Poison control may be contacted for **INFORMATION ONLY**. Treatment modalities are given within these protocols. Further treatments will be received through Medical Control. • Consider IO if other access unavailable and patient significantly symptomatic.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Ensure personnel protection from toxin and/or unruly patient.
3. Supportive care.
4. Obtain Pulse Oximetry while providing 2 minutes ventilatory support with BVM.
5. Utilize Naloxone Auto-injector or Naloxone 0.4 IN, if needed. If utilizing pre-filled delivery systems, dose per manufacturer's instructions.
6. Glucose check.
7. EKG Monitor – 12 Lead, transmission if applicable.

EMT STOP HERE

8. IV NS TKO or INT PRN.
9. Check fingerstick glucose level. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
10. Naloxone 0.4 mg IV/IO/IM/IN titrated to adequate ventilation ***(peds 0.1 mg/kg IVP/IN)*** if narcotic use is suspected. If utilizing pre-filled delivery systems, dose per manufacturer's instructions. May repeat dose. Synthetic opiates may require significantly larger doses of Naloxone. Physically manage airway if no response after 8 mg Naloxone.

AEMT STOP HERE

11. Observe EKG carefully for evidence of hyperkalemia and treat accordingly. Note: Patients with significant Opiate overdoses have been known to develop rhabdomyolysis.
12. If actively seizing:

    Adults:

    Diazepam (Valium) SLOW IVP/IO 2-5 mg or Midazolam (Versed) 2-5 mg IV/IO/IM may repeat if seizure continues, OR

    LORazepam 1-2 mg IV, every 5 minutes or; 2-4 mg IM, every 10 minutes (maximum dose 8 mg)

    Peds:

    a. ***Diazepam (Valium) 0.1 mg/kg or Midazolam (Versed) 0.1 mg/kg IV/IO, OR***
    b. ***Midazolam (Versed) IM 0.2 mg/kg IM (max single dose 6 mg). Repeat once if seizure activity persists after 10 minutes. Contact MEDICAL CONTROL if seizure activity persists after repeat dose.***
    c. ***Midazolam (Versed) IN 0.3 mg/kg IN (max single dose 10 mg) with maximum total dose of 0.4 mg/kg.***
    d. ***LORazepam (peds 0.1 mg/kg IV/IO, max single dose 4 mg, may repeat in 5 minutes if seizure activity continues; not to exceed 0.2 mg/kg total (maximum of 8 mg).***

    If seizure persists for 4 minutes repeat medication once. Contact Medical Control to consider Ketamine.

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 203 Electrocution / Lightning Injuries

Signs and Symptoms
Presence of signs and symptoms of electrical injury

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for patient's condition.
2. Spinal protection if electrocution/lightning over 1,000 volts or suspicion of spinal injury.
3. Treat burn per burn protocol.
4. Supportive care.
5. Control any gross hemorrhage and dress wounds.
6. Pulse oximetry.
7. EKG Monitor and transmit 12 Lead EKG if applicable.

EMT STOP HERE

8. IV LR – if signs of shock 20 cc/kg bolus of fluid ***(peds 20 cc/kg bolus).***

AEMT STOP HERE

9. Consider 2nd IV enroute to hospital.
10. Consider pain medication per protocol.

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 204 Hyperthermia

Signs and Symptoms	Notes:
History of exposure to warm temperature Usually seen with increased exertion Febrile May have hot and dry **or** warm and moist skin May be hypotensive Determine history of therapeutic drug use (antipsychotics); history of substance abuse (cocaine, amphetamines, etc.) Poor skin turgor Signs of hypovolemic shock History of infection or illness Drug use Dark urine – suggests muscle break-down and possible kidney damage Tachycardia, hyperventilation, hypertension Neurologic – light headedness, confusion to coma, seizures	1. Time is of the essence in decreasing the patient's body temperature. 2. **DO NOT** use IV iced saline for cooling patient. Use of fluids cooled slightly below ambient temperature is appropriate. 3. Hyperthermia may be caused by the following: • Antipsychotic Medications and major tranquilizers: Phenothiazine (Thorazine®), Butyrophenones (Haldol®) • SSRI (Selective Serotonin Reuptake Inhibitors): Citalopram (Celexa®), Escitalopram (Lexapro®), Fluoxetine (Prozac®), Paroxetine (Paxil®, Pexeva®), Sertraline (Zoloft®), Vilazodone (Viibryd®) • Cyclic antidepressants such as: Elavil®, Norpramin®, Tofranil® • Amphetamines • Monoamine Oxidase Inhibitors (MAOI) such as: Nardil®, Marplan® • Anticholinergic drugs such as: Atropine, Cogentin, Scopolamine • Illicit drugs: Cocaine, PCP, LSD, Ecstasy (MDMA)

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Remove clothing: apply wet linen or wet abdominal pads to groin/axillary area
    a. Expose to circulating air
    b. **DO NOT** cool patient to the point of shivering
3. Move patient to protected environment (shade, AC, etc.).
4. Pulse Oximetry.
5. Glucose check.
6. Monitor – 12 Lead EKG and transmit if applicable.

EMT STOP HERE

7. IV NS or LR 20 cc/kg bolus ***(peds 20 cc/kg bolus).*** Note: **DO NOT** use chilled IV fluids.
    a. Repeat second bolus of fluid if needed
    b. Oral rehydration if patient able to maintain airway
8. GENTLY massage extremities to prevent cold induced vasoconstriction.
9. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

10. EKG Monitor – Observe for hyperkalemia.

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 205 Hypothermia

Signs and Symptoms	Notes:
History of exposure to cold temperature including duration Core body temperatures < 95°F Drug/Alcohol use CNS depressants Examine for associated trauma Immersion in cold water Predisposing medical condition Signs – Vital signs, Bradycardia, Hypotension, Cold extremities, Neurologic (confusion, altered LOC, coma)	• If patient is alert and responding appropriately, rewarm actively: ○ Heat packs or warm water bottles to the groin, axillary, and cervical areas • If patient is unresponsive, rewarm passively: ○ Increase the room temperature gradually, cover with blankets • The following signs and symptoms are found at varying body core temperature: ○ 95°F – amnesia, poor judgment, hyperventilation, bradycardia, shivering ○ 90°F – loss of coordination (drunken appearance), decreasing rate and depth of respirations, shivering ceases or bradycardia ○ 85°F – decreased LOC, slow respirations, atrial fibrillation, decreased BP, decreased heart rate, ventricular irritability

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Remove the patient from the cold environment.
3. Remove wet clothing and cover with warm dry blankets.
4. Evaluate pulse for one full minute (DO NOT perform CPR until NO PULSE is confirmed).
5. Handle patient gently *(aggressive handling may trigger V-Fib).*
6. Do not allow patient to walk or exert themselves.
7. Do not massage extremities.
8. Glucose check.
9. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

10. IV NS warmed if possible ***(peds 20 cc/kg bolus then 4 cc/kg/hr).***
11. Check fingerstick glucose level. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
12. If patient in coma, Naloxone (Narcan) 0.4 mg IV/IO/IM/IN titrated to adequate ventilation ***(peds 0.1 mg/kg slow IVP/IN/IO/IN. If no response, may repeat q 2-3 min with maximum single dose of 2 mg).*** If utilizing pre-filled delivery systems, dose per manufacturer's instructions. May repeat dose.

AEMT STOP HERE

13. EKG monitor, no CPR if Bradycardic rhythm exists.
14. If body temperature > 85°F – follow normal arrest protocols.
15. If body temperature < 85°F and patient in V-fib:
    - ***a.*** Defibrillate @ 100 j, if no change, begin CPR ***defib at 2 min intervals, increase joules at each interval until 200 j max*** (120 j, 150 j, 200 j) ***(peds 2 j/kg then 4 j/kg)***
    - b. Withhold medications and further shocks until patient warmed to >85°F
    - c. Continue CPR and rewarming attempts

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 206 Drowning/ Near Drowning

Signs and Symptoms	Notes:
History compatible with near drowning Suspect hypothermia in “cold water” near drowning Suspect cervical spine injury	• Reinforce the need to transport and evaluation for all patients with a submersion incident. • Consider C-Spine protection

Treatment Pathway

1. Oxygen and airway maintenance appropriate to the patient’s condition:
    - Heimlich Maneuver may be indicated for airway obstruction
    - Gastric decompression may be necessary to ensure adequate respirations or ventilations; if necessary, ventilations may be started prior to patient’s removal from the water
2. Remove patient from the water, clear airway while protecting the C-Spine ASAP.
3. If patient is unconscious and pulseless – refer to the appropriate cardiac arrest protocol.
4. If hypothermic – go to hypothermia protocol.
5. Supportive care.
6. Pulse oximetry.
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. INT or IV NS TKO, if hypotensive give 20 cc/kg bolus of fluid ***(peds 20 cc/kg).***

AEMT STOP HERE

9. EKG Monitor and treatment specific for arrhythmia.

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 208 Venomous Snake Bite

Signs and Symptoms	Notes
Protect yourself from the exposure of snakebite. Snakes can envenomate up to one hour after death. Determine type of snake if possible, time of bite, and changes in signs and symptoms since occurrence If possible, obtain pictures of the snake for identification Paresthesia (numbing or tingling of mouth, tongue, or other areas) Local pain Peculiar or metallic taste Chills, nausea and vomiting, headache, dysphagia Hypotension Fever Local edema, blebs (blister or pustule jewel), discoloration Bite wound configuration	**DO NOT USE** ice, tourniquets, hemorrhage control clamps, or constricting bands at the bite site or proximal to bite site. If already applied, remove. **DO NOT** place IV or IO in affected extremity if possible.

TREATMENT PATHWAY

1. Remove rings and bracelets from patient immediately.
2. Oxygen and airway maintenance appropriate to patient's condition.
3. Immobilize affected area keeping extremities in neutral position.
4. Mark progression of swelling at the time of initial assessment and q 5 minutes.
5. Supportive care.
6. Pulse oximetry.
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. INT or IV NS TKO, if hypotensive 20 mL/kg ***(peds 20 mL/kg)*** NS bolus.

AEMT STOP HERE

9. EKG Monitor. Assess and treat arrythmias.

Treatment – Protocol:

Diazepam (Valium) or Midazolam (Versed) may be indicated if anxiety is overwhelming. Contact Medical Control prior to initiating therapy. ***(peds Diazepam (Valium) 0.1 mg/kg IV, Midazolam (Versed) 0.1 mg/kg IV for anxiety)***

Consider use of pain management protocol.

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 209 Radiation / HazMat Exposure

Signs and Symptoms	Notes
Extent of radiation/chemical exposure (number of victims, skin vs. inhalation exposure) Nature of exposure Symptoms exhibited by patient Neurologic status (LOC, pupil size) General appearance (dry or sweaty skin, flushed, cyanotic, singed hair) Associated injuries Decontamination prior to treatment	Personnel safety is the highest priority. Do not handle the patient unless they have been decontaminated. All EMS treatment should occur in the Support Zone after decontamination of the patient. Appropriate PPE will be utilized.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient condition.
2. If eye exposure, irrigate for a minimum 20 minutes.
3. Treat associated injuries (LSB, limb immobilization, wound treatment).
4. Supportive care.
5. Treat burn per burn protocol.
6. Pulse oximetry (keep sats > 94%).
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. INT or IV NS/LR, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

9. EKG Monitor. Assess and treat arrythmias.

PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 210 Carbon Monoxide Exposures

Signs and Symptoms	**Notes:**
Known or suspected CO exposure (Active Fire Scene) Suspected source/duration exposure Known or possible pregnancy Measured atmospheric levels Past medical history, medications Altered mental status/dizziness Headache, nausea/vomiting Chest pain/respiratory distress Neurological impairments Vision problems/reddened eyes Tachycardia/tachypnea Arrhythmias, seizures, coma	• If monitoring responders at fire scene, proceed with Scene Rehabilitation Protocol (SOP 203.09). • *Fetal hemoglobin has a greater attraction for CO than maternal hemoglobin. Females who are known to be pregnant or who could be pregnant should be advised that EMS measure SpCO levels reflect the adult's level, and that fetal COHb levels may be higher. Recommend hospital evaluation for any CO exposed pregnant person. • The absence (or low detected levels) of COHb is not a reliable predictor of firefighter or victim exposure to other toxic byproducts of fire. • In obtunded fire victims, consider HazMat Cyanide treatment protocol. • The differential list for CO toxicity is extensive. Attempt to evaluate other correctible causes when possible. • Utilize a device with the ability to monitor CO, these functions should be utilized any time Carbon Monoxide Exposure is suspected. • Transport patients with CO/CN toxicity to the burn center for evaluation.

TREATMENT PATHWAY

Measure Carbon Monoxide COHb% (SpCO)

If SpCO is 0% - 5% no further medical evaluation of SpCO is required*

SpCO < 15% **and** SpO_2 > 90%

If patient has **NO** symptoms of CO and/or Hypoxia no treatment for CO exposure is required*
Recommend that smokers seek smoking cessation treatment
Recommend evaluation of home/work environment for presence of CO

SpCO < 15% **and** SpO_2 > 90% that show symptoms of CO and/or Hypoxia transport

> 15% Oxygen by NRB and transport to ED
If cardiac/respiratory/neurological symptoms are also present, go to the appropriate protocol

Paramedic: Consider use of Cyanide Antidote if available.

EMT AEMT PARAMEDIC STOP

ENVIRONMENTAL EMERGENCY
SOP # 211 Riot Control Agents

Signs and Symptoms	Notes
Exposure to an identifiable riot control agent not intended to cause significant injury or fatality: CS (o-chlorobenzylidene malononitrile, tear gas), CN (2-chloroacetophenone, Mace®), OC (oleoresin capsicum, pepper spray); other lacrimators / harassing agents Eyes: tearing, pain, conjunctivitis, blurred vision Nose/mouth/throat: rhinorrhea, burning/pain, trouble swallowing, drooling Lungs: chest tightness, coughing, choking sensation, wheezing, dyspnea Skin: burning, redness, dermatitis GI: nausea and vomiting are rare and may be post-tussive	Excludes exposure to chlorine, phosgene, ammonia, or other agents intended to cause significant injury or fatality, and exposure to an UNKNOWN agent (SOP 201). Symptoms begin within seconds, are self-limited, and are best treated by removing the patient from ongoing exposure. They frequently decrease over 15–45 minutes after exposure ends.

TREATMENT PATHWAY

1. Assess scene safety: evaluate hazards to EMS personnel, patient, and bystanders.
    a. Determine the riot control agent being used
    b. Don appropriate PPE
    c. Determine the number of patients
2. Move affected individuals from the contaminated environment into fresh air if possible.
3. Remove contaminated clothing as able. Have the patient remove contact lenses if appropriate.
4. Irrigate with water or saline – recommended for decontamination of dermal and ocular exposure.
5. Note symptoms exhibited; examine as appropriate to complaints; assess lung sounds and for other traumatic injuries (e.g., from hose/stream under pressure, projectile, or grenade).
6. Pulse oximetry.
7. If the patient is in respiratory distress or wheezing, treat per SOP 314 (Respiratory Distress).
8. For persistent pain of the eye or skin, treat per SOP 214 (Topical Chemical Burn).
9. Exposed individuals who remain persistently symptomatic warrant further evaluation and treatment – contact Medical Control / transport per local standards.

Patient Safety Considerations

1. Toxicity is related to duration of exposure and concentration of the agent (exposure in a non-ventilated space).
2. Patients with pre-existing pulmonary conditions (asthma, COPD) may be prone to more severe respiratory effects.
3. Traumatic injury may result when exposed individuals are in proximity to the device used to disperse the agent (hose/stream under pressure, projectile, grenade).

Key Considerations

1. CN, CS, and OC are the most commonly encountered riot control agents. All three have a high safety ratio (high median lethal concentration and low median effective concentration).

Documentation

Type of riot control agent if known; symptoms being treated; treatment provided; response to treatment; PPE used by responders; agent identified before patient contact.

ENVIRONMENTAL EMERGENCY
SOP # 212 Dive (SCUBA) Injury / Accidents

Signs and Symptoms	Notes
Recent (within 48 hours) SCUBA diving with potential signs and/or symptoms of dive-related illness/injury, regardless of dive table compliance. Complications may occur anywhere, particularly when divers travel by air within 24 hours of diving. **Barotrauma / arterial gas embolism:** Pulmonary barotrauma, arterial gas embolism, pneumothorax, pneumomediastinum, subcutaneous emphysema; ear/sinus/dental barotrauma; dysrhythmias; skin mottling or erythema; neurologic signs and symptoms **Decompression sickness:** Joint pain, mental status change, other neurologic symptoms including paralysis **Nitrogen narcosis:** Confusion, intoxication Other: head or spine injury, marine envenomation, hypothermia, drowning/near-drowning	Excludes patients without a history of SCUBA diving within the past 48 hours. DO NOT disassemble, turn off, or modify any dive equipment. The dive computer may provide a clue about the patient's exposure to depth. Check for multiple patients (group dive table errors, contaminated dive gas).

TREATMENT PATHWAY

1. Scene safety / rescue: if the patient is still in the water, use the safest and most rapid means of removal (within scope of training) while minimizing risk of further injury. Seek assistance early for special rescue/extrication and transportation needs. Coordinate all responding agencies.
2. Obtain history: circumstances leading to the complaint, mechanism of injury, time under water, depth of dive, compliance with dive tables/decompression stops, gas mixture used, water temperature (if available).
3. If the accident includes drowning/near-drowning, treat per SOP 206.
4. Manage the airway as indicated and provide 100% oxygen. Patients with symptoms suspicious for decompression illness should receive supplemental oxygen regardless of saturation to enhance washout of inert gases. Otherwise, administer oxygen with a target SpO2 of 94–98%.
5. If air embolism is suspected, place in the left lateral recumbent position (left side down, knees drawn upward, flat). Trendelenburg is sometimes recommended to trap air in the dependent right ventricle, but it may increase cerebral edema.
6. Monitor vital signs including oxygen saturation and cardiac rhythm (if possible).
7. Use positive pressure ventilation (CPAP) carefully in patients for whom pulmonary barotrauma is a consideration.
8. Assess for hypothermia and treat per SOP 205. Assess for other associated injury (head/spine per SOP 608, marine envenomation).
9. Contact Medical Control early to discuss the need for hyperbaric treatment and primary transport to a facility with hyperbaric oxygen therapy (HBOT) capability. Include submersion time, greatest depth achieved, ascent rate, and gas mix.
10. If air medical transport is necessary, transport with the cabin pressurized to the lowest possible altitude. If an unpressurized aircraft is used, fly at the lowest safe altitude possible.

EMT STOP HERE

11. Establish IV access.
12. Fluid bolus as indicated.

AEMT STOP HERE

13. If signs or symptoms of tension pneumothorax are present, perform needle decompression per SOP 418.

PARAMEDIC STOP

Key Considerations

1. SCUBA accidents/incidents can result in barotrauma, air embolism, and decompression illness.
2. Decompression illness may have a variety of presentations depending on the system affected (skin, joint(s), pulmonary, neurologic) and can occur even when a diver does not exceed dive table limits.

Documentation

Water temperature if available; dive history (number of dives in recent days, bottom time, dive profiles, maximum depth, rate of ascent, safety stops, dive gas – air vs. Nitrox, Heliox, or Trimix); timing of symptom onset; history of altitude exposure after diving (air travel); associated injuries or exposures; vital signs, neurologic, and respiratory findings; subcutaneous emphysema.

ENVIRONMENTAL EMERGENCY

SOP # 213 Acetylcholinesterase Inhibitor Exposure (Carbamates, Nerve Agents, Organophosphates)

Signs and Symptoms	Notes
DUMBELS toxidrome: Diarrhea; Urination; Miosis / Muscle weakness; Bronchospasm / Bronchorrhea / Bradycardia ("the killer Bs"); Emesis; Lacrimation; Salivation / Sweating Heart rate may be normal, bradycardic, or tachycardic CNS: headache, confusion, generalized muscle weakness, seizures, lethargy or unresponsiveness **Estimated severity:** MILD – miosis alone (primary sign in vapor exposure, may not be present in all exposures); miosis and severe rhinorrhea MILD TO MODERATE – (plus mild) localized swelling, muscle fasciculations, nausea and vomiting, weakness, shortness of breath SEVERE – (plus mild to moderate) unconsciousness, convulsions, apnea or severe respiratory distress requiring assisted ventilation, flaccid paralysis Large exposures (any route): sudden loss of consciousness, seizures, copious secretions, apnea, death	Aliases: carbamate, organophosphate, nerve agent (GA, GB, GD, GF, VX), insecticide, pesticide, WMD. Personnel safety is the highest priority. Do not handle the patient until decontaminated. Accidental carbamate exposure rarely requires treatment. Clinical improvement = drying of secretions and easing of respiratory effort, NOT heart rate or pupil response.

TREATMENT PATHWAY

1. Don appropriate PPE.
2. Remove the patient's clothing and wash the skin with soap and warm water (agents are absorbed through skin; contaminated clothing is a source of continued exposure).
3. Rapidly assess respiratory status, mental status, and pupillary status. Obtain exposure history: time, route, quantity, alcohol or other intoxicants, cardiovascular history.
4. Administer ATROPINE immediately for confirmed or suspected exposure, using auto-injector (EMT) per the severity tables below. Repeat until secretions diminish; additional auto-injectors may be given. A dose calculation is not necessary before using an auto-injector.
5. Administer oxygen with a target SpO2 of 94–98% and provide airway management.
6. Pralidoxime (2-PAM) 600 mg auto-injector is given concurrently with atropine when possible and early – it has minimal effect if delayed.
7. Seizures: administer a benzodiazepine per Table 4 (auto-injector diazepam where available).
8. Continuous reassessment. Transport emergently to the closest appropriate facility as directed by Medical Control. Mass exposure: notify the incident commander early – local antidote stock is usually insufficient; request additional/CHEMPACK assets.

EMT STOP HERE

9. Establish IV access (IM administration of antidotes is preferred – do not delay for IV/IO access).
10. Apply cardiac monitor.

AEMT STOP HERE

11. Atropine IM/IV/IO in large, repeated doses per Tables 1–3 until secretions resolve and respiratory effort eases. Doses may exceed 20 mg in organophosphate or severe poisoning. Give until the agent has been metabolized.
12. Pralidoxime chloride IM/IV per Tables 2–3.
13. Seizures: diazepam or midazolam per Table 4 ***(peds dose by weight)***.

PARAMEDIC STOP

Auto-Injector and Antidote Information

1. Atropine is the primary antidote. Pralidoxime chloride is secondary.
2. DuoDote® and ATNAA®: one auto-injector contains atropine 2.1 mg and pralidoxime chloride 600 mg. Because they contain pralidoxime, do not use them for additional atropine beyond the recommended pralidoxime dose; use other atropine formulations (Atro-Pen® 0.25 mg yellow, 0.5 mg red, 1 mg blue, 2 mg green).
3. CANA® auto-injector: diazepam 10 mg.
4. Mark 1 kits, DuoDote®, and ATNAA® are not FDA-approved for pediatrics but may be considered for initial treatment of children of any age with severe symptoms, especially if other atropine formulations are unavailable. A pralidoxime 600 mg auto-injector may be given to an infant weighing more than 12 kg.
5. All antidotes may be given IV in the same doses as IM; IM is highly recommended to avoid delay. IO is acceptable but IM remains preferable.
6. Midazolam: there is emerging evidence that the intranasal route may be preferable to IM; IM absorption may be more effective in the presence of significant rhinorrhea.
7. CHEMPACK: federal antidote cache managed by CDC. EMS containers hold enough antidote for roughly 454 patients; hospital containers for roughly 1,000. Deployment is reserved for events that will deplete local or regional antidote supply.

Patient Safety Considerations

1. Continuous and ongoing reassessment is critical. Response is shown by drying of secretions and easing of respiratory effort. Do not base initiation or ongoing treatment on heart rate or pupillary response.
2. Excess pralidoxime: pediatrics – profound neuromuscular weakness and respiratory depression; adults, especially geriatric – severe systolic and diastolic hypertension, neuromuscular weakness, headache, tachycardia, visual impairment. In geriatric patients with impaired kidney function or hypertension, consider the lower recommended dose.
3. Low-dose chronic exposures may present late. Onset can be immediate after a large exposure; after liquid exposure there is usually an asymptomatic interval of minutes; vapor exposure effects occur almost immediately.

Key Considerations

1. Clinical effects result from inhibition of acetylcholinesterase, allowing excess acetylcholine to accumulate in the nervous system (muscles, glands, nerves).
2. Organophosphate insecticides can be legally purchased by the public, penetrate tissues, and bind to body fat, producing prolonged illness and ongoing toxicity even during aggressive treatment.
3. Nerve agents are extremely toxic, rapidly fatal by any route, and can persist in the environment.

Dosing Tables (IM or via auto-injector unless noted)

Table 1. MILD Exposure – Atropine

Patient (weight)	Atropine IM or via auto-injector
Infant 0–2 years	0.05 mg/kg (i.e., 0.25 and/or 0.5 mg auto-injector[s])
Child 3–7 years (13–25 kg)	1 mg (one 1 mg or two 0.5 mg auto-injectors)
Child 8–14 years (26–50 kg)	2 mg (one 2 mg or two 1 mg auto-injectors)
Adolescent / Adult	2 mg
Pregnant women	2 mg
Geriatric / frail	1 mg

Table 2. MILD TO MODERATE Exposure

Patient (weight)	Atropine IM or via auto-injector	Pralidoxime chloride IM or via 600 mg auto-injector
Infant 0–2 years	0.05 mg/kg (0.25 and/or 0.5 mg auto-injector)	15 mg/kg IM
Child 3–7 years (13–25 kg)	1 mg (one 1 mg or two 0.5 mg auto-injectors)	15 mg/kg IM OR one auto-injector (600 mg)
Child 8–14 years (26–50 kg)	2 mg (one 2 mg or two 1 mg auto-injectors)	15 mg/kg IM OR one auto-injector (600 mg)
Adolescent / Adult	2–4 mg	600 mg IM OR one auto-injector (600 mg)
Pregnant women	2–4 mg	600 mg IM OR one auto-injector (600 mg)
Geriatric / frail	2 mg	10 mg/kg IM OR one auto-injector (600 mg)

Table 3. SEVERE Exposure

Patient (weight)	Atropine IM or via auto-injector	Pralidoxime chloride IM or via auto-injector
Infant 0–2 years	0.1 mg/kg (0.25 and/or 0.5 mg auto-injector)	45 mg/kg IM
Child 3–7 years (13–25 kg)	0.1 mg/kg IM OR 2 mg via auto-injector (one 2 mg or four 0.5 mg)	45 mg/kg IM OR one auto-injector (600 mg)
Child 8–14 years (26–50 kg)	4 mg (two 2 mg or four 1 mg auto-injectors)	45 mg/kg IM OR two auto-injectors (1200 mg)
Adolescent 14 years or older	6 mg (three 2 mg auto-injectors)	Three auto-injectors (1800 mg)
Adult	6 mg (three 2 mg auto-injectors)	Three auto-injectors (1800 mg)
Pregnant women	6 mg (three 2 mg auto-injectors)	Three auto-injectors (1800 mg)
Geriatric / frail	2–4 mg (one to two 2 mg auto-injectors)	25 mg/kg IM OR two to three auto-injectors (1200–1800 mg)

Table 4. Seizures Secondary to Acetylcholinesterase Inhibitor Exposure

Patient	Diazepam	Midazolam
Infant (0–2 y/o)	0.2–0.5 mg/kg IM, repeat q 2–5 min; or 0.2–0.5 mg/kg IV q 15–30 min, may repeat twice. Total max 5 mg	0.2 mg/kg IM, repeat prn in 10 min, may repeat once. Total max 0.4 mg/kg
Child (3–13 y/o)	0.2–0.5 mg/kg IM, repeat q 2–5 min; or 0.2–0.5 mg/kg IV q 15–30 min, may repeat twice. Total max 5 mg if less than 5 years; 10 mg if 5 years or older. 1 CANA® auto-injector	0.2 mg/kg IM, not to exceed 10 mg, repeat prn in 10 min, may repeat once. Total max 0.4 mg/kg, not to exceed 20 mg
Adolescent (14 y/o or older)	2–3 CANA® auto-injectors; or 5–10 mg IV q 15 min. Total max 30 mg	0.2 mg/kg IM (total max single dose 10 mg), repeat prn in 10 min, may repeat once. Total max 20 mg
Adult / pregnant women / geriatric	2–3 CANA® auto-injectors; or 5–10 mg IV q 15 min. Total max 30 mg	10 mg IM, repeat prn in 10 min, may repeat once. Total max 20 mg

Documentation

Time to recognize initial signs and symptoms; number of repeated atropine doses required for secretions to diminish and respirations to improve; patient reassessments and response to therapy; decontamination measures; measures taken to protect clean environments from contamination.

ENVIRONMENTAL EMERGENCY
SOP # 214 Topical Chemical Burn

Signs and Symptoms	Notes
Exposure to a chemical that can cause a topical chemical burn (alkalis, acids, mustard agent, lewisite, hydrofluoric acid); presentation may be immediate or delayed Severity depends on: class of agent (alkali or acid); concentration (higher = greater risk); pH (alkali: increased risk at pH 11 or greater; acid: increased risk at pH 3 or less); onset (immediate vs. delayed, e.g., hydrofluoric acid) Estimated total body surface area (TBSA) involved Ocular or oropharyngeal exposure – evaluate for airway compromise (spasm, direct injury) Some acid and alkali agents cause systemic effects	Do NOT attempt to neutralize an acid with an alkali or an alkali with an acid – an exothermic reaction will cause serious thermal injury. Burn center consideration for significant TBSA, or burns of eyes, face, hands, feet, or genitals. Contact CHEMTREC or Poison Control (information only).

TREATMENT PATHWAY

1. Don appropriate PPE. Take measures to protect EMS clinicians and others from contamination and to prevent further contamination of the patient.
2. Remove the patient's clothing as necessary; place contaminated clothing in double bags.
3. Dry chemical: carefully brush off solid chemical BEFORE flushing (irrigating solution may activate a chemical reaction).
4. Wet chemical: flush the skin (and eyes, if involved) with copious amounts of water or normal saline.
5. Eye exposure: continuous flushing with irrigation fluid (Morgan lens may facilitate).
6. Gather information on the chemical on scene, including the safety data sheet (SDS/MSDS) if available. Communicate all data about the chemical to the receiving facility.
7. Calculate estimated TBSA involved. Take measures to minimize hypothermia.
8. Early airway intervention for airway compromise or bronchospasm associated with oropharyngeal burns.
9. If deemed necessary and manpower permits, transport using EMS clinicians who did not participate in decontamination and a vehicle not exposed to the chemical.
10. Hydrofluoric acid (HF) exposure, confirmed or suspected: vigorously irrigate all affected areas with water or normal saline for a minimum of 15 minutes, then apply calcium gluconate gel (see below).

EMT STOP HERE

11. IV access; fluid resuscitation if necessary to obtain hemodynamic stability (Parkland formula per SOP 407).
12. Analgesia per the Pain Management Protocol.
13. Cardiac monitor for oral or large dermal HF exposures.

AEMT STOP HERE

14. Topical anesthetic eye drops (e.g., tetracaine) for chemical burns of the eye.
15. HF ingestion or large dermal exposure: consider IV calcium gluconate 1–2 g of 10% solution (symptomatic hypocalcemia can develop rapidly – muscle spasms, seizures, hypotension, ventricular arrhythmias, QT prolongation).

PARAMEDIC STOP

Hydrofluoric Acid (HF) – Topical Calcium Preparations

1. HF readily penetrates intact skin; pain is typically out of proportion to visible findings. Low concentrations may cause delayed pain. Oral or large dermal exposure can cause significant systemic hypocalcemia with

QT prolongation and cardiovascular collapse. Consider inhalation exposure for any dermal exposure involving the face and neck, or clothing soaked in the product.
2. Apply commercially manufactured calcium gluconate gel generously to exposed skin. Leave in place for at least 20 minutes, then reassess; repeat as needed.
3. If gel is not available, prepare topical gel by combining 150 mL (5 oz) of a sterile water-soluble gel (e.g., Surgilube® or KY® jelly) with ONE of: 35 mL of calcium gluconate 10% solution; 10 g of calcium gluconate tablets (e.g., Tums®); or 3.5 g calcium gluconate powder.
4. If calcium gluconate is unavailable: 10 mL of calcium chloride 10% in 150 mL of sterile water-soluble gel.
5. Fingers: apply gel to the hand, squirt additional gel into a surgical glove, and insert the hand into the glove.
6. IV analgesics may be less effective but should be added to calcium gel for pain control.

Key Considerations

1. Obtain as much information as possible about the chemical (type, concentration, pH): transport the sealed container and/or a copy of the SDS to the receiving facility; contact CHEMTREC® to identify the agent.
2. Decontamination is critical for acids and alkalis. Removal of acids is more easily accomplished than alkalis, which penetrate and bind to deeper tissues.
3. Some chemicals also cause local and systemic signs, symptoms, and bodily damage.

Documentation

Burn site; body surface area involved; identification of the chemical; reported or measured pH; acquisition and transfer of SDS/container to the receiving facility; response to fluid resuscitation and analgesia; measures taken to prevent further contamination.

ENVIRONMENTAL EMERGENCY
SOP # 215 Beta Blocker Poisoning / Overdose

Signs and Symptoms	Notes
Bradycardia Hypotension Altered mental status Weakness Shortness of breath Possible seizures Hypoglycemia (especially pediatric) Examples: acebutolol, atenolol, betaxolol, bisoprolol, carteolol, esmolol, metoprolol, nadolol, nebivolol, penbutolol, pindolol, propranolol, timolol, sotalol; alpha/beta blockers carvedilol and labetalol	Early airway protection is required – mental status may deteriorate rapidly. A single pill can kill a toddler. Bring ALL suspect medications to the ED. Pacing may not capture, and may not correct hypotension when it does.

TREATMENT PATHWAY

1. Assess ABCDs; if indicated, expose and then cover to retain body heat. Vital signs including temperature.
2. Oxygen and airway maintenance appropriate to the patient's condition. Monitor pulse oximetry and EtCO2 for respiratory decompensation.
3. Cardiac monitor; examine the rhythm strip for arrhythmias; 12-lead EKG.
4. Check blood glucose on all patients, especially pediatric patients (beta blockers can cause hypoglycemia in children).
5. Identify the medication taken (immediate-release vs. sustained-release), time of ingestion, and quantity. Obtain cardiovascular history and other prescribed medications. Collect suspect medications for transport.

EMT STOP HERE

6. IV/IO access.
7. Fluid challenge 20 mL/kg for hypotension with associated bradycardia.

AEMT STOP HERE

8. If available, activated charcoal without sorbitol 1 g/kg PO ONLY if within the first hour of ingestion. If there is risk of rapidly decreasing mental status, do not give oral agents without adequately protecting the airway.
9. Symptomatic bradycardia: atropine – Adult 1 mg IV q 5 minutes to a maximum of 3 mg. ***(peds 0.02 mg/kg [0.5 mg maximum] q 5 minutes, maximum total dose 1 mg)***
10. Symptomatic patients with cardiac effects (hypotension, bradycardia): GLUCAGON – Adult 5 mg IVP, may repeat in 5–10 minutes for a total of 10 mg. ***(peds 1 mg IVP [25–40 kg] every 5 minutes as necessary; 0.5 mg IVP [less than 25 kg] q 5 minutes as necessary)***. Consider ondansetron prophylaxis (glucagon increases vomiting at these doses).
11. Vasopressors after adequate fluid resuscitation (1–2 L crystalloid) for the hypotensive patient, per the Shock SOP.
12. Transcutaneous pacing if refractory to initial pharmacologic interventions.
13. Widened QRS (100 msec or greater): sodium bicarbonate 1–2 mEq/kg IV; repeat as needed to narrow the QRS.
14. If seizure, treat per SOP 315.

PARAMEDIC STOP

Patient Safety Considerations

1. Transcutaneous pacing may not always capture, nor correct hypotension when capture is successful.
2. Aspiration of activated charcoal (if used) can make airway management nearly impossible. Do not give to any patient who may have worsening mental status.

Key Considerations

1. Atropine may have little or no effect (more helpful in mild overdoses). Hypotension and bradycardia may be mutually exclusive; blood pressure may not respond to correction of bradycardia.
2. Propranolol crosses the blood-brain barrier and can cause altered mental status, seizure, and widened QRS similar to TCA toxicity.
3. Acebutolol and propranolol may increase QRS duration; acebutolol and pindolol may produce tachycardia and hypertension; sotalol can increase QTc and cause ventricular dysrhythmias.
4. Frequent reassessment is essential – deterioration can be rapid and catastrophic.

Documentation

Repeat evaluation of signs/symptoms and vital signs; possible etiology; time of symptom onset and time of exposure-specific treatment; therapy and response; time of ingestion, route, quantity, alcohol or other intoxicants.

ENVIRONMENTAL EMERGENCY
SOP # 216 Calcium Channel Blocker Poisoning / Overdose

Signs and Symptoms	Notes
Bradycardia Hypotension Decreased AV nodal conduction (sinus bradycardia to complete heart block) Cardiogenic shock Hyperglycemia Examples: amlodipine, dilTIAZem, felodipine, isradipine, nicardipine, niFEdipine, nisoldipine, verapamil	Early airway protection is required – mental status may deteriorate rapidly. Hyperglycemia helps differentiate CCB from beta blocker overdose. Dihydropyridines (nifedipine, amlodipine) may cause early reflex tachycardia, then bradycardia as intoxication worsens. A single pill can kill a toddler. Bring ALL suspect medications to the ED.

TREATMENT PATHWAY

1. Assess ABCDs; if indicated, expose and then cover to retain body heat. Vital signs including temperature.
2. Oxygen and airway maintenance appropriate to the patient's condition. Monitor pulse oximetry and EtCO2 for respiratory decompensation.
3. Cardiac monitor; examine the rhythm strip for arrhythmias; 12-lead EKG.
4. Check blood glucose.
5. Identify the medication taken (immediate-release vs. sustained-release), time of ingestion, and quantity. Obtain cardiovascular history and other prescribed medications. Collect suspect medications for transport.

EMT STOP HERE

6. IV/IO access.
7. IV fluid bolus (NS or LR) 20 mL/kg, up to 2 liters.

AEMT STOP HERE

8. If available, activated charcoal without sorbitol 1 g/kg PO ONLY if within the first hour of ingestion. If there is risk of rapidly decreasing mental status, do not give oral agents without adequately protecting the airway.
9. Symptomatic bradycardia: atropine – Adult 1 mg IV q 5 minutes to a maximum of 3 mg. ***(peds 0.02 mg/kg [0.5 mg maximum] q 5 minutes, maximum total dose 1 mg)***
10. Calcium – consider calcium gluconate OR calcium chloride:
    a. Calcium gluconate: Adult 2–6 g slow IVP over 10 minutes. ***(peds 60 mg/kg IVP over 10 minutes)***
    b. Calcium chloride: Adult 0.5–1 g slow IVP (50 mg/minute). ***(peds 20 mg/kg [0.2 mL/kg] slow IVP over 10 minutes, maximum dose 1 g or 10 mL; calcium gluconate is preferred in pediatrics – calcium chloride has greater risk of tissue damage)***
11. Vasopressors after adequate fluid resuscitation for the hypotensive patient, per the Shock SOP.
12. If atropine, calcium, and vasopressors have failed in the symptomatic bradycardic patient, consider GLUCAGON: Adult 5 mg IVP, then 1 mg IVP q 5 minutes (may require 5–15 mg to see effect). ***(peds 1 mg IVP [25–40 kg] q 5 minutes as necessary; 0.5 mg IVP [less than 25 kg] q 5 minutes as necessary)***. Consider ondansetron prophylaxis.
13. Transcutaneous pacing if refractory to initial pharmacologic interventions.
14. If seizure, consider midazolam (benzodiazepine of choice) per SOP 315.

PARAMEDIC STOP

Patient Safety Considerations

1. Transcutaneous pacing may not always capture, nor correct hypotension when capture is successful.

Key Considerations

1. Avoiding calcium in patients on cardiac glycosides (digoxin) is felt to be a historical belief and not supported.
2. Glucagon increases vomiting at these doses – consider ondansetron prophylaxis.
3. Atropine may have little or no effect (more helpful in mild overdoses). Hypotension and bradycardia may be mutually exclusive; blood pressure may not respond to correction of bradycardia.
4. Hyperglycemia results from L-type calcium channel blockade in the pancreas and may correlate with severity of ingestion.
5. Close monitoring of EKG changes and dysrhythmias. Serial frequent assessments are essential – rapid deterioration with profound hypotension is common.

Documentation

Repeat evaluation of signs/symptoms; possible etiology; time of symptom onset and time of exposure-specific treatments; therapy and response; time of ingestion, route, quantity, alcohol or other intoxicants.

MEDICAL EMERGENCY
SOP # 300 Medical Complaint Not Specified Under Other Protocols

Signs and Symptoms
Pertinent history to complaint
Allergies/medications taken or prescribed
Onset, type, and duration of pain
Provocation
Quality of pain/discomfort
Relieved by?

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Patient positioning appropriate for condition.
3. Supportive Care.
4. Pulse oximetry.
5. Glucose check, PRN.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
8. If indicated, Consider INT or IV NS TKO unless signs of shock, then 20 cc/kg fluid bolus.

AEMT STOP HERE

9. EKG monitor as indicated.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 301 Abdominal/GU Pain (non-traumatic) / Nausea and Vomiting

Signs and Symptoms
Description of pain, onset, duration, location, character, radiation
Aggravating factors, last menstrual period and/or vaginal bleeding in females
Recent trauma
History of abdominal surgery or problems
Blood in urine, vomitus, or stool
Nausea, vomiting, diarrhea
Fever, diaphoresis, jaundice
Abdomen – tenderness, masses, rigidity, hernia, pregnancy, distension, guarding

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Allowing patient to assume comfortable position or place patient supine, with legs elevated with flexion at hip and knees unless respiratory compromise or a procedure contraindicates.
3. Supportive care.
4. Pulse Oximetry.
5. Glucose check.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS 20 cc/kg if signs of shock ***(peds 20 cc/kg bolus).***
8. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

9. EKG Monitor for arrythmias.
10. Ondansetron (Zofran) 2-4 mg IV ***(peds >20 kg, 0.15 mg/kg IV, max single dose 8 mg)*** if intractable nausea and persistent vomiting and no signs of shock. Ensure QT is not prolonged prior to ondansetron IV administration.
    May use Zofran ODT as an alternative to Zofran IV for the treatment of nausea in the prehospital setting. In situations where IV access is unavailable or IV fluids are not necessary, consider the use of Zofran ODT at the following doses:
    a. Adults and Pediatrics > 31kg, give 8mg PO as a one-time dose.
       Pediatrics 15-30kg, give 4mg PO as a one-time dose.
       Pediatrics < 15kg, contact medical control.
       Use lower dose initially especially in the elderly.
11. Consider second IV en route if patient exhibits signs of shock.
12. Contact Medical Control regarding initiation of pain management.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 302 Acute Pulmonary Edema / CHF

Signs and Symptoms
Focus assessment on airway, breathing, and circulation
Shortness of breath
Cyanosis
Pedal edema
Profuse sweating, or cool and clammy skin
Erect posture
Distended neck veins (engorged, pulsating) – late sign
Bilateral rales/wheezes
Tachycardia (rapid pulse, > 100 bpm)
History of CHF or other heart disease, or renal dialysis
Lasix or Digoxin on medication list

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition. If respiration is less than 10/min, or greater than 30/min, consider assisting breathing with BVM and 100% Oxygen. ***(peds: NRB or 4 L/min BNC or assist with BVM as needed. Contact medical control if CHF suspected in a pediatric patient).***
2. Keep patient in upright seated position.
3. Consider Albuterol 2.5 mg/3 cc NS via nebulizer or MDI if wheezing.
4. Pulse oximetry.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. INT.
7. If Systolic BP is > 100 mmHg. Assess for crackles, wheezes or rales, JVD, peripheral edema, cyanosis, diaphoresis, respiratory rate > 25/min or < 10/min then:
    i. One Nitroglycerine spray or tablet sublingually, repeat Nitroglycerine q 5 minutes after initial dose, discontinue therapy if systolic BP < 100 mmHg
    ii. Use caution in patients taking erectile dysfunction medications profound hypotension may occur
    iii. Albuterol 2.5 mg/3 cc NS via nebulizer q 5 minutes to maximum of 3 doses
8. If systolic BP < 100 mmHg.
    Continue oxygen and initiate rapid transport, see hypotension protocol. Contact Medical Control immediately.
9. If respiratory distress and no contraindications, begin CPAP.

AEMT STOP HERE

10. May continue Nitroglycerine spray or tablet and apply 1" of Nitropaste to chest wall. Discontinue therapy if systolic BP < 100 mmHg.
11. 12 Lead EKG. Assess for Ischemia.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 303 Anaphylactic Shock

Signs and Symptoms
Contact with a known allergen or with substances that have a high potential for allergic reactions
Sudden onset with rapid progression of symptoms
Dyspnea, presents with an audible wheeze, generalized wheeze on auscultation, decreased air exchange on auscultation
Generalized urticarial, erythema, angioedema especially noticeable to face and neck
Complaint of chest tightness or inability to take a deep breath

TREATMENT PATHWAY

1. Position of comfort, reassure.
2. Pulse oximetry, Oxygen and airway maintenance appropriate for patient's condition.
3. Administer patient prescribed Beta Agonist MDI if available.
4. Administer EPINEPHrine autoinjector for anaphylaxis if available.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. IV NS or LR, large bore @ TKO – If hypotensive 20 mL/kg bolus ***(peds 20 mL/kg bolus).***
7. Administer EPINEPHrine. 1:1,000 (now 1 mg/mL) - 0.3-0.5 mg IM, ***(peds EPINEPHrine 1:1,000 (now 1 mg/mL) - 0.01 mg/kg IM, max dose is 0.3 mg).***
8. Albuterol Inhalation Treatment if wheezing is present and persists post EPINEPHrine IM.

AEMT STOP HERE

9. EKG Monitor. Assess for Ischemia.
10. EPINEPHrine 1:1,000 (now 1 mg/mL) - 0.3-0.5 mg IM or IV/IO EPINEPHrine 1: 10,000 (now 0.1 mg/mL), ***(peds EPINEPHrine 1:1,000 (now 1 mg/mL) - 0.01 mg/kg IM, max dose is 0.3 mg).*** IV/IO route should be reserved for unstable patients, especially pediatric.
    a. If repeat IM doses are required, consider:
        i. EPINEPHrine 2–20 mcg/min ***(peds 0.1-1 mcg/kg/min.)***
11. DiphenhydrAMINE (Benadryl) 25-50 mg IV or deep IM ***(peds 1 mg/kg IVP).***
12. Methylprednisolone (Solu-Medrol) 62.5 mg (if small in stature, sensitive to steroids, on chronic steroid therapy) or 125 mg IVP ***(peds contact Medical Control).***

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 304 Cerebrovascular Accident (CVA) / Stroke

Signs and Symptoms
Altered level of consciousness (coma, stupor, confusion, seizures, delirium)
Intense or unusually severe headache of sudden onset or any headache associated with decreased level of consciousness or neurological deficit unusual and severe neck or facial pain
Aphasia/Dysphasia (unable to speak, incoherent speech or difficulty speaking)
Facial weakness or asymmetry (paralysis of the facial muscles, usually noted when the patient speaks or smiles); may be on the same side or opposite side from limb paralysis
In-coordination, weakness, paralysis, or sensory loss in one or more limbs; usually involves one half of the body particularly in the hand
Ataxia (poor balance, clumsiness, or difficulty walking)
Visual loss (monocular or binocular); may be a partial loss of visual field
Intense vertigo, double vision, unilateral hearing loss, nausea, vomiting, photophobia or phonophobia

TREATMENT PATHWAY

1. Oxygen at 2 – 6 LPM BNC and airway maintenance appropriate to patient's condition.
2. Continually monitor airway due to decreased gag reflex and increased secretions.
3. Conduct a brief targeted history and physical exam. Establish time of onset. Document witness to time of onset and contact information.
4. Maintain body heat, protect affected limbs from injury, and anticipate seizures.
5. Pulse oximetry.
6. If trauma is suspected, spinal stabilization, elevate head 30° if no evidence of spinal injury.
7. Glucose check and treat patient appropriately.
8. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

9. IV NS TKO (30 cc/hr) or INT.
10. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
11. Naloxone 0.4 mg IV/IO/IM/IN titrated to adequate ventilation, ***(peds 0.1 mg/kg up to 2 mg titrated to adequate ventilation)*** if Narcotic use suspected. If utilizing pre-filled delivery systems, dose per manufacturer's instructions. May repeat dose.
12. Complete a Cincinnati Pre-Hospital Stroke Scale (CPSS)

AEMT STOP HERE

13. EKG Monitor, 12 Lead EKG.
14. Complete thrombolytic screening protocol.
15. Complete the C-STAT Stroke Assessment Tool.
16. If LVO Score 2 or greater, recommend transport to stroke center.
17. If possible, obtain Blood sample tubes.
18. Contact Medical Control if SBP > 220 mmHg or DBP > 140 mmHg. If authorized give Nitro spray q 5 min. The goal is to reduce initial blood pressure by no more than 15%.

PARAMEDIC STOP

REFERENCE
REF-01
C-STAT Evaluation Tool

CINCINNATI PREHOSPITAL STROKE SEVERITY SCALE		
CPSSS Exam Domain	**Scoring Method**	**Possible Score**
Conjugate Gaze Deviation	Normal; no gaze deviation	0
	Partial gaze palsy	1
	Forced deviation	2
LOC Questions and LOC Commands	Normal responses to both LOC Questions and to both LOC Commands	0
	Incorrect responses to at least 1 of 2 LOC Q **and** at least 1 of 2 LOC C	1
Arm Drift	Holds arms above the bed for 10 seconds	0
	Cannot hold arm (L, R or Both) up for 10 seconds before arm(s) falls to bed	1
Total Score		0-4

Katz BS, et al. Stroke. 2015; 46: 1508-1512.
doi: 10.1161/STROKEAHA.115.008804..

The C-STAT score more accurately identifies patients with Large Vessel Occlusion (L.V.O.)
If C-STAT Score ≥ 2, patient should be transported to a stroke center with interventional capability.

REFERENCE

REF-02 Pre-hospital Screen for Thrombolytic Therapy

- Complete this report for all patients symptomatic for a myocardial infarct or CVA
- Report to the Emergency Department Physician/Nurse any positive findings
- Document all findings in the patient's ePCR

EMT AEMT PARAMEDIC

Time of onset of symptoms: _____		
Witness/Next of Kin Contact Info: _____		
Systolic BP > 240 mmHg	☐ Yes	☐ No
Diastolic BP > 110 mmHg	☐ Yes	☐ No
Right arm vs. Left arm Systolic BP difference > 15 mmHg	☐ Yes	☐ No
History of recent brain/spinal cord surgery, CVA, or injury	☐ Yes	☐ No
Recent trauma or surgery	☐ Yes	☐ No
Bleeding disorder that causes patient to bleed excessively	☐ Yes	☐ No
Prolonged CPR (> 10 minutes)	☐ Yes	☐ No
Pregnancy	☐ Yes	☐ No
Taking Coumadin, Aspirin, or other blood thinners	☐ Yes	☐ No

MEDICAL EMERGENCY
SOP # 305 Croup

Signs and Symptoms
History
Viral infections resulting in inflammation of larynx, trachea
Seasonal – Late fall / early winter
Children under 6 years old with cold symptoms for 1 – 3 days
Hoarseness
Barking, seal-like cough
Stridor, NOT wheezes
Low grade fever
No history of obstruction, foreign body, trauma

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Allow patient to assume comfortable position or place patient supine.
3. Supportive care.

EMT AEMT STOP HERE

4. Nebulized EPINEPHrine 1:1,000.
    a. 1 mg diluted to 2.5 – 3 cc with saline flush, nebulized (mask or blow-by)
    b. May repeat up to 3 total doses
    c. If the patient has significant distress, 3 ml (3 mg) diluted with 2.5 to 3 cc saline flush may be administered as an initial aerosol.
5. Contact Medical Control for subsequent aerosols.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 306 Family Violence

Signs and Symptoms	Assessment
• Injury to soft tissue areas that are normally protected • Bruise or burn in the shape of an object • Bite marks • Rib fracture in the absence of major trauma • Multiple bruising in various stages of healing	Fear of household member Reluctance to respond when questioned Unusual isolation, unhealthy / unsafe living environment Poor personal hygiene / inappropriate clothing Conflicting accounts of the incident History inconsistent with injury or illness Indifferent or angry household member Household member refused to permit transport Household member prevents patient from interacting openly or privately Concern about minor issues but not major ones Household with previous violence Unexpected delay in seeking treatment

TREATMENT PATHWAY

Treatment – Standing Order

1. Patient care is first priority.
2. If possible, remove patient from situation and transport.
3. Police assistance as needed.
4. If sexual assault, follow sexual assault protocol.
5. Obtain information from patient and caregiver.
6. Do not judge.
7. Report suspected abuse to hospital after arrival. Make a verbal and written report.

Direct questions to ask when alone with patient and if time available

- Has anyone at home ever hurt you?
- Has anyone at home touched you without your consent?
- Has anyone ever made you do things you didn't want to?
- Has anyone taken things that were yours without asking?
- Has anyone scolded or threatened you?
- Are you afraid of anyone at home?

NOTE: National Domestic Violence Hotline 1-800-799-SAFE (7233)

EMT AEMT PARAMEDIC STOP HERE

MEDICAL EMERGENCY
SOP # 307 Headache

Notes	**Always consider:**
• Rapidly assess and manage life-threatening injuries • Pertinent history to complaint • Allergies/medications taken or prescribed • Onset, type, and duration of complaint • Provocation • Quality of pain/discomfort • Relieved by? • Signs and symptoms	• Heat Illness • Carbon Monoxide • Intracranial catastrophe • Infection • Altitude Illness • Hypertension/CVA • Ocular sources (Glaucoma) • Preeclampsia/eclampsia • Medications • Migraine

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Patient positioning appropriate for condition.
3. Supportive Care.
4. Pulse oximetry.
6. Glucose check, PRN.
7. EKG Monitor prn.

EMT STOP HERE

8. IV NS/LR TKO. If systolic BP < 90mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients. If not hypotensive, avoid administering more than 500 cc crystalloid.

AEMT STOP HERE

9. Perform Stroke Screening.
10. Manage Fever if appropriate.
11. Manage Blood Pressure per appropriate protocol.
12. Contact Medical Control for pain management authorization.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 308 HEENT (Head, Eyes, Ears, Nose, Throat) Complaints

Notes

- Rapidly assess and manage life-threatening injuries
- Pertinent history to complaint
- Allergies/medications taken or prescribed
- Onset, type, and duration of complaint, signs and symptoms
- Provocation
- Quality of pain/discomfort
- Relieved by?
- Signs and symptoms

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Patient positioning appropriate for condition.
3. Supportive Care.
4. Pulse oximetry.
5. Glucose check, PRN.
6. IV NS/LR TKO. If systolic BP < 90mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients. If not hypotensive, avoid administering more than 500 cc crystalloid.
7. EKG Monitor prn.

EMT STOP HERE

AEMT STOP HERE

8. Address any need for airway intervention, hemorrhage control, foreign body, need for suctioning.
9. Manage Fever if appropriate.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 309 Hyperglycemia Associated with Diabetes

Signs and Symptoms
History of onset
Altered level of consciousness
Pulse: Tachycardia, thready pulse
Respirations (Kussmaul-Kien – air hunger)
Hypotension
Dry mucous membranes
Skin may be cool (consider hypothermia)
Ketone odor on breath (Acetone smell)
Abdominal pain, nausea and vomiting
History of polyuria, or polydipsia (excessive urination or thirst)
Blood glucose determination

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition; suction airway as needed.
2. Supportive care.
3. Pulse oximetry.
4. Glucose check.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. IV NS TKO or INT. If BS >250 mg/dl, start 10-20cc/kg NS bolus if patient with signs of dehydration, vomiting, or DKA. ***(Peds 4 cc/kg/hr max 150cc/hr. No Bolus).***

AEMT STOP HERE

7. EKG Monitor Consider 12 lead EKG.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 310 Hypertensive Crisis

Signs and Symptoms
Decreased / Altered LOC
Headache, blurred vision, dizziness, weakness
Elevated blood pressure (systolic BP > 220 mmHg and/or diastolic BP > 140 mmHg) with:
Dyspnea, peripheral or pulmonary edema
Cardiac dysrhythmia, neurological deficits, or signs of end organ dysfunction

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Position of comfort, elevation of head is preferred.
3. Keep patient calm, reassure.
4. Pulse oximetry.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. INT or IV NS TKO.
7. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

8. Assess for Ischemia.
9. Evaluate cardiac rhythm for dysrhythmia and treat appropriately with medical direction. Contact Medical Control prior to initiation of anti-arrhythmic therapy.
10. If motor/neuro deficits present, go to stroke protocol.
    If NO motor/neuro deficits:
    a. If systolic BP < 220 mmHg, contact Medical Control, monitor patient changes.
    b. If systolic BP > 220 mmHg and/or diastolic BP > 140 mmHg:
    Nitroglycerine one spray 0.4mg SL q 3 – 5 min until noted decrease in BP by 15%. May use Nitro Paste 1 inch to chest wall, remove if BP drops 15% from the original reading. Use with caution in patients taking erectile dysfunction medications as profound hypotension may occur.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 311 Hypoglycemia

Signs and Symptoms
History of onset of event
History of insulin excess (overdose, missed meal, exercise, vomiting or diarrhea)
Confusion, agitation, headaches, or comatose
Pulse rate (normal to tachycardia)
Respirations (shallow, slow)
Skin (sweaty, often cool)
Flaccid muscle tone
Grand Mal seizures
Fecal, urinary incontinence

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition. (Snoring respirations is a sign of an INADEQUATE airway)
2. Supportive care.
3. If patient is hypoglycemic and is conscious with an intact gag reflex, administer one tube of instant glucose and reassess.
4. Pulse oximetry.
5. Glucose check, monitor.

EMT STOP HERE

6. IV NS TKO.
7. If patient is unresponsive, infuse 250cc bag D10 until patient responds. May repeat once.
8. If blood sugar is <80 mg/dL and symptomatic: titrate D10 slowly until patient responds. Try to avoid large swings in serum glucose levels, ***(peds 2 mL/kg D_{25} IV/IO; if needed an admixture of D_{50} and Normal Saline can be obtained through mixing 1 mL to 1 mL for the treatment of symptomatic hypoglycemia in pediatric patients)***. Reassess blood sugar level q 15 min.

AEMT STOP HERE

9. EKG Monitor. Consider 12 Lead EKG.
10. Search for explanation of hypoglycemia (noncompliance, Infection, STEMI, Stroke, etc.).

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 312 Medications at Schools

Signs and Symptoms
The patient must exhibit the signs and symptoms for which the medication is prescribed.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Other treatments will be in accordance with the BLS / ALS SOPs.

EMT AEMT STOP HERE

3. Necessary medication(s) administration as requested by caregiver(s):
    a. Schools must provide the medication(s) to be administered
    b. Schools must provide a written copy of the physician order and care plan for attachment to the patient care report
    c. This documentation by the patient's primary physician should list the following:
        i. Name of the patient
        ii. Name of the primary physician
        iii. Document must be signed by the primary physician
        iv. Contact phone number of the primary physician
        v. Name of the medication(s)
        vi. Signs and symptoms for which the medication(s) is prescribed
        vii. Dosage of the medication(s)
        viii. Number of repeat doses of the medication(s)
        ix. Route(s) of administration(s)
        x. Potential side-effects of medication(s)
4. Medication(s) will only be administered if the patient means the signs and symptoms for that medication.
5. Copies or picture of the care plan and physician order must be attached to the patient care report.
6. If the medication(s) is not administered, documentation must include those reasons for withholding.
7. Whenever medication is administered under these circumstances, transport is mandatory.

Note: If you have additional questions or concerns, please contact Medical Control.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 313 Non-Formulary Medications

To provide authorization for the use of mediations not commonly used within the current guidelines. For Emergency Use Only.

Notes	**Signs and Symptoms**
If you have any additional questions or concerns, please contact Medical Control.	The patient must exhibit the signs and symptoms for which the medication is prescribed.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Other treatments will be in accordance with the BLS / ALS SOPs.

EMT AEMT STOP HERE

3. Necessary medication(s) administration as requested by caregiver(s):
    a. Caregiver must provide the medication(s) to be administered
    b. Caregiver must provide a written copy of the physician order and care plan for attachment to the patient care report
    c. This documentation by the patient's primary physician should list the following:
        i. Name of the patient
        ii. Name of the primary physician
        iii. Document must be signed by the primary physician
        iv. Contact phone number of the primary physician
        v. Name of medication(s)
        vi. Signs and symptoms for which the medication(s) is prescribed
        vii. Dosage of the medication(s)
        viii. Number of repeat doses of the medication(s)
        ix. Route(s) of administration(s)
        x. Potential side-effects of mediation(s)
4. Medication(s) will only be administered if the patient meets the signs and symptoms for that medication.
5. Copies or picture of the care plan and physician order must be attached to the patient care report.
6. If the medication(s) is not administered the documentation must include those reasons for withholding.
7. Whenever medication is administered under these circumstances, transport the patient or complete declination of care per protocol.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 314 Respiratory Distress (Asthma/COPD)

Signs and Symptoms
Mild Attack – Slight increase in respiratory rate. Mild wheezes. Good skin color.
Moderate Attack – Marked increase in respiratory rate. Wheezes easily heard. Accessory muscle breathing.
Severe Attack – Respiratory rate more than twice normal. Loud wheezes or so tight no wheezes are heard, patient anxious, grey or ashen skin color.
Hx – COPD, emphysema, asthma, or other restrictive lung disease.
Respiratory rate greater than 25 per minute or less than 10 per minute.
Labored respiration, use of accessory muscles or tripoding
Breath sounds: Bilaterally diminished, dry crackles, wheezing
Cyanosis / Diaphoresis
Use of short sentences
Unilateral breath sounds

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. If the patient has a prescribed Albuterol inhalation treatment, assist the patient with 2.5 mg/3 ml NS and start the oxygen flow rate at 6 LPM or until the appropriate mist is achieved.
3. Pulse oximetry.
4. If the patient uses an MDI, assist patient with one dose.
6. Administer Albuterol 2.5 mg/ 3 mL NS ***(peds 2.5 mg/ 3 mL NS q 5-15 min)*** and start the oxygen flow rate at 6 LPM or until the appropriate mist is achieved.
7. CPAP if no contraindications.

EMT STOP HERE

8. INT or IV NS TKO.
9. Administer Albuterol 2.5 mg/ 3 mL NS ***(peds 2.5 mg/ 3 mL NS q 5-15 min)*** and start the oxygen flow rate at 6 LPM or until the appropriate mist is achieved.
10. EPINEPHrine 1:1,000 (now 1 mg/mL) 0.3-0.5 mg IM (***peds 1: 1,000 (now 1 mg/mL) - 0.01 mg/kg IM, max dose is 0.3 mg)*** for patients in severe distress. Be mindful of cardiac side effects.
11. EKG monitor.

AEMT STOP HERE

11. Epinephrine 1:1,000, 0.3 – 0.5 mg IM ***(peds 1: 1,000 0.01 mg/kg IM, max dose is 0.3 mg***) for patients in severe distress. Be mindful of cardiac side effects.
12. In severe cases consider Solumedrol 62.5 mg (if small in stature, sensitive to steroids, or on chronic steroid therapy) or 125 mg IV ***(peds dosing – contact Medical Control).***
13. If asthma exacerbation, consider Magnesium 1-2 gms IV slowly. Magnesium is not recommended in COPD exacerbations.

Note:

Peds: consult Medical Control prior to administering Methylprednisolone (Solu-Medrol)

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 315 Seizures

Signs and Symptoms	Notes
Seizure (onset, duration, type, post-seizure, level of consciousness) Medical (diabetes, headaches, drugs, alcohol, seizure history) Physical (seizure activity, level of consciousness, incontinence, head and mouth trauma, vital signs) Trauma (head injury or hypoxia secondary to trauma)	• Specifically evaluate for: active bleeding, trauma, eye deviation, pupil equality, mouth or tongue bleeding, Urinary or fecal incontinence, lack of arm or leg movement or tone. • The goal of Naloxone therapy is to restore adequate ventilation. Patients, particularly those on chronic opiate therapy, often need very small doses of Naloxone in the event of overdose. Larger doses of Naloxone usually create more agitation and behavioral symptoms.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Protect patient from injury during active seizures.
3. If patient is actively seizing, consider therapy if:
   - Unstable ABC's exist.
   - Patient has been actively seizing for 5 or more minutes.
   - Patient has underlying disease or condition that will be adversely affected if seizures continue (trauma, COPD, pregnancy, severely hypertensive).
4. C-Spine precautions if appropriate, consider Naloxone if Opiates suspected.
5. If febrile, cool as per hyperthermia protocol and monitor.
6. Pulse oximetry.
7. Glucose check.
8. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

9. IV NS TKO or INT.
10. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
11. If no IV available and blood glucose levels are <80 mg/dl, consider Glucagon 1-2 mg IM (peds – Glucagon 0.5 mg/dose IM/IV if <20 kg, or 1 mg/dose IM/IV if 20 kg or greater.)
12. If narcotic overdose, Naloxone (Narcan) 0.4 mg IV/IO/IM/IN titrated to adequate ventilation ***(peds 0.1 mg/kg, titrated to adequate ventilation)***. If utilizing pre-filled delivery systems, dose per manufacturer's instructions. May repeat dose. Synthetic opiates may require larger Naloxone doses. Physically manage the airway if no response after 8 mg Naloxone.

AEMT STOP HERE

13. Adults – If actively seizing:
    - Diazepam (Valium) SLOW IVP/IO 2-5 mg or Midazolam (Versed) 2-5 mg IV/IO/IM may repeat if seizure continues.
      OR
    - LORazepam 1-2 mg IV, every 5 minutes or; 2-4 mg IM, every 10 minutes (maximum dose 8 mg).

Consider Ketamine (if available) for refractory seizures unresponsive to benzodiazepines.

14. <u>Peds:</u>

a. <u>Diazepam (Valium) 0.1 mg/kg or Midazolam (Versed) 0.1 mg/kg IV/IO</u>

b. ***Midazolam (Versed) IM 0.2 mg/kg IM (max single dose 6 mg) Repeat once if seizure activity persists after 10 minutes. Contact MEDICAL CONTROL if seizure activity persists after repeat dose.***
c. ***Midazolam (Versed) IN 0.3 mg/kg IN (max single dose 10 mg) with maximum total dose of 0.4 mg/kg.***
d. ***LORazepam (peds 0.1 mg/kg IV/IO, max single dose 4 mg, may repeat in 5 minutes if seizure activity continues; not to exceed 0.2 mg/kg total (maximum of 8 mg)***
e. ***If seizure persists for 4 minutes repeat medication once.***
f. ***Additional Alternative: Diazepam rectal gel (DiaStat) is 0.2-0.5 mg/kg depending on age. See the dosing table for specific recommendations.***

Because Diazepam rectal gel is provided as unit doses of 2.5, 5, 7.5, 10, 12.5, 15, 17.5, and 20 mg, the prescribed dose is obtained by rounding upward to the next available dose. The following table provides acceptable weight ranges for each dose and age category, such that patients will receive between 90% and 180% of the calculated recommended dose.

PARAMEDIC STOP

Age (years)	Recommended Dose
2 through 5	0.5 mg/kg
6 through 11	0.3 mg/kg
12 and older	0.2 mg/kg

2 - 5 Years 0.5 mg/kg		6 - 11 years 0.3 mg/kg		12+ Years 0.2 mg/kg	
Weight (kg)	Dose (mg)	Weight (kg)	Dose (mg)	Weight (kg)	Dose (mg)
6 to 10	5	10 to 16	5	14 to 25	5
11 to 15	7.5	17 to 25	7.5	26 to 37	7.5
16 to 20	10	26 to 33	10	38 to 50	10
21 to 25	12.5	34 to 41	12.5	51 to 62	12.5
26 to 30	15	42 to 50	15	63 to 75	15
31 to 35	17.5	51 to 58	17.5	76 to 87	17.5
36 to 44	20	59 to 74	20	88 to 111	20

MEDICAL EMERGENCY
SOP # 316 Sexual Assault

Signs and Symptoms
Traumatic Injuries

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition
2. Be calm and assuring with sensitivity toward the patient, do not evaluate/pass judgement of the credibility of the circumstances of the assault, and take a medical history, not a history of the assault. By nature of this event, any touch may be traumatic for this patient. Overtly and repeatedly explain what you are doing to try to lessen the impact of procedures and touching.
3. DO NOT make unnecessary physical contact with the patient and do not disturb the crime scene.
4. If possible, have an attendant the same gender as the victim present at all times.
5. If needed, wrap a linen sheet around the victim.
6. DO NOT inspect genitals unless evidence of uncontrolled hemorrhage, trauma, or severe pain is present.
7. Advise the patient not to eat, drink, smoke, bathe, change clothing or go to the bathroom if it all possible to preserve any forensic evidence. If they must urinate, request that they do not wipe.
8. Collect patient's clothing when possible
    a. Place clothing in paper bags with ID labels.
    b. Leave all sheets placed in paper bag with patient at facility.
    c. Notify all staff of clothing samples.
9. Transport patient to appropriate facility for treatment and examination.
10. Contact dispatch to notify Rape Crisis Agency of possible Sexual Assault. Law Enforcement notification should be offered to the victim but should not be pushed or forced upon them.

EMT AEMT PARAMEDIC STOP HERE

MEDICAL EMERGENCY
SOP # 317 Sickle Cell Crisis

Notes:	**Signs and Symptoms**
• Use caution in administering narcotics to a patient with SpO_2 < 95% • All patients who receive narcotic medication must be transported for further evaluation or have documented capacity for decision making to decline. • The goal of Naloxone therapy is to restore adequate ventilation; larger doses, especially in patient on chronic opiate therapy, often need very small doses in the event of overdose. Larger doses of Naloxone usually create more agitation and behavioral symptoms.	History of Sickle Cell Anemia Signs of infection Hypoxia Dehydration Painful joints Limited movement in joints

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Supportive care.
3. Pulse oximetry (Keep sats > 94%).

EMT STOP HERE

4. IV NS bolus 20 cc/kg ***(peds 20 cc/kg bolus).***

AEMT STOP HERE

5. EKG Monitor prn.
6. If pain persists, administer pain medications per pain protocol.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 318 Unconscious / Unresponsive / Altered Mental Status

Signs and Symptoms	Note
Unconscious or unresponsive with vital signs Any patient not responding appropriately to verbal or painful stimulus Altered level of consciousness with vital signs Assess for head trauma Assess for Hypothermia or Hyperthermia, hemiparesis, fever, OD, hypoglycemia ***Peds*** – less commonly associated with intussusception (fold of one intestine into another), intracranial catastrophe, metabolic disorder	The goal of Naloxone therapy is to restore adequate ventilation. Patients, particularly those on chronic opiate therapy, often need very small doses of Naloxone in the event of overdose. Larger doses of Naloxone usually create more agitation and behavioral symptoms except in the case of synthetic opioids.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Assess for underlying causes: head trauma, hypovolemia, hypothermia, hemiparesis, and fever and treat accordingly.
3. Consider Naloxone if opiates suspected.
4. Pulse oximetry.
5. Glucose check.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS TKO or INT.
8. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.
9. Administer Naloxone (Narcan) 0.4 mg IV/IM/IN/IO titrated to adequate ventilation ***(peds 0.1 mg/kg IV, max single dose 2 mg)***. If utilizing pre-filled delivery systems, dose per manufacturer's instructions. May repeat dose, with the exception of synthetic opioids which may require larger doses of Naloxone. Physically manage the airway if no response after 8mg of Naloxone.

AEMT STOP HERE

10. Contact Medical Control for further orders; 20 cc/kg NS fluid challenge ***(peds 20 cc/kg).***
11. If hypoglycemic and unable to maintain airway, and CVA is not suspected, and the patient has a history of diabetes:
    a. If blood sugar is < 80 mg/dL and symptomatic, infuse 250cc bag D10 until patient responds. ***(peds 2 mL/kg D25 IV/IO; IV Infusion of 10% Dextrose in 250mL for Hypoglycemia Max dose 250mL. Titrate to patient's response/condition. Not to exceed 25g (1g/10ml))***
    b. Reassess blood sugar level q 15 min. If unable to obtain IV access, consider Glucagon 1-2 mg IM ***(peds Glucagon 0.5 mg/dose IM/IV if <20 kg, or 1 mg/dose IM/IV if 20 kg or greater.***
    c. 20 cc/kg NS fluid challenge ***(peds 20 cc/kg).***

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 319 Syncope

Signs and Symptoms	**Note**
Loss of consciousness with recovery Lightheadedness, dizziness Palpitations, slow or rapid pulse, irregular pulse Decreased blood pressure	Consider repeat EKGs. A single normal EKG does not rule out a cardiac etiology.

TREATMENT PATHWAY

1. Oxygen at 2 – 6 LPM and airway maintenance appropriate to patient's condition.
2. Supportive care.
3. Consider Naloxone if Opiates suspected.
4. Pulse oximetry.
5. Glucose check.
6. Monitor, 12 Lead EKG and Transmission if applicable.

EMT STOP HERE

7. INT or IV NS TKO – if hypotensive 20 cc/kg bolus (peds 20 cc/kg bolus).
8. If glucose <60, titrate D10 slowly per hypoglycemia guidelines.

AEMT STOP HERE

9. Treat any cardiac dysrhythmia per appropriate protocol. Consider repeat EKGs.
10. Assess neuro status; if abnormal refer to appropriate protocol.

PARAMEDIC STOP

MEDICAL EMERGENCY
SOP # 320 Brief Resolved Unexplained Event (BRUE) & Acute Events in Infants

Signs and Symptoms	Notes
Infant less than 1 year old Event reported by a bystander as sudden, brief (less than 1 minute), unexplained, and completely resolved upon EMS arrival that includes one or more of the following: – Breathing change (absent, decreased, or irregular) – Color change (central cyanosis or pallor) – Marked change in muscle tone (hyper- or hypotonia) – Altered level of responsiveness (increased, irritability, or decreased) Aliases: Apparent Life-Threatening Event (ALTE)	BRUE is a group of symptoms, not a disease process. The determination of a BRUE is made only after hospital evaluation, not in the field. ALL infants with a history of signs or symptoms suggestive of BRUE are transported to an ED, even if well-appearing. Avoid using “BRUE”, “ALTE”, “SIDS”, or “near-miss SIDS” with the parent/guardian.

TREATMENT PATHWAY

1. Confirm the patient meets the definition of a possible BRUE. If the infant is not completely well-appearing on EMS arrival, this is NOT a possible BRUE – treat and transport per the appropriate protocol. Exclude if any of the following are present:
    a. Signs or symptoms of acute illness or injury on EMS evaluation: abnormal vital signs for age (including fever), vomiting, signs of trauma, noisy or labored breathing
    b. Identifiable cause: gastric reflux (spitting up), swallowing dysfunction, nasal congestion or excessive secretions, periodic breathing of the newborn, breath-holding spell, change in tone associated with choking/gagging/crying/feeding, seizure, hypoglycemia, significant past medical history (congenital heart disease, pulmonary disease, VP shunt, seizure disorder), or need for IV medication administration
    c. History or exam concerning for child abuse or neglect
    d. Color change involving only redness (e.g., face) or isolated hand/foot cyanosis
2. Obtain history: circumstances and symptoms before, during, and after the event (duration, interventions done, color, tone, breathing, feeding, position, location, activity, level of consciousness); concurrent symptoms (fever, congestion, cough, vomiting, diarrhea, rash, labored breathing, fussiness, poor feeding); prior BRUE (ever, including past 24 hours); past medical history (prematurity, birth complications, reflux, congenital heart disease, developmental delay, airway abnormalities, hospitalizations, surgeries, injuries); family history of sudden unexplained death or arrhythmia; social history (household stressors, toxin/drug exposures, sick contacts).
3. Consider possible child abuse (multiple/changing versions of the story, implausible mechanism for developmental stage) – treat per Family Violence protocol (SOP 306).
4. Exam: full set of vital signs; signs of respiratory distress or increased work of breathing (tachypnea, grunting, nasal flaring, retractions, head bobbing); color (central and peripheral) and capillary refill; mental status (alert, tired, lethargic, unresponsive, irritable); head-to-toe exam for trauma or neglect, pupillary response and anterior fontanelle.
5. Continuous pulse oximetry and cardiac monitoring for ALL patients with a possible BRUE; serial observations during transport for change in condition.
6. Check blood glucose; treat symptomatic hypoglycemia per SOP 311.
7. Oxygen for signs of respiratory distress or hypoxemia – escalate from nasal cannula to simple face mask to non-rebreather mask as needed. Suction excessive nose/mouth secretions with bulb syringe or suction catheter.
8. Transport to the appropriate facility even if the infant appears well or has returned to baseline.

9. Contact Medical Control if the parent/guardian refuses care and/or transport, especially if any high-risk criteria are present.

EMT STOP HERE

10. Routine IVs should NOT be placed on all suspected BRUE patients. Place an IV/IO only for clinical concern of shock or to administer IV medications.

AEMT STOP HERE

11. Treat shock or other emergent conditions per the appropriate SOP.

PARAMEDIC STOP

Patient Safety Considerations

1. Regardless of well appearance, all infants with a history suggestive of BRUE should be transported for further evaluation.
2. Transport to a facility with at least baseline pediatric readiness (appropriate equipment, resources, and trained staff for initial emergency care and stabilization of pediatric patients), if feasible.
3. Consider transport to a facility with pediatric critical care capability for any high-risk criteria:
    a. Less than 2 months of age
    b. History of prematurity (32 weeks gestation or less)
    c. More than one BRUE, now or in the past
    d. Event duration greater than 1 minute
    e. CPR or resuscitation by caregivers or trained rescuers

Key Considerations

1. EMS clinicians play a unique and important role in obtaining an accurate history soon after the event and in observing, documenting, and reporting environmental, scene, and social indicators that may point to an alternate diagnosis (abuse, neglect, non-accidental trauma, unsafe sleeping practices).
2. High-risk patients with a possible BRUE have worse outcomes and may require ED or inpatient testing, intervention, and/or follow-up. A few of these infants will die even after hospital evaluation and treatment.

Documentation

Event: breathing (apnea or respiratory distress), color change (central and/or peripheral), change in muscle tone, level of responsiveness, event duration, witnessed or not. Pre-event circumstances (feeding or other activity, prematurity, prior BRUE, past medical history). Caregiver resuscitation efforts. Post-event symptoms. Full vital signs, temperature, SpO2, respiratory effort, mental status (pupils, pediatric GCS or AVPU), color, capillary refill, signs of abuse/trauma/neglect, and environmental or social clues.

MEDICAL EMERGENCY
SOP # 321 Back Pain (Non-Traumatic)

Signs and Symptoms	Notes
Back pain or discomfort related to a non-traumatic cause Midline back tenderness Back erythema or swelling Motor and/or sensory loss in arms or legs Loss of perianal (saddle) sensation Absence of, or significant inequality of, femoral or distal arterial pulses in the lower extremities Hyper- or hypothermia Rectal bleeding or hematemesis	Excludes: back pain from spinal trauma (SOP 412, SOP 608); sickle cell pain crisis (SOP 317); suspected labor (SOP 500). Back and abdominal pain often coexist with similar disease processes (SOP 301).

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Vital signs including pulse, blood pressure, respiratory rate, neurologic status assessment, pulse oximetry, and temperature.
3. Assess for life-threatening causes of back pain:
    a. Spinal cord compression (spinal epidural abscess, malignancy, spinal epidural hematoma in patients on anticoagulants):
        i. Urinary and/or bowel incontinence
        ii. Inability to walk due to weakness
        iii. New neurologic deficits in extremities
        iv. Loss of sensation in saddle distribution
    b. Aortic dissection or ruptured abdominal aortic aneurysm:
        i. Unequal femoral or distal lower extremity pulses
        ii. "Pulsatile" abdominal mass
        iii. Associated abdominal pain and/or chest pain
        iv. Known history of abdominal aortic aneurysm or dissection
    c. Pyelonephritis:
        i. Fever; nausea, vomiting; urinary frequency/urgency; dysuria; hematuria; abdominal pain; costovertebral angle tenderness to percussion
4. Assess for signs of shock. If shock is present, treat per the appropriate Shock SOP (SOP 404, 406, or 410).
5. Assess for other non-life-threatening causes, e.g., kidney stone (unilateral flank pain, nausea/vomiting, possible hematuria, history of kidney stones).
6. Transport to an appropriate receiving facility. Consider specialty destination centers for conditions such as suspected aortic emergency.

EMT STOP HERE

7. IV/IO access as necessary to provide analgesia and/or fluid resuscitation.
8. Provide analgesia per the Pain Management Protocol.
9. Administer antiemetics per SOP 301.

AEMT STOP HERE

10. Treat hypotension/shock per the appropriate SOP.
11. Reassess vital signs and response to therapeutic interventions throughout transport.

PARAMEDIC STOP

Key Considerations

1. Assess for life-threatening causes of back pain; provide appropriate treatment for pain, vomiting, and shock.
2. Identify patients on anticoagulants – higher risk for spinal epidural hematoma or retroperitoneal hemorrhage, which can present as back pain.
3. Identify patients with a history of intravenous drug abuse and/or impaired immune system – higher risk for spinal epidural abscess.
4. Identify patients with a history of cancer, or one suspicious for cancer – spinal metastases can cause spinal cord compression.
5. Identify older adults and patients with prolonged corticosteroid use – at risk for vertebral body compression fracture.

Documentation

Assessment of back and abdomen including palpation/percussion findings, presence or absence of masses, and presence and nature of tenderness/pain. Initial and changing neurologic status. Initial and changing perfusion/pulses.

SHOCK/TRAUMA
SOP # 401 Air Ambulance Transport

In the absence of an on-scene Paramedic or EMS Lieutenant, the most qualified FF/EMT or AEMT shall have the responsibility of determining the need for an air ambulance transport. Request for an Air Ambulance must be through the Incident Commander.

A scene flight by air ambulance MAY be indicated IF:

The Level I trauma patient's condition warrants immediate and extreme action **and** the extrication **and/or** transport time is greater than **30** minutes **and** if patient **is not** in trauma full arrest.

Transport time is defined as the length of time beginning when the emergency unit would leave the scene transporting until time of arrival at the trauma center.

Additional Criteria:

- Multi-system blunt or penetrating trauma with unstable vital signs
- Greater than 25% TBSA burns
- Paralysis or spinal injury
- Amputation proximal to wrist or ankle
- Flail or crushed chest
- Consider need for blood transfusion or advanced airway when not available by ground crew

Situational Criteria:

- High energy mechanisms
- Prolonged entrapment
- Multiple casualty incident

Patients will be categorized according to the current Tennessee Trauma Destination Determinates.

DO NOT call for air ambulance transport if patient is in traumatic cardiopulmonary arrest. If the patient has no vital signs, they are a trauma full-arrest.

The Paramedic in charge of the patient shall have the authority through the Incident Commander to disregard the response of the air ambulance.

The Paramedic will coordinate with the Incident Commander to ensure the helicopter receives patient information and landing zone location.

The following may impact transport by helicopter:

- Adults who have a traction splint(s) applied
- Patients over 6' 4" (relative limitation)
- Patients whose girth exceeds 27"
- Any splint or device that exceeds the boundary of the long spine board

Note: Medical responsibility will be assumed by the medical flight crew personnel upon arrival at the scene.

EMT AEMT PARAMEDIC STOP HERE

SHOCK/TRAUMA
SOP # 402 Abdominal / Pelvic Trauma

Signs and Symptoms
Abdominal / retroperitoneal abrasions / contusions
Penetrating injuries
Hypotension
Abdominal evisceration(s)
Abdominal pain on palpitation
Hematuria, bloody stool
Altered bowel sounds
Vomiting blood
History of abdominal injury / trauma
Suspected injury secondary to mechanism of trauma

TREATMENT PATHWAY

1. Oxygen 100% and airway maintenance appropriate for the patient's condition.
2. C-Spine protection as appropriate.
3. Stop any life-threatening hemorrhaging.
4. Supportive care.
5. Pulse oximetry.
6. Systolic BP or peds normal for age:
    a. If systolic BP > 90 mmHg place patient supine with legs elevated and flexed at knees and hips if no c-spine concerns, contact Medical Control.
    b. Patient pregnant:
        i. IF patient is not past 1st trimester: place patient supine with legs elevated and flexed at knees and hips if no c-spine concerns, contact Medical Control.
        ii. If patient is past 1st trimester: place patient in left lateral recumbent position.
    c. Penetrating object:
        i. If no penetrating object: place patient supine with legs elevated and flexed at knees and hips if no c-spine concerns, contact Medical Control.
    d. Evisceration:
        i. If present: place patient supine with legs elevated and flexed at knees and hips if no c-spine concerns, contact Medical Control, cover evisceration(s) with saline soaked trauma dressing.
7. EKG Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)*** target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

9. EKG monitor. Assess for arrythmias, ischemia.
10. The KED may be inverted and used as a Pelvic Stabilization Device.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 403 Avulsed Teeth

Signs and Symptoms
Avulsed teeth may be handled in the same manner as small parts; i.e. rinse in normal saline (do not rub or scrub) and place in gauze moistened with saline
Do not cool tooth/teeth with ice

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient condition.
2. C-spine protection as appropriate.
3. Treat other associated injuries.
4. Pay attention to the airway, bleeding and avulsed teeth may cause airway obstruction.
5. Supportive care.

EMT AEMT STOP HERE

6. Re-implantation is recommended at the scene as this creates maximum possibility of reattachment if possible. The following guidelines pertain to re-implantation at the scene:
    a. Applicable only for permanent teeth (i.e., with patients over 6.5 years of age)
    b. Applicable when only one or two teeth are cleanly avulsed, and the entire root is present
    c. Applicable only to anterior teeth (front 6, upper and lower)
    d. The patient must be conscious
    e. Should be attempted within the first 30 minutes (the sooner performed, the greater the success rate).
    f. Do not force re-implantation, gentle insertion is all that is necessary, slight incorrect positioning can be corrected later
7. If re-implantation is not feasible and the patient is a fully conscious adult, then the best procedure is to place the tooth in mouth, either under the tongue or in the buccal vestibule. This is not recommended for children.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 404 Cardiogenic Shock

Signs and Symptoms
Frequently associated with tachy/brady dysrhythmia, AMI, or blunt chest trauma
Neck vein distention in sitting position
Moist sounding lungs (rales, rhonchi)
Peripheral edema (if chronic heart failure)
Determine if cardiac dysrhythmia exists
Consider tension pneumothorax
Consider cardiac tamponade
Increased heart rate
Decreased BP
Altered LOC

TREATMENT PATHWAY

1. Semi-fowlers or position of comfort.
2. Oxygen and airway maintenance appropriate to patient's condition.
3. Pulse oximetry.
4. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

5. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)*** target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

6. Evaluate cardiac rhythm and treat appropriately.

Contact Medical Control, consider: EPINEPHrine 2-10 mcg/min, ***(peds 0.1-1 mcg/kg/min).***

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 405 Eye Trauma

Signs and Symptoms
Impaled object
Inability to open eye(s)
Swollen, edematous eye(s)
Photophobia
Visual defects, loss of vision
Redness

TREATMENT PATHWAY

EMT AEMT PARAMEDIC

Treatment – Standing Order

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. C-Spine protection PRN.
3. If thermal or chemical:
    a. Flush eye(s) with NS or water for 15 minutes
    b. Cover both eyes
    c. Transport
4. Penetration:
    a. Stabilize
    b. Do not apply tight dressing to penetrating eye injury, simply cover with eye shield
    c. Consider covering unaffected eye
    d. Transport
5. Blunt trauma:
    a. Consider covering both eyes
    b. Transport
6. Is loss of vision present?
    a. No – contact Medical Control
    b. Yes – If loss of vision was sudden, painless and non-traumatic, consider Retinal Artery Occlusion. Contact Medical Control and:
        i. Apply cardiac monitor and assess for changes (EMT and above only)
        ii. Apply vigorous pressure using heel of hand to affected eye for 3 – 5 seconds, then release (patient may perform this procedure and may be repeated as necessary

SHOCK/TRAUMA
SOP # 406 Hypovolemic Shock

Signs and Symptoms	Note
Blood loss due to penetrating injuries to torso or other major vessel Fracture of femur or pelvis G.I. bleeding, vaginal bleeding, or ruptured ectopic pregnancy Dehydration cause by vomiting, diarrhea, inadequate fluid intake, excessive fluid loss due to fever, uncontrolled diabetes, or burns Pulse may be greater than 120 beats per minute Blood pressure may be less than 90 mmHg systolic Orthostatic (Tilt) changes in vital signs (consider possible spinal injury) pulse increase of 20 beats per minute, BP decrease of 10 mmHg systolic Severe shock (hypovolemia) is defined as a decreased level of consciousness, absent radial pulse, capillary refill greater than 2 seconds, no palpable blood pressure	Cervical spine immobilization is not necessary in patients suffering penetrating trauma (stab or gunshot wound) below the nipple line **and** no evidence of spinal or head injury. Do not delay transport of patients meeting these criteria for immobilization.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Consider spinal protection.
3. Trendelenburg patient if no suspected injury.
4. Pulse oximetry.
5. Control gross hemorrhage – consider tourniquet or hemorrhage control clamp.
6. Monitor, 12 Lead EKG, transmission if applicable.

EMT STOP HERE

7. IV NS or LR x 2 large bore titrated to restore patient's vital signs (in patients with ongoing blood loss maintain patient's systolic blood pressure 90 – 110 mmHg).
8. ***Pediatrics –***
   a. ***IV/IO NS 20 cc/kg bolus***
   b. ***Reassess patient***
   c. ***Repeat fluid bolus 20 cc/kg if no improvement***
   d. ***Place a second IV as needed***
   e. ***Maintain temperature > 97°***

AEMT STOP HERE

If nonresponsive to volume resuscitation, consider: Adults EPINEPHrine 2–20 mcg/min ***(peds EPINEPHrine 0.1-1 mcg/kg/min).***

To prepare EPINEPHrine 10 mcg/ml:
   a. Draw up 9 ml of normal saline into 10 ml syringe
   b. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) (1:10,000)
   c. Label syringe

Adult Push Dose Pressor IV Administration:
Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension. Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 407 Major Thermal Burns

Signs and Symptoms	Note
Look for burns of the nares, oropharyngeal mucosa, face or neck Listen for abnormal breath sounds Note if burn occurred in closed space Determine extent of injury (including associated injuries) Cardiac monitor for all major burn patients Respiratory distress ETOH / drug use Associated injuries / trauma Hypotension Past medical history	Remove clothing from affected parts **DO NOT** pull material out of the burn site. Cut around it Major Burn: • Greater than 20% BSA, partial thickness surface involvement • Greater than 10% BSA, full thickness burn • Full thickness burns of the head, face, feet, hands, or perineum • Inhalation burns or electrical burns • Burns complicated by fractures or other significant injury • Elderly, pediatric, or compromised patients

TREATMENT PATHWAY

1. Stop the burn process with tepid water or normal saline solution and remove any smoldering clothing.
2. High flow oxygen and airway maintenance appropriate to patient's condition:
    a. Edema may cause patient's airway to close without warning signs
    b. Be prepared to assist ventilation with BVM
3. Monitor all vital signs and continue reassessment with emphasis on the respiratory rate, peripheral pulses (circulation) and level of consciousness.
4. Remove any jewelry and document.
5. Cover burned area with dry sterile dressing or burn sheet. Attempt to keep blisters intact.
6. DO NOT use Water-jel or any other commercially manufactured burn products. DO NOT remove if applied prior to arrival.
7. Monitor to prevent hypothermia.
8. Stabilize all associated injuries (e.g., chest, potential spine injury, fractures, dislocations, etc.).
9. Pulse oximetry.
10. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

11. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)*** target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

12. For major burns, administer pain medication per pain management protocol (contact Medical Control in multi-system trauma/pregnancy), transport (all additional doses must be approved by Medical Control).
13. If extremely injured, cover open fractures/lacerations/injuries with sterile dressing, splint fractures PRN, avoid unnecessary movement, transport.
14. Consider cyanide poisoning and smoke inhalation in all burn patients.
15. Patients with significant possibility of smoke inhalation or exposure to superheated air should be transported to the Regional Burn Center.
16. Consider contacting Medical Control for sedating/pain management agents especially in pediatric patients.

Administer IV fluids using the following guide:

- 500 mL per hour for patients over 15 years old.
- ***250 mL per hour for patients 5 – 15 years old.***
- ***125 mL per hour for patients under 5 years old***

Excessive or overly aggressive amounts of fluid administration may increase third spacing shock.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 408 Musculoskeletal Trauma

Signs and Symptoms	Notes
Deformity, swelling, tenderness, crepitus, open or closed fractures Hemorrhaging, lacerations, ecchymosis, instability Decreased function, pulses Loss of sensation of distal extremities ETOH/drug use Mechanism of injury	• May also utilize patient controlled Nitrous Oxide for pain management • Cervical spine protection is not necessary in patients suffering penetrating trauma if no evidence of neurological injury

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. C-spine protection PRN.
3. Control any life-threatening hemorrhaging.
4. Splint PRN, stabilize penetrating objects.
5. Consider tourniquet or hemorrhage control clamp.
6. Pulse oximetry.
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)*** Target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

9. Assess and manage Perfusion Status, Injuries, and Pain:
    a. If systolic BP > 90 mmHg or peds normal range for age
        i. Consider pain medications per pain management protocol.
        ii. Cover open fractures/lacerations, check distal motor/sensory/pulse pre/post splinting, avoid unnecessary movement.
    b. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg ***(peds 20 cc/kg).***
    c. If patient pregnant: isolated extremity trauma only
        i. If past 1st trimester and systolic BP > 90 mmHg contact Medical Control.
        ii. If systolic BP <90 mmHg place patient in left lateral recumbent position, IV NS/LR 20 cc/kg.
10. Consider Pain Management per protocol

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 409 Neurogenic Shock

Signs and Symptoms	**Note**
Associated with spinal cord injuries, closed head injuries, and overdoses Signs of hypovolemic shock without pale diaphoretic skin (warm shock)	Consider occult bleeding and treat as Hypovolemic Shock protocol. Hypotension and Hypoxemia are profoundly harmful in traumatic brain injury patients. AVOID THEM.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Establish and maintain c-spine protection.
3. Supportive care.
4. Pulse oximetry.
5. Hemorrhage control - Consider tourniquet use or hemorrhage control clamp.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

8. If refractory hypotension:

EPINEPHrine 2-10 mcg/min *(peds 0.1-1 mcg/kg/min).*

To prepare EPINEPHrine 10 mcg/ml:

1. Draw up 9 ml of normal saline into 10 ml syringe
2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) from a cardiac syringe
3. Label syringe

Adult Push Dose Pressor IV Administration:

Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 410 Septic Shock

Signs and Symptoms
Hot and dry or cool and clammy skin
Poor capillary refill with Tachycardia / Hypotension
End Tidal CO_2 levels below normal
Potential for underlying infection

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Obtain and record an oral or axillary temperature if possible.
3. Pulse oximetry.
4. Maintain body temperature above 97°F.
5. Glucose check.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS/LR TKO, If systolic BP < 90 mmHg, IV NS/LR **30** cc/kg bolus ***(peds 20 cc/kg bolus)***.
8. If hypoglycemic, titrate D10 slowly until patient responds. ***(peds – see glucose dosing).***

AEMT STOP HERE

9. Obtain Blood Sample tubes if available.
10. Notify receiving hospital of Sepsis Alert.

Treatment – Protocol

If no improvement after two boluses of IV fluids, contact Medical Control and consider:
Adult EPINEPHrine 2-10 mcg/min ***(peds 0.1-1 mcg/kg/min).***

To prepare EPINEPHrine 10 mcg/ml:

1. Draw up 9 ml of normal saline into 10 ml syringe
2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) (1:10,000)
3. Label syringe

Adult Push Dose Pressor IV Administration:
Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.
Ensure body substance isolation precautions.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 411 Soft Tissue / Crush Injuries

Signs and Symptoms
Hypotension
Deformity, swelling, tenderness, crepitus, open or closed fractures
Hemorrhaging, lacerations, ecchymosis, instability
Decreased function, pulses
Loss of sensation of distal extremities
ETOH / drug use
Mechanism of injury
Notes
Cervical spine protection is not necessary in patients suffering penetrating trauma (stab or gunshot wound) below the nipple line **and** no evidence of spinal or head injury. Do not delay transport of patients meeting these criteria for immobilization.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. C-spine protection PRN.
3. Control any life-threatening hemorrhaging.
4. Other splints PRN, stabilize penetrating objects.
5. Consider hemorrhage control clamp, iTClamp may be used on scalp lacerations as well.
6. Pulse oximetry.
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

9. Trauma:
    a. If systolic BP > 90 mmHg or peds normal range for age;
        i. Consider pain medications per pain management protocol.
        ii. Cover open fractures/lacerations, check distal motor/sensory/pulse pre/post splinting, avoid unnecessary movement.
    b. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg ***(peds 20 cc/kg)***
    c. If patient pregnant: isolated extremity trauma only
        i. If past 1st trimester and systolic BP > 90 mmHg, contact Medical Control.
        ii. If systolic BP < 90 mmHg, place patient in left lateral recumbent position, IV NS/LR 20 cc/kg.

Consider Ketamine/opiate per pain management protocol.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 412 **Spinal Cord Injury**

Signs and Symptoms
Hypotension without actual volume loss
Warm/flushed skin despite hypotension
Paralysis
Loss of reflexes
Posturing
Priapism
Diaphragmatic breathing

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. C-Spine protection.
3. Control hemorrhaging.
4. Pulse oximetry.
5. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

6. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

7. If signs of Hypotension/ inadequate perfusion:

   Adult EPINEPHrine 2-10 mcg/min ***(peds 0.1-1 mcg/kg/min).***

   To prepare EPINEPHrine 10 mcg/ml:
   1. Draw up 9 ml of normal saline into 10 ml syringe
   2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) (1:10,000)
   3. Label syringe

   Adult Push Dose Pressor IV Administration:
   Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 413 Trauma - Blunt

Notes
- Rapidly assess and manage life-threatening injuries
- Safely move the patient to prevent worsening injury severity
- Rapidly and safely transport to the appropriate level of trauma care.
- Hypotension and Hypoxemia are profoundly harmful in traumatic brain injury patients. AVOID THEM.

TREATMENT PATHWAY

1. Initiate in-line C-Spine protection while simultaneously evaluating and controlling the patient's ABCs. Incorporate the mechanism of injury into the patient care scheme.
2. Control any hemorrhage and simultaneously provide oxygen and airway maintenance appropriate to patient's condition.
3. Spine injuries in the adult population may be present at more than one level simultaneously. Spinal Motion Restriction (SMR), when indicated, should apply to the entire spine. An appropriately sized cervical collar is a critical component of SMR and should be used to limit movement of the cervical spine whenever SMR is employed. The remainder of the spine can be stabilized using an ambulance cot, a vacuum mattress, a long back board or a similar device.
4. Pulse oximetry.
5. Consider tourniquet use or hemorrhage control clamp.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS/LR TKO. If systolic BP < 90mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients. If not hypotensive, avoid administering more than 500 cc crystalloid.

AEMT STOP HERE

8. If refractory hypotension:
   EPINEPHrine 2-10 mcg/min *(peds 0.1-1 mcg/kg/min).*
   To prepare EPINEPHrine 10 mcg/ml:
   1. Draw up 9 ml of normal saline into 10 ml syringe
   2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) (1:10,000)
   3. Label syringe

Adult Push Dose Pressor IV Administration:
Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction.

9. Contact Medical Control for pain management authorization.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP #414 Adult Trauma Arrest (Blunt)

This protocol establishes criteria for the appropriate management for handling blunt traumatic cardiac arrest.

TREATMENT PATHWAY

If patient meets criteria for Death or No resuscitation/DNR/ POST Form OR has all the following clinical findings:

- Apnea
- Pulselessness
- Asystole or PEA <40

If patient does not meet all the above criteria, Initiate resuscitation per protocol.

Consider chest decompression in these patients prior to discontinuing resuscitative efforts.

NOTES

Withholding resuscitative efforts with blunt and penetrating trauma victims who meet criteria is appropriate.

First arriving EMS personnel should make the assessment concerning agonal respirations, pulselessness, asystole or PEA < 40.

If mechanism not consistent with traumatic arrest and cardiac could have been primary cause, then the arrest should be worked in conjunction with that protocol.

Efforts should be directed at high quality and continuous compressions with limited interruptions and early defibrillation when indicated. Consider early IO placement if available and difficult IV anticipated.

DO NOT HYPERVENTILATE: If no advanced airway (BIAD, ETT) compressions to ventilations are 30:2. If advanced airway in place ventilate 8 – 10 breaths per minute.

Lightning strike, drowning or in situations causing hypothermia, resuscitation should be initiated.

Where multiple lightning strike victims are found used Reverse Triage: Begin CPR where apneic / pulseless.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 415 Trauma - Penetrating

Notes

- Rapidly assess and manage life-threatening injuries
- Safely move the patient to prevent worsening injury severity
- Rapidly and safely transport to the appropriate level of trauma care
- Hypotension and Hypoxemia are profoundly harmful in traumatic brain injury patients. AVOID THEM
- Cervical spine protection is not necessary in patients suffering penetrating trauma if no evidence of neurological injury

TREATMENT PATHWAY

1. Initiate in-line C-Spine protection while simultaneously evaluating and controlling the patient's ABCs. Incorporate the mechanism of injury into the patient care scheme.
2. Control any hemorrhage and simultaneously provide oxygen and airway maintenance appropriate to patient's condition.
3. Spine injuries in the adult population may be present at more than one level simultaneously. Spinal Motion Restriction (SMR), when indicated, should apply to the entire spine. An appropriately sized cervical collar is a critical component of SMR and should be used to limit movement of the cervical spine whenever SMR is employed. The remainder of the spine can be stabilized using an ambulance cot, a vacuum mattress, a long back board or a similar device.
4. Pulse oximetry.
5. Consider tourniquet use or hemorrhage control clamp.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS/LR TKO. If systolic BP < 90mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients. If not hypotensive, avoid administering more than 500 cc crystalloid.

AEMT STOP HERE

Cont'd…

8. If refractory hypotension:
   EPINEPHrine 2-10 mcg/min *(peds 0.1-1 mcg/kg/min).*

To prepare EPINEPHrine 10 mcg/ml:

1. Draw up 9 ml of normal saline into 10 ml syringe
2. Into this syringe, draw up 1 ml of EPINEPHrine 0.1 mg/ml (1 mg/10 ml) (1:10,000)
3. Label syringe

Adult Push Dose Pressor IV Administration:

Using above EPINEPHrine preparation (EPINEPHrine 10mcg/ml) 10 ml syringe, administer 5-10 mcg (0.5-1 ml) slow IV push (over 30-60 seconds) every 2-5 minutes as needed as a temporizing measure for severe hypotension.

Note: Monitor HR and BP continuously while administering/titrating EPINEPHrine, as it may cause significant tachycardia and tachyarrhythmia in addition to the desired vasoconstriction

9. Contact Medical Control for pain management authorization.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 416 **Adult Trauma Arrest (Penetrating)**

This protocol establishes criteria for the appropriate management for handling penetrating traumatic cardiac arrest.

TREATMENT PATHWAY

If patient meets criteria for Death or No resuscitation/DNR/ POST Form OR has all the following clinical findings:

- Apnea
- Pulselessness
- Asystole or PEA <40
- Absence of Pupillary Reflexes
- No Spontaneous movements

If patient does not meet all the above criteria, Initiate resuscitation per protocol

NOTES

Withholding resuscitative efforts with blunt and penetrating trauma victims who meet criteria is appropriate.

First arriving EMS personnel should make the assessment concerning agonal respirations, pulselessness, asystole or PEA < 40, pupillary reflexes and spontaneous body movements.

Efforts should be directed at high quality and continuous compressions with limited interruptions and early defibrillation when indicated. Consider early IO placement if available and difficult IV anticipated.

DO NOT HYPERVENTILATE: If no advanced airway (BIAD, ETT) compressions to ventilations are 30:2. If advanced airway in place ventilate 8 – 10 breaths per minute.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 417 **Adult Traumatic Cardiac Arrest**

Signs and Symptoms	**Note**
Cardiac arrest secondary to trauma	Upon arrival at scene, patients in traumatic cardiac arrest should be placed on a cardiac monitor. If no cardiac activity, consider cessation of resuscitative efforts.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. CPR.
3. Pulse oximetry.
4. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

5. IV NS/LR give 20 cc/kg bolus.
6. Consider second IV access.

AEMT STOP HERE

7. Treat cardiac rhythms per specific protocols.
8. If suspected pneumothorax, perform needle chest decompression.
9. Consider viability of patient prior to transport.

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 418 **Tension Pneumothorax**

Signs and Symptoms	**Note**
Acute respiratory distress, cyanosis Unilaterally decreased breath sounds or absent breath sounds Hyper-resonance of chest unilaterally Jugular vein distention Subcutaneous Emphysema Acute traumatic chest injury, ecchymosis or obvious rib fractures History of COPD or other chronic lung disease which predisposes patient to spontaneous pneumothorax Hypotension Tracheal deviation away from the affected side Arrhythmia Oxygen saturation - < 90% Mechanism of injury	Patient must meet **AT LEAST THREE** of these Signs/Symptoms to qualify for this standing order; otherwise, contact Medical Control.

TREATMENT PATHWAY

1. Oxygen 12 – 15 LPM NRB and airway maintenance appropriate to patient's condition.
2. Perform frequent evaluation of the breath sounds and blood pressure.
3. Control any life-threatening hemorrhaging.
4. Consider initiation of the multiple trauma protocol if indicated. Remember this order may be indicated for the medical patient as well.
5. If the traumatic tension pneumothorax is secondary to a sucking chest wound, apply an occlusive dressing and treat appropriately.
6. Pulse oximetry.
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg bolus ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patient.

AEMT STOP HERE

9. If tension pneumothorax suspected, perform needle decompression. Use 14 g 3.5" needle ***(peds may use smaller 18 g needle).***

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 419 Traumatic Amputation(s)

Signs and Symptoms
Hypotension
Past medical history
Deformity, swelling, tenderness, crepitus, open or closed fractures
Hemorrhaging, lacerations, ecchymosis, instability
Decreased function, pulses
Loss of sensation of distal extremities
ETOH / Drug use
Mechanism of injury

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. C-spine protection PRN.
3. Control any life-threatening hemorrhaging.
4. Other splints PRN.
5. Amputated part: if recovered rinse with NS, wrap in moist dressing, place in plastic bag, and transport with patient.
6. Consider use of tourniquet or hemorrhage control clamp, if appropriate.
7. Pulse oximetry.
8. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

9. IV NS/LR TKO. If systolic BP < 90 mmHg, IV NS/LR 20 cc/kg ***(peds 20 cc/kg bolus)***. Target SBP is 90 – 110 mmHg in adult trauma patients.

AEMT STOP HERE

10. Amputation:
    a. If present and systolic BP > 90 mmHg, consider pain management per protocol
    b. Cover open fractures/lacerations, check distal motor/sensory/pulse pre/post splinting, avoid unnecessary movement

PARAMEDIC STOP

SHOCK/TRAUMA
SOP # 420 Blast Injuries

Signs and Symptoms	Notes
Exposure to explosive force. Injuries may include any or all of the following: – Blunt trauma – Penetrating trauma (including shrapnel) – Burns (thermal, chemical, airway) – Pressure-related injuries (barotrauma): blast lung, pneumothorax/tension pneumothorax, tympanic membrane perforation – Toxic chemical contamination – Chemical, biological, radiological, nuclear, and explosive (CBRNE) devices or agents	Scene safety is of paramount importance. Always consider the possibility of subsequent explosions. Structural safety, toxic chemical contamination, poisonous gases, and other hazards may delay patient extraction. Tympanic membrane perforation/deafness may complicate evaluation of mental status and ability to follow commands.

TREATMENT PATHWAY

1. Ensure scene safety. Identify ongoing threats; consider secondary devices and structural, chemical, and poisonous-gas hazards. Remove the patient from the scene as soon as practical and safe.
2. Hemorrhage control: assess for and stop severe hemorrhage (tourniquet or hemorrhage control clamp per procedure).
3. Airway: assess patency. Consider thermal or chemical burns to the airway; if suspected, early airway management is vital. Secure the airway using airway maneuvers and adjuncts.
4. Breathing: evaluate respiratory effort, oxygenation, lung sounds, and chest wall integrity. Continually reassess for blast lung injury. Administer oxygen with a target SpO2 of 94–98% and assist respirations as needed. Cover any open chest wounds with a semi-occlusive dressing.
5. Circulation: look for hemorrhage; assess BP, pulse, skin color/character, and distal capillary refill for signs of shock.
6. Disability: assess responsiveness (AVPU) and level of consciousness (GCS), pupils, and gross motor movement of extremities. Monitor GCS during transport for changes. Apply spinal precautions per SOP 608.
7. Exposure: rapidly evaluate the entire skin surface, including the back (log roll), to identify blunt or penetrating injuries. Keep the patient warm to prevent hypothermia.
8. If chemical, biological, or radiological contamination is suspected, manage per SOP 201 and SOP 209.
9. Transport: primary transport to a trauma or burn center is preferable whenever possible. If the patient has sustained burns (thermal, chemical, or airway), consider transport to a specialized burn center.

EMT STOP HERE

10. Establish IV access with two large bore IVs or IOs.
11. Resuscitative fluids per SOP 413/415. Minimize IV fluid resuscitation in patients without signs of shock. If the patient is burned, administer NS or LR per SOP 407.
12. Supraglottic airway as needed for airway control.

AEMT STOP HERE

13. If thermal or chemical burn to the airway is suspected, secure the airway early using a supraglottic device or endotracheal tube.
14. If the patient has evidence of tension pneumothorax (hypotension or other signs of shock with decreased or absent breath sounds, JVD, and/or tracheal deviation), perform needle decompression per SOP 418.

15. If evidence of head injury, treat per SOP 413.

PARAMEDIC STOP

Key Considerations

1. Patients may sustain complex, multi-system injuries, including blunt and penetrating trauma, shrapnel, barotrauma, burns, and toxic chemical exposure.
2. Consideration of airway injury, particularly airway burns, should prompt early and aggressive airway management.
3. Minimize IV fluid resuscitation in patients without signs of shock. Consider injuries due to barotrauma.

Documentation

Evidence of multi-system trauma, especially airway injury/burn, barotrauma to the lungs, and toxic chemical contamination. Initial and changing GCS.

SHOCK/TRAUMA
SOP # 421 High Threat Considerations / Active Shooter Scenario

Signs and Symptoms	Notes
High threat environment – conditions greater than normal exist that could cause threat to the clinician or patient (e.g., active shooter, unstable building collapse, improvised explosive device, hazardous material threat) **Zone definitions:** Hot Zone / Direct Threat Zone – an area within the inner perimeter where an active threat and active hazards exist. Warm Zone / Indirect Threat Zone – an area within the inner perimeter where security and safety measures are in place. This zone may have potential hazards, but no active hazards exist.	Not applicable when no significant threat exists to clinician or patient, allowing routine care. Clinician and patient safety must be considered simultaneously. Risks taken must be weighed against the expected benefit to the patient.

TREATMENT PATHWAY

HOT ZONE / DIRECT THREAT CARE

1. Mitigate the threat as able to minimize risk to patients and clinicians. Move to a safer position and recognize that threats are dynamic and may be ongoing, requiring continuous assessment of threat.
2. Defer in-depth medical interventions while engaged in an ongoing direct threat.
3. Defer triage until no longer in a hot zone / direct threat zone.
4. Prioritization for extraction is based on resources available and the situation encountered.
5. Encourage patients to provide self-first aid, or instruct uninjured bystanders to provide aid.
6. Consider hemorrhage control:
    a. Tourniquet application is the primary “medical” intervention to be considered in the Hot Zone. Tourniquet choice should be guided by expected ability to perform in the desired patient population (pediatrics).
    b. If no tourniquet is available (or application is not feasible), consider instructing the patient to apply direct pressure to the wound.
    c. Consider quickly placing, or directing the patient to be placed, in a position that protects the airway if not immediately moving the patient.

WARM ZONE / INDIRECT THREAT CARE

7. Maintain situational awareness.
8. Ensure safety of both responders and patients by rendering equipment and environment safe (firearms, vehicle ignition).
9. Conduct a primary survey per SOP 413/415 and initiate appropriate life-saving interventions:
    a. Hemorrhage control: tourniquet; wound packing if feasible
    b. Maintain airway and support ventilation
10. Maintain body temperature and prevent hypothermia.
11. Do not delay patient extraction and evacuation for non-life-saving interventions.
12. Consider establishing a casualty collection point if multiple patients are encountered.
13. Unless in a fixed casualty collection point, triage in this phase of care is limited to the following categories:
    a. Uninjured and/or capable of self-extraction
    b. Deceased / expectant
    c. All others

Patient Safety Considerations

1. Anticipate unique threats based on the situation.
2. During high threat situations, clinician safety should be considered in balancing the risks and benefits of patient treatment.

Key Considerations

1. In high threat situations clinician and patient safety must be considered simultaneously.
2. An integrated response with other public safety entities may be warranted.
3. Risks taken and threats to responder safety must be weighed in relation to the expected benefit to patient safety and outcome.
4. Maintaining communications and incident management concepts may be crucial to maximizing efficiency and mitigating dangers.

Documentation

Traditional documentation may not be appropriate during Hot Zone and Warm Zone care. Key interventions should be relayed to the receiving team: time of tourniquet application; GCS for patients with suspected head injury.

REFERENCE
OBSTETRICAL EMERGENCIES
APGAR Scoring

EMT AEMT PARAMEDIC

The APGAR score should be calculated after birth of the infant. The five (5) clinical signs are evaluated according to the scoring system detailed above. Each sign is assigned points to be totaled. A total score of 10 indicates that the infant is in the best possible condition. A score of 4 to 6 indicates moderate depression and a need for resuscitative measures.

- **DO NOT delay resuscitation efforts to obtain APGAR score.**
- Obtain APGAR at 1 and 5 minutes after delivery.

Clinical Sign	**0 points**	**1 point**	**2 points**
Appearance	Blue/Pale	Body Pink Extremities Blue	Completely Pink
Pulse	Absent	Below 100/minute	Above 100/minute
Grimace	No response	Grimace	Cries
Activity	Limp	Some flexion of extremities	Action motion
Respiratory	Absent	Slow/irregular	Good strong cry

OBSTETRICAL EMERGENCIES

SOP # 500 Obstetrical / Gynecological Complaints (Non-Delivery or Gynecological Only)

Signs and Symptoms

Patient Para (number of births) and Gravida (number of pregnancies)
Term of pregnancy in weeks, EDC, Multiple births expected, or history
Vaginal bleeding (how long and approximate amount)
Possible miscarriage / products of conception
Pre-natal medications, problems, and care
Last menstrual cycle
Any trauma prior to onset
Lower extremity edema

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.
2. Patient positioning appropriate for condition.
3. Control hemorrhage as appropriate.
4. Pulse oximetry.
5. Glucose check.

EMT STOP HERE

6. INT or IV NS TKO unless signs of shock, then 20 cc/kg fluid bolus.

AEMT STOP HERE

7. EKG monitor PRN.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 501 Normal Delivery

Signs and Symptoms	Notes
Patient Para (number of births and Gravida (number of pregnancies) Term of pregnancy in weeks, EDC Vaginal bleeding Pre-natal medications, problems, and care Membranes ruptured Lower extremity edema	• The greatest risks to the newborn infant are airway obstruction and hypothermia. Keep the infant warm (silver swaddler), dry, covered, and the infant's airway maintained with a bulb syringe. Always remember to squeeze the bulb prior to insertion into the infant's mouth or nose. • The greatest risk to the mother is post-partum hemorrhage. Watch closely for signs of hypovolemic shock and excessive vaginal bleeding. • Spontaneous or induced abortions may result in copious vaginal bleeding. Reassure the mother, elevate legs, treat for shock, and transport. • Record a blood pressure and the presence or absence of edema in every pregnant woman you examine, regardless of chief complaint. • Complete individual patient care reports on **BOTH** mother and child.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate for the patient's condition.

Mother:

2. Pulse oximetry.
3. Check mother for crowning PRN.
4. Use gentle pressure to control delivery. When head delivers, suction airway and check for cord around neck.
5. After delivery, keep mother and infant on same level, clamp cord at 8 and 10 inches from the baby and cut between clamps.
6. Dry infant and wrap to keep warm, maintain airway.
7. Check APGAR at 1 and 5 minutes post delivery.
8. Allow placenta to deliver:
    a. Massage uterine fundus (lower abdomen)
    b. Observe and treat signs of shock with increased delivery of oxygen and IV fluids
    c. Be alert to the possibility of multiple births
9. Re-evaluate vaginal bleeding.

Infant:

1. Protect against explosive delivery.
2. After delivery, suction airway (mouth first then nose) and check for cord around neck.
3. After delivery, clamp cord at 8 and 10 inches from baby and cut between clamps.
4. Dry infant and wrap to keep warm (silver swaddler). Maintain airway, suction PRN.
5. Oxygen and airway maintenance appropriate to patient's condition.
6. Check APGAR score at 1 and 5 minutes after delivery.
7. Re-evaluate cord for bleeding, if bleeding, add additional clamp and re-evaluate.

EMT STOP HERE

8. NT or IV LR TKO if patient in active labor defined as regular contractions q 3 – 5 min with 30 – 60 second duration.

AEMT STOP HERE

9. EKG monitor PRN.
10. Evaluate for postpartum hemorrhage.
11. If available, administer Oxytocin (Pitocin) 10 units per ampule/vial IM. If no IV access or 30-40 units in 1 liter free flow.
12. For severe Postpartum hemorrhage, if available, give either:

    Methergine 0.2mg IM 1 dose only. Do not give if SBP>160 or DBP>110,

    OR

    Misoprostol (Cytotec) 400 mcg buccal or sublingual

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 502 Abruptio Placenta

Signs and Symptoms
Multiparity
Maternal hypertension
Trauma
Drug Use
Increased maternal age
History
Vaginal bleeding with no increase in pain
No bleeding with low abdominal pain

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Position patient in left lateral recumbent position.
3. Pulse oximetry.
4. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

5. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

6. EKG monitor for arrythmias.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 503 Amniotic Sac Presentation

Signs and Symptoms
Amniotic sac visible
Membranes not broken
Fetus may or may not be visible
Pre-natal medications, problems, and care
Usually third trimester
Applies to greater than 20 weeks gestation
Abdominal pain
Indications of immediate delivery

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Place patient in position of comfort.
3. Pulse oximetry.
4. Amniotic sac:
    a. If no fetus visible, cover presenting part with moist, sterile dressing
    b. If head of the fetus has delivered, tear sac with fingers and continue steps for delivery
5. Contact Medical Control ASAP.
6. Monitor.

EMT STOP HERE

7. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

8. EKG monitor for arrhythmias.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 504 Breech or Limb Presentation

Signs and Symptoms
Patient Para (number of births and Gravida (number of pregnancies)
Term of pregnancy in weeks, EDC
Vaginal bleeding
Pre-natal medications, problems, and care
Water broken
Buttock, arm, or leg presentation

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.

   Breech Presentation:
   Treatment – Standing Order

   a. Allow the delivery to progress spontaneously – DO NOT PULL!
   b. Support the infant's body as it delivers.
   c. If the head delivers spontaneously, deliver the infant as noted in 'Normal Delivery'.
   d. If the head does not deliver within **3** minutes, insert a gloved hand into the vagina to create an airway for the infant.
   e. DO NOT remove your hand until relieved by a Higher Medical Authority.

   Limb Presentation:

   Treatment – Standing Order
   Position the mother in a supine position with head lowered and pelvis elevated.
   Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

2. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

3. EKG monitor for arrythmias.
4. Transport ASAP.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 505 Meconium Stain

Signs and Symptoms
Patient para (number of births and Gravida (number of pregnancies)
Term of pregnancy in weeks, EDC
Vaginal bleeding
Pre-natal medications, problems, and care
Membranes ruptured
Amniotic fluid that is greenish or brownish-yellow
Fecal material expelled with the amniotic fluid

TREATMENT PATHWAY

1. Do not stimulate respiratory effort before suctioning the oropharynx.
2. Suction the **mouth then nose** (using a meconium aspirator) while simultaneously providing Oxygen by blow-by method and while maintaining the airway appropriate to the patient's condition.
3. Obtain an APGAR score after airway treatment priorities. Score at one minute after delivery and at five minutes after delivery (Time permitting).
4. Repeat initial assessment and complete vital signs until patient care is transferred to the appropriate ED staff.
5. Pulse oximetry.
6. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

7. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

8. EKG monitor for arrythmias.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 506 Placenta Previa

Signs and Symptoms	Note
Painless bleeding which may occur as spotting or recurrent hemorrhage Bright red vaginal bleeding usually after 7th month History Multiparity Increased maternal age Recent sexual intercourse or vaginal exam Patient para (number of births) and gravida (number of pregnancies) Term of pregnancy in weeks Pre-natal medications, problems, and care History of bed rest Placenta protruding through vagina	Any painless bleeding in the last trimester should be considered Placenta Previa until proven otherwise. If there are signs of eminent delivery membranes rupture is indicated followed by delivery of the baby. The diagnosis of eminent delivery depends on the visual presence of the baby's body part through the membrane.

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Position of comfort.
3. Pulse oximetry.
4. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

5. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

6. EKG monitor for arrythmias.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 507 Prolapsed Umbilical Cord

Signs and Symptoms
Cord emerges from the uterus ahead of the baby
With each uterine contraction, the cord is compressed between the presenting part and the pelvis
Pulse on exposed cord may or may not be palpable
Patient para (number of births) and gravida (number of pregnancies)
Term of pregnancy in weeks, EDC
Vaginal bleeding
Pre-natal medications, problems, and care
Membranes ruptured

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's condition.
2. Position the mother with hips elevated:
    a. Knee to chest
    b. Hips elevated as much as possible on pillows
3. Palpate pulses in the cord.
4. Instruct mother to pant with each contraction, which will prevent her from bearing down.
5. Check for a pulse in the cord:
    a. If no pulse – insert a gloved hand into the vagina and gently push the infant's head off the cord, while pressure is maintained on the head cover the exposed cord with a sterile dressing moistened in saline; transport immediately and **DO NOT** remove your hand until relieved by hospital staff
    b. If pulse present – cover exposed cord with moist dressing
6. Contact Medical Control as soon as possible if time and patient condition allows.
7. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

8. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

9. EKG monitor for arrythmias.

PARAMEDIC STOP

OBSTETRICAL EMERGENCIES
SOP # 508 Pre-eclampsia and Eclampsia

Signs and Symptoms	Note
Patient para (number of births) and gravida (number of pregnancies) Term of pregnancy in weeks, EDC Vaginal bleeding Pre-natal medications, problems, and care Membranes ruptured Usually begins after the 20th week of pregnancy Most often affects women during their first pregnancy May have a history of chronic hypertension and/or diabetes May experience headaches, blurred vision, and abdominal pain May experience seizures which indicates a progression from pre-eclampsia to eclampsia	Record a blood pressure and the presence or absence of edema in every pregnant woman you examine regardless of chief complaint. This condition can occur several weeks after delivery!!

TREATMENT PATHWAY

1. Oxygen and airway maintenance appropriate to patient's saturation >95%.
2. Place patient in left lateral recumbent position.
3. Pulse oximetry.
4. Monitor, 12 Lead EKG and transmission if applicable.

EMT STOP HERE

5. IV NS TKO, if hypotensive 20 cc/kg ***(peds 20 cc/kg).***

AEMT STOP HERE

6. EKG Monitor for arrythmias.
7. Administer Magnesium Sulfate 6 g IV slowly (over 20 minutes).
8. If seizures continue, administer additional 2g of Magnesium Sulfate, then Diazepam (Valium) 5 mg slow IV or Midazolam (Versed) 2-5 mg IVP/IM per seizure protocol if persistent generalized seizure activity.
9. Contact Medical Control.

Maternal hypertension management (Indicated for SBP>160 or DBP>110):
If available, administer:

Labetalol 20 mg IV over 2 minutes, doubling dose every 10 minutes to max of 80mg until SBP<140 or DBP<90, OR

Hydralazine 5-10 mg IV over 2 minutes, repeating aver 20 minutes until SBP<140 or DBP<90, OR

Immediate release oral nifedipine, 10mg po, may repeat in 20 minutes if not below 140/90. OR

Utilize Nitrates sublingual/ repeated

PARAMEDIC STOP

MISCELLANEOUS
SOP # 601 Discontinuation / Withholding of Life Support

Notes

- Once life support has been initiated in the field, **non-ALS** personnel **CANNOT** discontinue resuscitative measures unless directed to do so by the on-scene physician, Paramedic or presented with a valid Physician Orders for Scope of Treatment (POST).
- Upon termination in the field any tubes, needles and IV lines will be left in place (IV lines to be tied off and cut with catheter left in place.
- Personnel shall give careful consideration when utilizing this standing order. Conditions such as overdose, electrical shock, hypothermia, and hypoglycemia may mimic some of the above signs and symptoms.
- All deaths **must** be confirmed by a Paramedic.

EMT AEMT PARAMEDIC

Withholding CPR – Standing Orders

1. If there is no CPR in progress, CPR may be withheld if one or more of the conditions are met:
    a. Obviously dead patients with dependent lividity, rigor mortis, or massive trauma (i.e., crushed chest, crushed head, etc.)
    b. Obviously dead patients with tissue decomposition
    c. Patients without vital signs who cannot be accessed for treatment due to entrapment for prolonged time (12 – 15 minutes or greater)
    d. Severe blunt trauma with absence of BP, pulse, respiratory effort, neurologic response, and pupillary response
    e. When presented a valid POST order or a copy as approved by the Tennessee Department of Health, DNR and POST orders not on the official state form can be accepted if it is documented in a medical record such as a nursing chart, hospice care, or home nursing
    f. Instructed to do so by the Paramedic on the scene

Discontinuing Life Support

Once life support has been initiated in the field, to discontinue life support, the following conditions must be met:

1. Asystole present on the EKG monitor in two leads **and;**
2. There is absence of pulse, respirations, and neurological reflexes, **and**
3. At least one of the following conditions are met:
    a. Appropriate airway management has been confirmed, the patient has been well ventilated with 100% oxygen and multiple (at least three administrations of medications have not been effective in generating an EKG complex)
    b. Transcutaneous pacing, if available, has not been effective in generating a pulse
    c. Obvious signs of death in the absence of hypothermia, cold water drowning, lightning strikes, or induced coma
    d. The Paramedic can document a lack of CPR for at least ten minutes
    e. Prolonged resuscitation in the field (25 minutes of resuscitation with agonal or asystolic rhythm) without hope for survival, and ETCO2 less than 10 mmHg
    f. Massive trauma such as complete evacuation of the cranial vault, etc.
    g. Severe blunt trauma with absence of vital signs and pupillary response
    h. End-tidal CO_2 less than 10 mmHg while performing effective CPR

MISCELLANEOUS
SOP # 602 Field Determination of Death

Signs and Symptoms
Pulseless, non-breathing with definitive signs of death
Rigor Mortis
Dependent lividity
Decomposition of body tissue
Devastating, un-survivable injury
Decapitation
Incineration
Separation of vital internal organ from the body or total destruction of organs
Gunshot wound to the head that crosses the midline (entrance and exit)

EMT AEMT PARAMEDIC

If the patient is pulseless, non-breathing without definitive signs of death:
Must receive resuscitation unless a properly executed DNR or POST form is present.
Treatment – Standing Order
DNR Orders:

- If family member or caregiver can produce a properly executed DNR or POST order, resuscitation can be withheld
- Treat patients with known DNR orders appropriately; do not initiate CPR if they develop cardiovascular or respiratory arrest
- When there is any doubt about what to do, begin resuscitative efforts with all skill and equipment available and consider contacting an EMS Supervisory Personnel or Medical Control.

Resuscitation has been initiated prior to EMS arrival:
Anytime CPR or an attempt at resuscitation has been initiated by anyone at the scene, resuscitative efforts will be continued until:

- Medical Control directs the team to stop (either online or on scene)
- It is determined the patient meets the criteria for “definitive signs” of death
- A properly executed DNR or POST form is presented

MISCELLANEOUS
SOP # 603 **Mandatory EKG**

EMT AEMT PARAMEDIC

EKGs will be mandatory under the following conditions:

1. Patients complaining of or presenting with:
    - Chest pain regardless of source (trauma or illness)
    - Abdominal pain
        - All males > 40 years old
        - All females > 50 years old
    - Nausea and vomiting
        - All males > 40 years old
        - All females > 50 years old
    - Shortness of breath – all new onset
    - All diabetics and smokers whose symptoms may be cardiac related
    - Known heart disease (including CHF, post-surgery, previous MI)
    - Weakness – new onset
    - Syncope
    - Unresponsive
    - Patients with sympathomimetically active drug use (cocaine, crack, methamphetamine, etc.)
2. Cardiac arrest with or without CPR in progress.
3. That are non-viable (other than those exhibiting body decomposition, dependent lividity, or rigor mortis, decapitation).

EKGs will have the following information printed on the recording:

- Name or report number
- Age (if possible)
- Unit number and date

EKGs will be appended appropriately to the patient care report.

Note: 12 Lead EKGs may be applied and transmitted by any EMT or higher on scene; however, treatment decisions may only be made by a Paramedic.

MISCELLANEOUS

SOP # 604 Patient Refusal or Declination of Care – Patient Non-Transport Situations

Signs and Symptoms
Determine presence of injury or illness and desire for transport
Identify the person who made the EMS call
Reason for refusal

EMT AEMT PARAMEDIC

Standing Orders:

1. Utilize the mini-mental status exam on any patient where you have concerns regarding the decision-making capacity of the patient.
2. Confirm and document the absence of intoxicating substance or injury.
3. Confirm patient is of legal age of majority, or emancipated minor.
4. Document mechanism of injury or circumstances of illness.
5. Document pertinent past history.
6. Perform at least two sets of vital signs (one on arrival and one before leaving) and problem directed exam, document if unable to obtain vital signs.

The following may not refuse transport:

- Patients with impaired judgment and decreased mental status (Utilize the mini-mental status exam to determine, document). Engage law enforcement for protective custody options.
- Minors (less than 18 years of age unless they are emancipated by the courts).
- All minors must have refusal from parent or guardian, not older sibling or other relative.
- Do not release minor on the scene without parent/guardian consent.

Reasons for Non-Transport:

- Minor illness or injury and acceptable alternative transportation available.

No patient found on scene:

- Patient Care Report is to be completed in detail as to why no patient was found (i.e., no person found on scene, person located with no complaint of injury/illness and denies needing medical assistance).

REFERENCE
Mini-Mental Status Exam

EMT AEMT PARAMEDIC

A score of 21 or better is considered mentally capable for a patient to make reasonable decisions.

1. Orientation to time – time of day, day, week, month, year	5 points max
2. Orientation to place – building, street, city, state, country	5 points max
3. Say “boy, dog, ball” and have them repeat it	3 points max
4. Ask the patient to spell would backward, or do serial 3s backward from 20	5 points max
5. Without repeating the words, ask them to repeat the previous 3 words (boy, dog, ball)	3 points max
6. Ask the patient to do the following after you have completed the request: “stick out your tongue and touch your right hand to your left ear.”	3 points max
7. Ask the patient to identify your pen and watch	2 points max
8. Ask the patient to read the following sentence and then does as it says “shut your eyes”	1 point
9. Ask the patient to write a sentence	1 point
10. Ask the patient to draw two overlapping pentagons (show them an example)	1 point

MISCELLANEOUS
SOP # 605 Physician On-Scene

EMT AEMT PARAMEDIC

If private physician intervenes by phone, the Firefighter/EMR, EMT, AEMT, or Paramedic shall:

- Request the physician contact Medical Control and relay any orders through them
- **NO ORDERS** will be taken over the phone from the private physician

Standing Orders:

1. No one will be recognized as a physician without proof of license. This must be in the form of a wallet card or visual personal recognition. NO ORDERS will be accepted until proof of license is verified.
2. Consider need for Law Enforcement if any difficulty with person occurs.
3. The EMT shall:
    a. Inform the physician that they must contact Medical/Trauma Control.
    b. Inform Medical/Trauma Control of the presence of a physician on scene.
4. Medical/Trauma Control may:
    a. Speak to the physician to determine qualifications.
    b. Request the EMT/Paramedic to verify licensure of the physician.
    c. Relinquish total responsibility for the patient to the on-scene physician.
5. Physician (intervening) may:
    a. Assist the EMT (or above) and allow you to operate under standing orders and protocols, offer assistance by allowing the EMS provider to remain under Medical/Trauma Control.
    OR
    b. Request to talk to Medical/Trauma control to offer advice and assistance.
    OR
    Take total responsibility for the care given by the EMS Provider if ok with Medical/Trauma Control, then physically accompany the patient to the Emergency Department where responsibility is assumed by the receiving physician; and shall:
        i. Sign for all instructions given to the EMS Provider.
        ii. Contact should be made with Medical/Trauma Control if this happens.
6. If private physician intervenes by phone or in person, the EMS Provider shall:
    a. Inform the physician that the EMS Provider must contact Medical/Trauma control.
    b. Request the physician contact Medical Control and relay any orders through them
    c. NO ORDERS should be taken over the phone from the private physician, at no time should any order be taken over the telephone except from Medical/Trauma control.

MISCELLANEOUS
SOP # 606 Bystanders on the Scene

EMT AEMT PARAMEDIC

Standing Order:
Bystander participation – you may use them at your discretion. However, YOU will be responsible for their actions and treatment. This includes other medical professionals. In any situation you need assistance you may utilize their expertise and skills.

Note: Request proof of their licensure by visualization of their current license, if possible. Remember YOU are responsible for the patient. If any bystander is trying to take over direction of patient care, other than a physician (Follow Physician on Scene SOP # 606 in this situation), you may have law enforcement remove the person for “Obstruction of Emergency Services”.

MISCELLANEOUS
SOP # 607 **Procedure for Deviation from Standing Orders**

EMT AEMT PARAMEDIC

NEVER simply disregard a standing order or protocol.
These standing orders have been established so that EMS personnel may provide the best care possible for our patients. Most of our patients will be covered by a single standing order. However, some patients may have illnesses and/or injury that are covered by more than one standing order or, in rare cases, following a standing order may not be in the best interest of the patient. In these cases, you must be aware that combining standing orders may not be in the best interest of the patient and that combining standing orders may lead to medication errors, overdose, and medication incompatibility. You are expected to use your judgment and to always make decisions that are in the best interest of the patient.
If you use more than one standing order when treating your patient, you must document your reasoning in the NARRATIVE SECTION of the Patient Care Report.
If in your judgment, following a standing order is not in the best interest of the patient, CONTACT MEDICAL CONTROL regarding your treatment. Document the rationale for deviation, and the name of the physician giving the order.

MISCELLANEOUS
SOP # 608 Spinal Protection

EMT AEMT PARAMEDIC

The intent of this guideline is to decrease injury and discomfort to patients caused by unnecessary spinal immobilization and use of long spine boards.

- Studies show that immobilizing trauma victims may cause more harm than good to the patient.
- Penetrating trauma victims benefit the most from rapid assessment and transport to a trauma center without **spinal motion restriction (SMR).**
- There is evidence that backboards result in harm by causing pain, changing the normal anatomic lordosis of the spine, inducing patient agitation, causing pressure ulcers, and compromising respiratory function.
- **Use of the backboard is recommended in the event of CPR.**

Spinal Injury Assessment
Introduction:

- Perform spinal motion restriction (**SMR)** for a patient who is suspected of having a traumatic unstable spinal column injury. Have a high index of suspicion for pediatrics and patients with degenerative skeletal/connective tissue disorders (i.e., osteoporosis, elderly, previous spinal fractures, etc.).
- Penetrating trauma such as a gunshot wound or stab wound should **NOT** be immobilized on a long board unless there are signs of spinal injury. Emphasis should be on airway and breathing management, treatment of shock, and rapid transport to a Level 1 trauma center.
- Determination that immobilization devices should be used or removed should be made by the highest-level provider on scene.
- If the immobilization process is initiated prior to the arrival and assessment by the highest level of provider, STOP and perform spine injury assessment to determine the best course of action.

Spinal Motion Restriction
The term spinal motion restriction (**SMR**) better describes the procedure used to care for patients with possible unstable spinal injuries. SMR includes:

- Reduction of gross movement by patient
- Prevention of duplicating the damaging mechanism to spine
- Regular reassessment of motor/sensory function

Procedure:

1. Assess the scene to determine the risk of injury. Mechanism alone should not determine if a patient requires cervical spinal immobilization. However, mechanisms that have been associated with higher risk of injury are the following:
    a. Motor vehicle collisions, including automobiles, all-terrain vehicles, and snowmobiles
    b. Axial loading injuries to the spine
    c. Associated, substantial torso injuries
    d. Falls >10 feet
2. Assess the patient in the position he/she was found, initial assessment should focus on determining whether or not a cervical collar needs to be applied.
3. Assess for mental status neurological deficits, spinal pain or tenderness, any evidence of intoxication, or other severe injuries.

Treatment and Interventions

1. Immobilize patient with cervical collar if there is **any** of the following:
    a. Patient complains of midline neck or spine pain
    b. Any midline neck or spinal tenderness with palpation
    c. Any abnormal mental status (including extreme agitation) or neurologic deficit
    d. Any evidence of alcohol or drug intoxication (which limits your ability to accurately assess the patient)
    e. Another severe or painful distracting injury is present

f. Torticollis in children
g. A communication barrier that prevents accurate assessment
h. If none of the above apply, patients should not have a cervical collar placed

2. Patients with penetrating injury to the neck should not receive spinal immobilization, regardless of whether they are exhibiting neurologic symptoms or not. Doing so can lead to delayed identification of injury or airway compromise and has been associated with increased mortality.
3. If extrication may be required:
    a. From a vehicle: After placing a cervical collar, if indicated, children in a booster seat and adults should be allowed to self-extricate, for infants and toddlers already strapped in a car seat with a built-in harness, extricate the child while strapped in his/her car seat
    b. Other situations requiring extrication: A padded long board may be used for extrication, using the lift and slide (rather than a logroll) technique
    c. Patients should not routinely be transported on long boards unless the clinical situation warrants long board use. An example of this may be facilitation of immobilization of multiple extremity injuries or an unstable patient where removal of a board will delay transport and/or other treatment priorities. In these rare situations, long boards should be padded or have a vacuum mattress applied to minimize secondary injury to the patient. Spine injuries in the adult population may be present at more than one level simultaneously
    d. SMR, when indicated, should apply to the entire spine, and an appropriately sized cervical collar is a critical component of SMR and should be used to limit movement of the cervical spine whenever SMR is employed.
    e. The remainder of the spine can be stabilized using an ambulance cot, a vacuum mattress, a long back board or a similar device

If a patient experiences negative effects of SMR methods used, alternative methods should be utilized.

1. If hard backboard is utilized for extrication, the patient should be removed from the backboard when possible and placed on the ambulance stretcher.
2. Patient positions and/or methods/tools to achieve SMR that are allowable (less invasive to more invasive):
    a. Patient position: supine, lateral, semi fowlers, fowlers.
    b. Tools/methods to achieve position of comfort include, but not limited to pillows, children's car seat, scoop, vacuum mattress.
3. Provide manual stabilization restricting gross motion. Alert and cooperative patients may be allowed to self-limit motion if appropriate with or without cervical collar.
4. Apply cervical collar; patients who are unable to tolerate cervical collar may benefit from soft collars, pillows, or other padding.
5. Considerations for patient movement when decision to SMR has been made:
    a. Keeping with the goal of restricting gross movement of spine and preventing increased pain and discomfort, self-extrication of the patient is allowable.
    b. If needed, extricate the patient limiting flexion, extension, rotation and distraction of spine.
    c. Pull sheets, other flexible devices, scoops, and scoop like devices can be employed, if necessary, hard backboards should only have limited utilization.
6. **Standing take downs of ambulatory patients are unnecessary.** Ambulatory patients who meet the above criteria for cervical immobilization should have c-collar applied and are allowed to sit onto the stretcher.
7. Apply adequate padding to prevent tissue ischemia and increase comfort. **Patients should be allowed to be in a position of comfort if they do not meet the requirements for immobilization.**
8. Place patient in position best suited to protect airway.
9. Regularly reassess motor/sensory function (include finger abduction, wrist/finger extension, plantar/dorsal flexion, and sharp/dull exam if possible.
10. Use SpO_2 and $EtCO_2$ to monitor respiratory function.
11. Delivery to hospital: movement of patient to hospital stretchers should be done while limiting motion of the spine.

Special Considerations:

- **Patients with acute or chronic difficulty breathing:** SMR has been found to limit respiratory function an average of 17% with the greatest effect experienced by geriatric and pediatric subjects restricted to a hard backboard. **USE SMR WITH CAUTION with patients presenting with dyspnea and position appropriately.**
- **Pediatric patients, < 9 years of age:**
    - Consider use of padded pediatric motion restricting board
    - Avoid methods that provoke increased spinal movement
    - If choosing to apply SMR to patient in car seat, ensure that proper assessment of patient posterior is performed
- **Combative patients:** Avoid methods that provoke increased spinal movement and/or combativeness.

Pediatric Patients and Car Seats

- **Infants restrained in a rear-facing car seat** and **children restrained in a car seat (with a high back – convertible or booster)** may receive SMR and be extricated in the car seat, the child may remain in the seat if the SMR is secure and his/her condition allows (no signs of respiratory distress or shock).
- **Children restrained in booster seat (without a back)** need to be extricated and receive standard SMR procedures.

Helmet Removal

Safe and proper removal of the helmet should be done following the steps outlined in an approved trauma curriculum.

Indications for football helmet removal:

- When a patient is wearing a helmet and not shoulder pads
- In the presence of head and/or facial trauma, and removal of the face piece is not sufficient
- Patients requiring advanced airway management when removal of the facemask is not sufficient
- When the helmet is loose on the patient's head
- In the presence of cardiopulmonary arrest (the shoulder pads must also be removed)

When helmet and shoulder pads are both on the spine is kept in neutral alignment. If the patient is wearing only a helmet or shoulder pads, neutral alignment must be maintained. Either remove the other piece of equipment or pad under the missing piece. *All other helmets must be removed in order to maintain spinal alignment.*

MISCELLANEOUS
SOP # 609 Stretcher Transport

EMT AEMT PARAMEDIC

Patients with the following conditions should be transported by stretcher or stair chair. Other patients may be transported ambulatory unless their condition warrants stretcher use.

1. Pregnant greater than 20 weeks.
2. Possible cardiac chest pain.
3. Shortness of breath.
4. Asthma.
5. Chronic Obstructive Pulmonary Disease (COPD).
6. Stroke.
7. Patients requiring spinal protection.
8. Penetrating trauma to the torso, neck, or head.
9. Lower extremity, pelvis trauma.
10. Low back trauma.
11. Unconscious, unresponsive patients.
12. Seizures within the past hour or actively seizing.
13. Generalized weakness.
14. Patients unable to ambulate, secondary to pain or weakness.
15. Altered level of consciousness, except psychiatric patients.
16. Behavioral Emergency patients requiring restraint.

MISCELLANEOUS
SOP # 610 Terminally Ill Patients

EMT AEMT PARAMEDIC

Standing Order:

1. Maintain a calm environment and avoid performing measures beyond basic life support.
2. Elicit as much information from persons present who are familiar with the patient's condition as possible.
3. Obtain and document the name and phone number of the patient's physician if possible.
4. Maintain BLS procedures and contact Medical Control as soon as possible. Provide full information on the patient's present condition, history, and the name and telephone number of the patient's physician.
5. Medical Control will direct management of the call.
6. Acceptable DNR/POST forms (**original or copy**):
    a. State approved forms.
    b. Signed order in patient's medical records: nursing home, hospice, or home care.

Note: If DNR/POST form is used to withhold or terminate resuscitation efforts, a copy must be attached to the PCR.

PEDIATRIC CARDIAC EMERGENCY
SOP # 613 Neonatal Resuscitation

Signs and Symptoms	Notes Pulse Oximetry (right hand preferred)
Newborn with respiratory or circulatory distress.	1 min 60-65 2 min 65-70 3 min 70-75 4 min 75-80 5 min 80-85 10 min 85-95

TREATMENT PATHWAY

1. Dry and place in face up head down position.
2. Keep infant level with mother until cord is clamped.
3. Suction mouth, then nose, if obvious obstruction to spontaneous breathing or requiring positive pressure ventilation.
4. Respirations:
    a. If spontaneous:
        i. Wait 1 – 2 minutes, then complete clamping cord and cut between clamps
        ii. Cover infant head
        iii. Wrap and keep infant warm
        iv. Provide oxygen only if HR <100
        v. Transport without delay
    b. If poor respirations, HR < 100:
        Dry, stimulate respirations: rub back, snap bottom of feet gently for 30 seconds
        i. Re-suction mouth, then nose
        ii. Ventilate with BVM at 30 /min, room air oxygen (21%)
        iii. Clamp cord and cut between clamps
        iv. Establish airway SGA or ETT)
        v. Transport immediately
5. Pulse: If pulse rate is less than 60 perform CPR at a rate of 90 compressions/min with 100% oxygen for full minute.
6. Pulse oximetry.
7. EKG monitor..

EMT STOP HERE

8. IV NS, if HR < 60bpm, bolus *10 cc/kg over 5 minutes.*
9. If pulse rate is > 60 keep warm, ventilate with BVM if necessary, transport.

AEMT STOP HERE

10. If HR remains <60 bpm, EPINEPHrine 1:10,000 (now 1 mg/mL) is ***0.02 mg/kg IV*** (0.2 ml/kg) given q 3-5 minutes and repeat until heart rate is above 60/minute. Refer to the length-based tape to confirm dosage.

PARAMEDIC STOP

BEHAVIORAL HEALTH EMERGENCIES
SOP # 701 Agitated or Combative Patient

Signs and Symptoms:	**Notes**:
An individual who displays excessive verbal or motor activity such as physical or verbal abuse, threatening gestures or language, physical destructiveness, and/or excessive verbalizations of distress.	• Providers should always be considerate of their own safety. Never underestimate the potential for violence or turn your back on a potentially violent patient. • Enough providers should be on the scene to adequately handle the situation. Secure the scene and use universal precautions. • Utilize additional personnel and Police. • Use least restrictive method of restraint.

TREATMENT PATHWAY

1. Assess ABCs.
2. Obtain vital signs, pulse oximetry and temperature if possible.
3. Establish IV access with 0.9% NS.
4. Monitor, 12 Lead EKG and transmission if applicable.

EMT AEMT STOP HERE

5. If mental health crisis, consider contacting Mental Health Response.
6. Consider sedation of the patient as necessary by administering Versed via MAD 2-5 mg at a time up to 10 mg total or 0.1 mg/kg to a maximum of 10 mg**.**
7. **Utilize Patient Restraint Medication Flow Chart as needed.**
8. If patient has been TASERED, had extensive muscle activity, or has elevated skin temperature, initiate 500 ml fluid bolus of cool 0.9% normal saline over 20 minutes with 25 mEq Sodium Bicarbonate in IV bag.
9. Use restraints if the patient is perceived to be a threat to themselves or others.

PARAMEDIC STOP

BEHAVIORAL HEALTH EMERGENCIES
SOP # 702 BEHAVIORAL HEALTH EMERGENCIES

Purpose:

To establish criteria for EMS assessment, triage and treatment of patients with potential behavioral/mental health emergencies and direct transport to Behavioral Health.

Definition:

Behavioral health encompasses behavioral factors in chronic illness care, care of physical symptoms associated with stress rather than diseases, and health behaviors, as well as mental health and substance abuse conditions and diagnoses.

Appropriate patients for protocol:

Voluntary patient or patient on police or mental health hold.
Primary 911 call or police request.
Age between 18-70 years.
Mental health complaint (depression, psychosis, suicide or homicidal ideation), substance abuse or behavioral disorder with no acute medical or traumatic condition requiring treatment.
No evidence of trauma other than minor abrasions.
Able to perform activities of daily living (ambulate, bathe, toileting, eat and drink) independently.
If BG is obtained, between 60 and 300 mg/dl.

Exclusion:

Possible drug overdose or acute intoxication significantly impairing ability to ambulate or perform activities of daily living.

Acute medical or traumatic condition including altered level of consciousness, chest or abdominal pain, significant bleeding, respiratory distress, or other acute illness or injury. Patients with abnormal physical findings or vital signs out of range:

HR 60-130.

O_2 sat > 90%.

Systolic BP 90-200 mmHg.

Diastolic BP <110 mmHg.

Temperature between 96.0 F and 100.4 F (38 C) if taken.

Patients who require chemical restraint

Signs/symptoms of acute drug/alcohol withdrawal (tachycardia, hypertension, tremor, visual hallucinations).

Central or peripheral IV lines.
Gastric or nasogastric tube feedings.
Pregnancy greater than 20 weeks.
Patients that require oxygen therapy.
Patients requiring dialysis therapy.

Any patient who demonstrates restlessness, agitation, confusion, or potentially violent behavior regardless of underlying diagnosis. Clinicians will assess the patient and take appropriate measures to sedate and restrain the patient prior to and during transport to ensure a safe and secure environment.

TREATMENT PATHWAY

Assess mental, emotional, and physical status thoroughly. Anticipate changes in attitude and behavior of patient.

If Crisis Intervention Team (CIT) is on scene, EMS assessment and intervention should not be delayed, however, police or the CIT may need to diffuse the situation in order to allow for EMS to safely assess the patient. EMS crews should get an initial report from the officer before approaching the patient. If EMS is first on scene, give an initial report to officer.

Approach the patient in a calm, slow, reassuring and honest manner. Multiple people attempting to intervene may increase the patient's confusion and agitation.

Protect the patient, bystanders and rescuers from injury. Consider restraint and follow ***Patient Restraint*** protocol, if indicated.

Obtain history, physical and mental status examination. Changes in behavior may have a physiologic or pharmacologic explanation. Attend to any medical conditions per EMS protocol and then determine if patient is eligible for transport to Crisis Center.

All patients will be assessed and evaluated by EMS regardless of transport status. Ensure that patient is not carrying weapons or other items which may be used as such (e.g., ballpoint pens).

If potential for agitation, make attempt to secure IV access and EKG Monitor.

AEMT STOP HERE

Medicate confused/combative patients as needed per Agitated Patient Protocol.

Notes:

- Remember that agitation may signal a physiologic deterioration of the patient and accompany hypoxia, hypoglycemia, cerebral edema, etc.
- If behavior compatible with safe transport cannot be achieved or predictably maintained, other transport modes MUST be considered.

PARAMEDIC STOP

BEHAVIORIAL HEALTH EMERGENCIES
SOP # 703 Delirium with HyperAgitation

Signs and Symptoms	Notes
Delirium with HyperAgitation: "a state of extreme mental and physiological excitement," characterized by exceptional agitation and hyperactivity, overheating, excessive tearing of the eyes, hostility, superhuman strength, aggression, acute paranoia, and "endurance without apparent fatigue." Individuals displaying this behavior may have been TASERED or restrained by law enforcement prior to EMS arrival	Using the acronym PRIORITY, EMS should look for the following: P – **P**sychological issues R – **R**ecent drug/alcohol use I – **I**ncoherent thought processes O – **O**ff (clothes) and sweating R – **R**esistant to presence/dialog I – **I**nanimate objects/shiny/glass – violent toward T – **T**ough, unstoppable, superhuman strength Y – **Y**elling

TREATMENT PATHWAY

1. Assess ABCs.
2. Obtain vital signs, EKG, pulse oximetry and temperature if possible.
3. Establish IV access with 0.9% NS.

EMT AEMT STOP HERE

4. Sedate the patient as necessary by administering Versed via MAD 2-5 mg at a time up to 10mg total or 0.1 mg/kg to a maximum of 10 mg**.**

   Utilize Patient Restraint Medication Flow Chart as needed.

5**.** If patient has been TASERED, had extensive muscle activity, or has elevated skin temperature initiate 500 ml fluid bolus of cold 0.9% normal saline over 20 minutes with 25 mEq Sodium Bicarbonate in IV bag.

6. Use restraints if the patient is perceived to be a threat to themselves or others

PARAMEDIC STOP

BEHAVIORAL HEALTH EMERGENCIES
SOP # 704 Physical Restraint

EMT AEMT PARAMEDIC

The following steps should be taken and documented in determining the need for physical restraints:

1. **Assessment of mental status** - Observe for uncontrolled agitation, combativeness, threats of violence to self or others, disorientation, altered mental status impeding medically necessary interventions, or pulling at necessary medical interventions (e.g., oxygen, IV lines, endotracheal tubes).

2. **Alternatives to physical restraint**- Reassurance, support of concerned parties (family, friends, coworkers, etc.), reorientation, diversionary activity, explanation of illness, injury, and medically necessary interventions.

3. **Justification for physical restraint**- Failure of alternatives to physical restraint, reduce likelihood of patient harm to self, reduce likelihood of patient harm to others, enable medically necessary interventions per EMS protocols.

4. **Inform patient and concerned parties of physical restraint use.**

Apply physical restraints

Restraints are to be soft and are not to impede airway patency, respiratory mechanics, or circulation. Patients will not be restrained prone unless an impaled object or airway patency necessitates such positioning. Restraints will be applied in an effective yet compassionate manner. Every effort should be made to avoid injury to the patient during application of physical restraints.

Humane restraints that reduce potential for patient injury from the restraints are those made from roll gauze, soft leather, and those designed as single-patient use, disposable foam with cloth ties. Restraints are to be non-locking unless applied by law enforcement officers in appropriate circumstances and able to be released rapidly if patient condition mandates.

During treatment and transport of a patient in law enforcement-instituted restraints (including handcuffs), EMS professionals should monitor for and advocate for change in restraints that compromise airway patency, respiratory mechanics, or circulation. Patients will not be transported with wrists cuffed to ankles either directly or indirectly (also referred to as “hog-tying”). These positions have been shown to impair respiratory mechanics and pose significant obstacles to definitive airway management if required enroute. During transport of patients in law enforcement-instituted locking restraints, a law enforcement officer should either accompany the patient in the ambulance or provide the treating EMS professionals means to unlock the restraints. This policy allows rapid restraint release should the patient deteriorate to a condition requiring restraint release to properly treat.

Patients restrained using this protocol should generally be restrained to a backboard. This facilitates patient transfer in the emergency department and in the case of airway secretions or vomiting, enables rapid positioning of the patient to reduce aspiration. Patients will not be transported “sandwiched” between two backboards; this positioning impedes patient care and increases risk of aspiration.

Once physical restraints are applied, they will be left in place until the patient is transferred to emergency department personnel. This policy prevents recurrent harm to self, harm to others, and

disruption of intact medical devices and treatment. Despite assurance from the patient that they will comply with treatment, restraints are to be left in place unless a direct order from OLMC is given to release the physical restraints. Such an order must be clearly documented on the patient care form.

Apply a surgical mask, oxygen mask at 10 LPM to patients who are spitting at providers after mechanical or pharmacologic interventions have been performed.

Handcuffs:

Law Enforcement may handcuff the patient at their own discretion. If handcuffs cannot be removed, a law enforcement officer MUST accompany ambulance staff during transport. If a law enforcement officer cannot be present during transport, a handcuff key must be immediately accessible to facilitate removal of handcuffs during an emergency. If a law enforcement officer cannot be present during transport, they must follow the ambulance at a close distance and be available immediately via radio communication.

Patient Restraint Medication Flow Chart

PARAMEDIC ONLY

If Immediate Threat to Safety

Midazolam 2.5-5 mg IV/IO or 5-10 mg IM/IN (Ativan and Diazepam are alternatives)

AND/OR

Haldol 5-10 mg IV/IO (Droperidol and Ketamine alternatives)

NO immediate Threat to Safety

If Alcohol/Drugs:

Midazolam 2.5-5 mg IV/IO or 5-10 mg IM/IN (Ativan and Diazepam are alternatives)

AND/OR

Haldol 5-10 mg IV/IO (Droperidol and Ketamine are alternatives)

Droperidol 5mg IM or slow IV push, For >65 years old, give 2.5mg

Ketamine 0.2-0.75mg/kg IV or 2-4 mg/kg IM

Repeat in 10 minutes if response inadequate.

If Psychiatric:

Haldol 5-10 mg IV/IO (Droperidol and Ketamine are alternatives)

AND/OR

Midazolam 2.5-5 mg IV/IO or 5-10 mg IM/IN (Ativan and Diazepam are alternatives)

Repeat in 10 minutes if response inadequate.

Interfacility Transfers: Request Sedation orders from Transferring Physician if IFT or Medical Control prior to transport. If unavailable, follow protocol above.

Pharmacological Restraint Protocol

Evaluate the personnel needed to safely restrain the patient.

If patient is agitated, attempt to determine cause of agitation (i.e., drug or alcohol intoxication or withdrawal, medical or psychiatric problem) and consider oral benzodiazepine or oral antipsychotic especially if interfacility transfer.
If patient is an immediate threat to responders, bystanders or patient: Administer midazolam (2.5-5 mg IV, IO or 5-10 mg IM/IN) PLUS Droperidol or haloperidol. Titrate midazolam 2.5-5 mg IV, IO or 5 mg IM/IN as needed every 5-10 minutes to control agitation.

Cause unknown or likely psychiatric: Administer Droperidol (5mg IM or Slow IV) or haloperidol (5-10 mg IV, IO, IM to a MAX dose of 10 mg). If initial dose of droperidol/haloperidol has no effect after 10 minutes, repeat droperidol/haloperidol (MAX dose of haloperidol is 10 mg IV, IO, IM). If 10 minutes after administration patient remains agitated, administer midazolam (2.5 mg IV/IO or 5mg IM/IN). May repeat once.

Cause likely drug ingestion (especially stimulants), withdrawal or postictal state: Administer midazolam (2.5-5 mg IV/IO or 5-10 mg IM/IN). If initial 2.5 mg IV or 5 mg IM/IN dose has no effect after 10 minutes, give an additional dose. (MAX dose is 5 mg IV/IO or 10 mg IM/IN). Consider and treat hypoxia, head injury or hypoglycemia.

If 10 minutes after administration of the second dose, the patient remains combative, administer either droperidol or haloperidol as described above. Assess vital signs within the first 5 minutes, if possible and thereafter as appropriate (at least every 10 minutes and before additional medication).

After administration of droperidol/haloperidol, consider diphenhydramine 25 mg IV or IM if the patient shows signs of acute dystonic reaction. May repeat once.

Monitor EKG, obtain 12-lead and start IV, if possible.

PARAMEDIC STOP

HAZARDOUS MATERIALS
SOP # 801 Ammonia

Ammonia is a colorless, water-soluble alkaline gas that is most commonly used a cleaning agent, fertilizer, and industrial refrigerant. The life threat of ammonia exposure is from pulmonary edema and hypotension.

PARAMEDIC ONLY

DECON:
Airway protection via SCBA and chemical protective clothing may be required of the rescuer and should be performed only by properly trained personnel. The patient should be removed from the contaminated area. Remove and bag their clothing and any jewelry. Brush away any dry particles and blot excess liquids. Wash patient with a mild soap and warm water.

Assessment:

The following are not all inclusive and may not be present in all patients but include the most common signs and symptoms.

Cardiovascular:

1. Ventricular Arrhythmias
2. Hypotension

Respiratory:

1. Laryngeal Edema
2. Pulmonary Edema
3. Bronchospasm
4. Stridor
5. Cough
6. Dyspnea

CNS:

1. Lethargy
2. Coma

Gastrointestinal:

1. GI Bleed

Eye:

1. Chemical Conjunctivitis

Skin:

1. Burns
2. Frostbite

Treatment-Standing Order:

1. 100% oxygen and airway maintenance appropriate for pt. condition
2. Pulse Oximetry
3. Cardiac Monitor
4. IV NS
5. Treat underlying signs and symptoms per ALS SOP’s
6. Tetracaine, 2 drops each affected eye, for eye exposure
7. Flush eyes for 15 min with sterile water or saline

HAZARDOUS MATERIALS
SOP # 802 Chlorine

The primary health concern with exposure to Chlorine is irritation of the respiratory system. Although it is unlikely, severe respiratory distress and pulmonary edema may occur with prolonged exposure or exposure to high quantities of chlorine. Also, Chlorine gas is highly corrosive when it contacts moist tissues such as the eyes, nose mouth, and respiratory system.

PARAMEDIC ONLY

DECON:
There is a risk of secondary exposure to EMS personnel from off-gassing of the affected person, especially if their clothing has been soaked with a liquid chlorine product. All persons exposed to Chlorine gas should have their clothing and jewelry removed and bagged. They should then be washed with a mild soap and water. If the exposure has occurred inside of a structure or an area with limited ventilation, the appropriate personnel should remove the victim from the area while wearing full PPE and SCBA.

Assessment: Signs and symptoms will vary according to the amount of Chlorine, route, and length of exposure:

Respiratory:
1. Nasal and throat irritation
2. Respiratory distress
3. Upper airway obstruction notes by cyanosis, wheezing, rales
4. Pulmonary edema

Cardiovascular:
1. Tachycardia
2. Hypertension followed by hypotension

Eyes:
1. Burning pain
2. Ocular spasms
3. Redness and Tearing
4. Corneal burns

Skin:
1. Burning pain
2. Inflammation
3. Blisters
4. Frostbite (if liquefied Chlorine below -30 degrees F)

Treatment – Standing Order
1. 100% oxygen and airway maintenance appropriate for pt. condition.
2. Administer sterile water via nebulizer.
3. Pulse oximetry.
4. Consider the need for BVM, intubation or CPAP.
5. Treat bronchospasms with Albuterol, 2.5mg in 3cc NS.
6. Cardiac Monitor.
7. Large bore IV of NS.
8. Tetracaine ophthalmic solution, 2 drops in each affected eye.
9. Treat respiratory, cardiovascular and other signs and symptoms as appropriate per SOP's.

Treatment-Protocol
1. If burning persists, titrate half strength adult sodium bicarbonate (3.75% or 4.2%) and administer 5 cc via the nebulizer. This is made by diluting 2.5-3 cc of adult strength sodium bicarbonate in 2.5 cc sterile water.
2. This is the only time a chemical will be neutralized in or on the body by field medical personnel.
3. 3ml Sodium Bicarb in 2ml NS nebulized for severe respiratory distress. **DO NOT MIX WITH BRONCHODILATOR.**

HAZARDOUS MATERIALS
SOP # 803 Cyanide Exposure

Signs and Symptoms	Notes
Suspect in smoke exposures (firefighting; byproduct of combustion of synthetic materials), occupational/industrial exposures (paper manufacturing, blueprinting, engraving, metal treatment, fumigants), natural catastrophes, suicide or murder attempts, chemical warfare/terrorism – whenever there are multiple casualties of unclear etiology Early / non-specific: anxiety, vertigo, weakness, headache, tachypnea, nausea, dyspnea, vomiting, tachycardia High concentration: markedly altered level of consciousness/rapid collapse, seizures, respiratory depression or arrest, cardiac dysrhythmias (other than sinus tachycardia) Rapid onset: early hypertension followed by hypotension, sudden cardiovascular collapse, seizure/coma, rapid death Cardiovascular: bradycardia, hypertension followed by hypotension, palpitations, ventricular arrhythmias, cardiac arrest Respiratory: rate and depth increase initially; respirations become slow and labored; pulmonary edema; respiratory arrest CNS: weakness, headache, confusion, lethargy, seizure, coma. GI: nausea and vomiting, excessive salivation Eye: redness, edema, dilated pupils. Skin: inflammation, ulcers; cyanosis may or may not be present Aliases: blood agent, hydrogen cyanide	One of the fastest acting poisons; enters the body by all routes (pale blue liquid, white crystal, or colorless gas). "Bitter almond" odor – only about 40% of people can smell it, and the olfactory response fades quickly. Pulse oximetry does not reflect tissue oxygenation in cyanide/smoke exposure – do not rely on it. After hydroxocobalamin, skin, tears, and urine turn red (not an allergic reaction) and pulse oximetry is no longer accurate.

TREATMENT PATHWAY

1. Remove the patient from the toxic environment. Airway protection via SCBA and chemical protective clothing may be required of the rescuer and should be performed only by properly trained personnel.
2. Decontamination: remove and bag clothing and jewelry; brush away dry particles and blot excess liquids; wash the patient with mild soap and warm water.
3. Assess ABCDs; expose and then re-cover the patient to retain body heat.
4. 100% oxygen via non-rebreather mask, CPAP, or BVM.
5. Vital signs including temperature and pulse oximetry (may not correlate with tissue oxygenation). Monitor for hypoxia (SpO2 less than 94%) and respiratory decompensation regardless of pulse oximetry reading.
6. Cardiac monitor and 12-lead EKG; EtCO2.
7. Blood glucose.
8. Identify the agent, time of ingestion/inhalation, and quantity/timing of exposure. Obtain cardiovascular history and medication history.

EMT STOP HERE

9. IV/IO access (NS).
10. Collect a pre-treatment blood sample for lactate and cyanide levels, if feasible.

AEMT STOP HERE

11. Treatment is based on history and signs/symptoms (there is no rapid confirmatory blood test). Treat according to the type of exposure and severity:
    a. EXPOSURE BY MEANS OTHER THAN SMOKE INHALATION: for a patient with an appropriate history and one or more significant signs or symptoms, administer the antidote (step 12).
    b. EXPOSURE BY SMOKE INHALATION – MILD (CAO, no serious signs or symptoms): 100% oxygen and airway maintenance, IV NS, cardiac monitor. Supportive care; antidote not indicated.

c. EXPOSURE BY SMOKE INHALATION – MODERATE TO SEVERE (altered level of consciousness, severe respiratory or cardiac symptoms, coma): 100% oxygen and airway maintenance, IV NS, cardiac monitor, and administer the antidote (step 12).

12. Antidote – administer ONE of the following:
    a. HYDROXOCOBALAMIN (Cyanokit®, preferred agent): Adult – starting dose 5 g (two 2.5 g vials) slow IV infusion over 15 minutes (reconstitute each 5 g with 200 mL LR, NS, or D5W = 25 mg/mL; infuse at 10–15 mL/min). Depending on the severity of the poisoning and clinical response, a second 5 g dose may be administered by IV infusion for a total of 10 g; the infusion rate of the second dose may range from 15 minutes (patients in extremis) to 2 hours based on patient condition. ***(peds 70 mg/kg IV, same 25 mg/mL concentration; maximum single dose 5 g)***
    b. SODIUM THIOSULFATE: Adult – 12.5 g IV (50 mL of 25% solution). ***(peds 0.5 g/kg IV [2 mL/kg of 25% solution])***
13. Several drugs and blood products are incompatible with Cyanokit; it may require a separate IV line for administration.
14. If seizure, treat per SOP 315.

PARAMEDIC STOP

Warnings and Precautions

1. Use caution in patients with known anaphylactic reactions to hydroxocobalamin or cyanocobalamin. Consideration should be given to use of alternative therapies, if available.
2. Allergic reactions may include anaphylaxis, chest tightness, edema, urticaria, pruritus, dyspnea, and rash.
3. Blood pressure increase: substantial increases in blood pressure may occur following Cyanokit therapy.

Adverse Reactions

1. Most common adverse reactions (greater than 5%) include transient chromaturia, erythema, rash, increased blood pressure, nausea, headache, and injection site reaction.

Patient Safety Considerations

1. Multiple casualties: wear appropriate PPE during rescue/evacuation from the toxic environment.
2. Ingested cyanide reacts with stomach acid and generates hydrogen cyanide gas. Maximize air circulation in closed spaces (ambulance); vomiting or belching may release gas.
3. Do NOT use nitrites with suspected carbon monoxide poisoning – they worsen hemoglobin oxygen-carrying capacity even more than CO.
4. Hydroxocobalamin is the only agent safe for treatment of cyanide poisoning in pregnant patients.

Key Considerations

1. Amyl nitrite and sodium nitrite are no longer used and no longer available in commercial kits.
2. Early and repeated assessment is essential; the patient may deteriorate rapidly.

Documentation

Repeat evaluation and documentation of signs/symptoms; possible etiology; time of symptom onset and time of exposure-specific treatments; therapy and response; accurate exposure history (time, route, quantity, alcohol or other intoxicants).

HAZARDOUS MATERIALS
SOP # 804 Heavy Metals

"Heavy Metals" is a loosely defined term used to include elements that exhibit metallic properties. Although there are many elements that can be defined as "heavy metals", these SOPs are intended to apply specifically to arsenic, mercury, lead and copper. You should provide supportive care and contact medical control if you encounter poisoning from any other metallic compound.

PARAMEDIC ONLY

DECON:
If the exposure has occurred inside of a structure or an area with limited ventilation, the appropriate personnel should remove the victim from the area while wearing full PPE and SCBA. Remove the patients' clothing and jewelry and place them in a bag. The patient should be washed with a mild soap and warm water.

Assessment:

The following are not all inclusive and may not be present in all patients but include the most common signs and symptoms.

Cardiovascular:

1. Tachycardia, Weak pulse
2. Hypotension
3. Ventricular arrhythmias
4. Prolonged QT segment and T wave changes (Arsenic)

Respiratory:

1. Cough
2. Acute bronchitis
3. Tachypnea
4. Dyspnea
5. Apnea
6. Chest Pain
7. Pulmonary edema

CNS:

1. Headache
2. Fatigue
3. Vertigo
4. Syncope
5. Anxiety
6. Seizure
7. Coma

Gastrointestinal:

1. Abdominal pain
2. Nausea, Vomiting
3. Cramps
4. Bloody diarrhea

Eyes:

1. Chemical conjunctivitis
2. Ocular edema

Skin:

1. Irritated, red
2. Pale, cool, clammy (Copper)
3. Cyanotic, cold (Arsenic)

Treatment – Standing Order:

1. 100% oxygen and airway maintenance appropriate for pt. condition.
2. Pulse oximetry.

3. Large bore IV NS.
4. Cardiac Monitor.
5. Treat shock and arrhythmias per SOP's.
6. Continuous flush of affected eyes with NS.
7. Give 4 – 8 oz. of water for ingestion.

Treatment – Protocol:

1. If patient is unstable, administer Dimercaprol (BAL), 3mg/kg deep IM.

HAZARDOUS MATERIALS
SOP # 805 Hydrogen Fluoride

Hydrogen fluoride is a colorless, fuming liquid or gas with a strong, irritating odor. Hydrogen Fluoride is used as a cracking catalyst in oil refineries, and for etching glass and enamel, removing rust, and cleaning brass and crystal. The primary life threat from Hydrogen Fluoride and Hydrofluoric Acid is from severe burns and pulmonary edema.

PARAMEDIC ONLY

DECON:

Airway protection via SCBA and chemical protective clothing may be required of the rescuer and should be performed only by properly trained personnel. The patient should be removed from the contaminated area. Remove and bag their clothing and any jewelry. Brush away any dry particles and blot excess liquids. Wash patient with a mild soap and warm water.

Assessment:

The following are not all inclusive and may not be present in all patients but include the most common signs and symptoms.

Cardiovascular:

1. Tachycardia
2. Weak Pulse, Hypotension
3. Arrhythmias

Respiratory:

1. Acute Bronchitis
2. Dyspnea
3. Pulmonary Edema

CNS:

1. Headache
2. Lethargy
3. Altered LOC

Gastrointestinal:

1. Nausea
2. Vomiting
3. Burns to the mouth and oropharynx

Eye:

1. Intense Pain
2. Chemical Conjunctivitis

Skin:

1. Severe Pain
2. Burns may or may not be visible
3. White areas of discoloration may be present

Treatment-Standing Order

1. 100% Oxygen and airway maintenance appropriate to pt. condition.
2. Pulse Oximetry.
3. I.V. NS in unexposed extremity if possible.
4. Cardiac Monitor – Watch for signs of hypocalcemia (prolonged QT interval).
5. Inhalation: Administer nebulized Calcium Gluconate, 2.5ml in 10cc NS if pt. is displaying signs and symptoms of inhalation (sore throat, coughing, bronchospasm).
6. Skin Exposure: make a mixture of 2.5g Calcium Gluconate and 100ml of water-soluble lubricant (KY Jelly) and massage onto affected area.
7. Ingestion: If pt. is conscious and gag reflex is present, administer 2-4 glasses of water.
8. DO NOT induce emesis.
9. Eye Exposure: Irrigate with 1% aqueous solution of Calcium Gluconate (50ml of 10% Calcium Gluconate in 450 ml of NS).

HAZARDOUS MATERIALS
SOP # 806 Hydrogen Sulfide

Hydrogen Sulfide is a colorless, flammable, highly toxic gas that is used in gas and crude oil operations. It is also a naturally occurring by-product of decaying organic matter (AKA sewer gas) and has the odor of rotten eggs to those who can smell it and be aware that the olfactory nerve may become fatigued and less responsive with exposure! It is heavier than air. This also is a chemical asphyxiant that interferes with cellular respiration. This is taken into the body through all routes.

PARAMEDIC ONLY

DECON:

Airway protection via SCBA and chemical protective clothing may be required of the rescuer and should be performed only by properly trained personnel. The patient should be removed from the contaminated area. Remove and bag their clothing and any jewelry. Brush away any dry particles and blot excess liquids. Wash patient with a mild soap and warm water.

Assessment:

The following are not all inclusive and may not be present in all patients but include the most common signs and symptoms.

Cardiovascular:

1. Tachycardia or Bradycardia
2. Arrhythmias
3. Circulatory Collapse

Respiratory:

1. Cough
2. Dyspnea
3. Tachypnea
4. Acute Bronchitis
5. Pulmonary Edema

CNS:

1. Headache
2. Confusion
3. Dizziness
4. Altered LOC
5. Seizure
6. Coma

Gastrointestinal:

1. Nausea
2. Vomiting
3. Profuse Salivation

Eye:

1. Chemical Conjunctivitis
2. Lacrimation
3. Photophobia

Skin:

1. Irritation
2. Local Pain
3. Excessive Sweating
4. Cyanosis

Treatment-Standing Order:

1. 100% Oxygen and airway maintenance appropriate for pt. condition.
2. Do Not induce vomiting.
3. Pulse Oximetry.

4. Cardiac Monitor.
5. IV NS.
6. Flush eyes with copious amounts of water for eye exposure.
7. Tetracaine, 2 drops each eye after flushing for eye exposure.
8. Follow Seizure protocol if seizures are present.

Treatment-Protocol:

1. Administer Amyl Nitrite, 1 ampule every 5- 10 minutes.
2. Administer Sodium Nitrite, 300mg I.V. over 5 minutes (Flush I.V. line after administration).

HAZARDOUS MATERIALS
SOP # 807 Methyl Bromide

Methyl Bromide is a colorless liquid or gas that is used as an insecticide and as a fumigant for grain elevators and greenhouses. It is also used in refrigerants and solvents. Methyl Bromide is a neurotoxin that can cause severe respiratory irritation, pulmonary edema, and respiratory failure as well as seizures, coma and death.

PARAMEDIC ONLY

DECON:

Airway protection via SCBA and chemical protective clothing may be required of the rescuer and should be performed only by properly trained personnel. The patient should be removed from the contaminated area. Remove and bag their clothing and any jewelry. Brush away any dry particles and blot excess liquids. Wash patient with a mild soap and warm water.

Assessment:

The following are not all inclusive and may not be present in all patients but include the most common signs and symptoms.

Cardiovascular:

1. Arrhythmias
2. Circulatory Collapse

Respiratory:

1. Throat Irritation
2. Tightness of the chest
3. Dyspnea
4. Tachypnea
5. Bronchospasm
6. Pulmonary Edema

CNS: (Symptoms may be delayed)

1. Headache
2. Weakness
3. Confusion
4. Dizziness
5. Slurred Speech
6. Seizures
7. Coma

Gastrointestinal:

1. Nausea
2. Vomiting
3. Abdominal Pain

Eye:

1. Chemical Conjunctivitis
2. Blurred Vision

Skin:

1. Chemical Burns
2. Cyanosis
3. Pain

Treatment- Standing Orders

1. 100% Oxygen and airway maintenance appropriate for pt. condition.
2. Pulse Oximetry.
3. Cardiac Monitor.
4. I.V. NS.
5. Irrigate eyes with sterile water or NS for 5 minutes, remove contact lenses, and apply 2 drops of Tetracaine in each affected eye if exposure to eyes has occurred.

There is no antidote for Methyl Bromide poisoning. EMS personnel should provide supportive measures for underlying signs and symptoms according to ALS SOP's and contact medical control for further guidance.

HAZARDOUS MATERIALS
SOP # 808 Nitrogen Oxides

Nitrogen Oxides are a mixture of gases that are composed of nitrogen and oxygen that are most commonly released into the air by vehicle motor exhaust, burning coal, oil, and natural gas. People are most often exposed to excessive nitrogen oxides levels by close proximity to combustion sources. These chemicals are also commonly found in fertilizers, paints, inks, and dyes and changes the hemoglobin into methemoglobin, which is non-oxygen carrying compound and leads to chemical asphyxiation.

PARAMEDIC ONLY

DECON:

Airway protection via SCBA and chemical protective clothing may be required of the rescuer and should be performed only by properly trained personnel. The patient should be removed from the contaminated area. Remove and bag their clothing and any jewelry. Brush away any dry particles and blot excess liquids. Wash patient with a mild soap and warm water.

Assessment:

The following are not all inclusive and may not be present in all patients but include the most common signs and symptoms.

Cardiovascular:

1. Rapid, Weak Pulse
2. Hypotension

Respiratory:

1. Dyspnea
2. Bronchospasm
3. Pulmonary Edema
4. Glottic Edema

CNS:

1. Fatigue
2. Altered LOC

Gastrointestinal:

1. Nausea
2. Vomiting
3. Abdominal Pain

Eye:

1. Chemical Conjunctivitis

Skin:

1. Irritation
2. Pallor
3. Cyanosis
4. Burns if exposed to liquefied NOx

Presentation:

Cyanosis, unresponsive to oxygenation, headache, nausea, vomiting, tachycardia, arrhythmias, syncope, dyspnea, seizures, coma.

Treatment-Standing Order

1. 100% Oxygen and airway maintenance appropriate for pt. condition.
2. Pulse Oximetry.
3. Cardiac Monitor.
4. I.V. NS.
5. Treat underlying signs and symptoms per ALS SOP's.
6. Administer Methylene Blue, 1-2mg/kg IV over 10 min. if pt. has severe hypoxia and cyanosis that does not respond to other treatments.

SOP # 810 Crush Syndrome

A crush injury results from muscle cell disruption due to compression. Compartment syndrome is crush injury caused by swelling of tissue inside the confining fibrous sheath of muscle compartments. Compartment syndrome symptoms include pain, paresthesia, pallor, poikilothermy, and pulselessness. Crush syndrome is the systemic manifestations of muscle crush injury and cell death. This occurs when the crushed muscle is released from compression. Crush injury syndrome should be suspected in patients with an extensive area of involvement of large muscle groups such as legs, buttocks, entire upper extremity and pectoral areas. The syndrome can begin within an hour if severe compressive forces are involved constricting the venous return. Time of onset is directly related to muscle mass involved versus force applied.

Medical treatment should be on a case-by-case basis looking at the history, muscle groups involved, and the time and pressure involved.

PARAMEDIC ONLY

Procedure:

1. Scene safety.
2. Primary patient assessment. Placement of appropriate hemodynamic monitoring equipment. If oxygen saturations are greater than 93% on room air, the use of high flow oxygen is discouraged due to free oxygen radical exchange. Caution should be used when introducing high flow oxygen into a confined environment. Risk/benefit analysis should be done with the rescue officer, safety officer and the incident commander prior to use.
3. Spinal immobilization as dictated by patient access/confinement.
4. Maintain patient in a dry, normothermic state. Hypothermia may cause a rapid deterioration in physiologic status as well as rapid utilization of glucose stores resulting in hypoglycemia. Hypoglycemia should be treated with administration of dextrose by the most appropriate route (IV, PO, NGT/OGT) as dictated by patient situation.
5. Intravenous access with large bore catheters, minimum of two sites.
6. Administration of normal saline 1000-2000 ml bolus (20 ml/kg) initially and then 1000 ml/hr. The aggressive administration of volume prior to extrication is important to minimize the potential for obstruction of the renal tubules with myoglobin. Lactated Ringer's should not be used due to its potassium content.
7. Administer Sodium bicarbonate 50 mEq IVP ***(pediatric 1meq/kg)***. A Sodium bicarbonate infusion of 150 mEq /1000ml D5W should be initiated. The total IVF rate (NS+D5W) should total 1000mllhr ***(pediatric 5ml/kg/hr.)***. The IV fluid rate should be guided by urine output. Sodium bicarbonate should not be mixed in normal saline due to sodium overload. Alkalization prevents precipitation of myoglobin in the renal tubules which causes acute tubular necrosis and acute renal failure. Myoglobin precipitates in an acidic environment. Myoglobinuria is noted when the urine is tea colored.
8. Analgesia and sedation should be administered per hemodynamic profile. This is also beneficial in facilitating ongoing rescue operations.
9. Prior to extricating the patient with moderate symptoms of crush injury from a confined space, the following medications should be administered:
    a. 50% Dextrose 25 grams IVP ***(pediatric 0.5 grams/kg)***
    b. Regular insulin 10 units IVP ***(pediatric 0.2 units/kg)***
10. Administer Albuterol up to 5 mg via nebulizer. Albuterol lowers serum potassium by driving it back into the cells.
11. Life threatening arrhythmias can occur following release of compressive force.
12. EKG changes due to hyperkalemia are listed below from elevated to high potassium levels:
    a. Tall, peaked T waves
    b. Prolonged PR interval

c. Small P wave, ST depression
d. A V block, Bundle Branch Block
e. Wide QRS with no P wave *
f. Ventricular Fibrillation *

*Life threatening arrhythmias such as wide QRS and ventricular fibrillation require immediate treatment with Calcium Chloride 1 gram IVP ***(pediatric dose 20mglkg)***.

13. Consider the following in situations with prolonged entrapment:
    a. The addition of Mannitol 1 gram/kg to the intravenous bag, Mannitol is thought to be useful in promoting diuresis of the circulating volume to reduce urine acidity.
    b. The use of the ISTAT blood analyzer which can be obtained from the Urban Search and Rescue Team.
    c. Field amputation kit available on site at rescue, this can be obtained with a physician from the local trauma center or the Urban Search and Rescue team.

HAZARDOUS MATERIALS
SOP # 811 Airway / Respiratory Irritants

Signs and Symptoms	Notes
Inhalation of gases, mists, fumes, aerosols, or dusts causing irritation or injury to the airways, pharynx, lung, asphyxiation, or other systemic effects Unusual odor/smell; tearing or itchy eyes; burning sensation and burns to nose, pharynx, respiratory tract; sneezing; general excitation; cough; chest tightness; nausea; dyspnea; wheezing; stridor; dyspnea on exertion; dizziness; change in voice Airway obstruction (laryngospasm, laryngeal edema); non-cardiogenic pulmonary edema; seizures; cardiopulmonary arrest Some agents have no warning properties and delayed onset of symptoms	Aliases: toxic inhalation, chemical respiratory injury. Heavier-than-air gases (especially hydrogen sulfide) accumulate in low areas – "knock-down" risk to unprotected rescuers. Off-gassing in the patient compartment: ventilate the ambulance. SCBA before entering suspected suicide-by-gas scenes.

TREATMENT PATHWAY

1. Don appropriate PPE – respiratory protection is critical. Make sure the scene is safe.
2. Remove the patient from the toxic environment. Remove clothing that may retain gases, or decontaminate if liquid or solid contamination. Flush irrigated/burned areas; early eye irrigation.
3. Rapidly assess respiratory status, mental status, and oxygenation. Assess ABCDs; expose then cover to retain body heat. Vital signs including temperature.
4. Administer humidified oxygen if available (target SpO2 94–98%). Support breathing if hypoventilation, toxic inhalation, or desaturation.
5. Pulse oximetry, EtCO2, cardiac monitor (consider 12-lead), blood glucose. Perform carboxyhemoglobin and cyanide device assessment if available.
6. Identify the suspected agent if possible. Obtain cardiovascular history and medications.
7. Wheezing: albuterol 2.5 mg nebulized for mild wheezing without distress; albuterol 5 mg nebulized (or 6 puffs MDI) for ALL patients in respiratory distress with bronchospasm, repeated at this dose with unlimited frequency for ongoing distress (BLS or ALS).
8. Severe respiratory distress or impending failure: CPAP; BVM ventilation in hypoventilation, respiratory failure, or arrest.
9. Continuous reassessment. Treat topical chemical burns per SOP 214. Cyanide: SOP 803. Carbon monoxide/smoke: SOP 210. Riot control agents: SOP 211. Acetylcholinesterase inhibitors: SOP 213.

EMT STOP HERE

10. IV access for LR or NS; obtain pre-treatment blood samples if feasible (glucose, lactate, cyanide). Fluid bolus 20 mL/kg for evidence of hypoperfusion.

AEMT STOP HERE

11. Ipratropium 0.5 mg nebulized, up to 3 doses, in conjunction with albuterol.
12. Assess for airway burns, stridor, or airway edema; if indicated, intubate EARLY. Avoid supraglottic airways (cricothyrotomy may rarely be required in severe cases).
13. Significant pain: morphine 0.1 mg/kg IV/IO or fentanyl 1 mcg/kg IV/IO.
14. Severe respiratory irritation (especially hydrogen sulfide) with altered mental status and no improvement after removal from the environment: oxygen (humidified if available) to SpO2 94–98%; consider consultation for transfer to a tertiary care hospital. If carbon monoxide is confirmed or suspected, a facility with hyperbaric oxygen capability is preferred.

PARAMEDIC STOP

Agent Groups (by water solubility)

1. High solubility / highly irritating – upper airway (particles greater than 10 micrometers): acrolein, ammonia, chloramine, ethylene oxide, formaldehyde, hydrogen chloride, methyl bromide, sodium azide, sulfur dioxide.
2. Intermediate solubility – bronchus/bronchioles (5–10 micrometers): chlorine.
3. Low solubility / less irritating – alveolar (less than 5 micrometers): cadmium fume, fluorine, hydrogen sulfide (rotten egg odor; olfactory fatigue), mercury fume, mustard gas (also delayed blistering), nickel carbonyl, ozone, phosgene.
4. Simple asphyxiants (oxygen below 19.5%): argon, carbon dioxide, ethane, helium, methane, natural gas, nitrogen, nitrogen dioxide (delayed onset).
5. Chemical asphyxiants: carbon monoxide, cyanide, hydrogen sulfide.
6. Inhalants of abuse (volatile solvents, aerosols/sprays, propellants/nitrous oxide): loss of consciousness, paint/residue around nose and mouth, slurred speech, dysrhythmias. “Sudden sniffing death” can follow a single use (butane, propane, aerosols).
7. Riot control agents – SOP 211. Anticholinesterase inhibitors – SOP 213.

Agent-Specific Presentations

1. **Ammonia**: sharp odor, nasal/pharyngeal burning, tearing, sneezing, altered mental status, cough, dyspnea, chest tightness, wheezing, voice change, upper airway obstruction (laryngospasm, laryngeal edema), corneal burns, skin burns, pharyngeal/tracheal/bronchial/esophageal burns; non-cardiac pulmonary edema with high concentration or prolonged exposure.
2. **Chlorine**: all of the above plus greater likelihood of bronchiole burns and bronchospasm; non-cardiac pulmonary edema within 6–24 hours of higher exposures.
3. **Phosgene**: often NO early symptoms for a half hour to several hours (only warning may be a “fresh mowed hay” odor), then up to 24-hour delay to exertional dyspnea, bronchospasm, hypoxia, severe non-cardiac pulmonary edema, cardiopulmonary arrest.
4. **Hydrogen sulfide**: direct neurotoxin, rapidly absorbed through the lung; rotten egg smell causes rapid olfactory fatigue; cough, dyspnea, rapid alterations in cognition or consciousness, pulmonary hemorrhage/hemoptysis, non-cardiac pulmonary edema; “knock-down” gas – sudden loss of consciousness at high concentrations; asphyxia and death.
5. **Nitrogen dioxide** (Silo Filler’s disease): heavier than air, displaces oxygen in low-lying areas and closed spaces; low concentration – ocular irritation, cough, dyspnea/tachypnea, fatigue; high concentration – altered mental status, cyanosis, vomiting, dizziness, loss of consciousness, cardiopulmonary arrest.

Patient Safety Considerations

1. Patients exposed to highly soluble irritants have usually self-extricated because of warning properties (smell, rapid onset of irritation).
2. Less soluble agents may generate only an odor and cause delayed serious symptoms (acute pulmonary edema, hypoxia, dyspnea with minimal exertion).

Key Considerations

1. Irritants can exacerbate reactive airway disease (asthma, COPD).
2. Removal from the toxic environment, oxygen, supportive care, bronchodilators, respiratory support, and rapid transport are the core of care – there are no specific antidotes for these inhaled agents.
3. Hydrogen sulfide stuns the cells responsible for smell – the victim stops perceiving the odor while continuing to absorb the gas.
4. Household bathroom, kitchen, and oven cleaners, when mixed, can release ammonia, chloramine, and chlorine. Chloramine (bleach plus ammonia) hydrolyzes in the distal airways and alveoli to ammonia and hypochlorous acid.

5. Inhaled agents have become a means of suicide. If suicide signage, hoses, or buckets of substances are visible, retreat to a well-ventilated area and don SCBA before entry.

Documentation

Respiratory rate, oxygen saturation, accessory muscle use or tracheal tugging, breath sounds, air entry/stridor, mental status, color, reduction of burning in airway/pharynx; decontamination provided; PPE used. Patients may describe a specific odor, but the human nose is a poor discriminator of scent.

Additional Optional Procedures and Medications

Protocols in this section may be implemented on an individual basis after approval by EMS Medical Director and EMS leadership.

PARAMEDIC

Lidocaine substitution for Amiodarone

Standing order for the use of Lidocaine when Amiodarone is not available.

Lidocaine 2% will be used for treating ventricular arrhythmias.

Dosing:

Adult

Cardiac Arrest from VT/VF:

Initial dose 1 to 1.5 mg/kg IV/IO

Repeat 0.5 to 0.75 mg/kg every 15 minutes with maximum total dose of 3mg/kg.

Perfusing Arrhythmia:

For stable VT, wide-complex tachycardia of uncertain type and significantly symptomatic ectopy:

Initial Dose 1 to 1.5 mg/kg IV/IO

May give additional 0.5 to 0.75 mg/kg every 15 minutes with maximum total dose of 3mg/kg for persistently symptomatic arrhythmia.

Pediatric Dosing

Initial *peds dose is 0.5mg/kg-1mg/kg*, followed by a repeat dose in 15 minutes. Repeat dose is appropriate in perfusing arrythmias only if the arrythmia persists and is resulting in symptoms.

PARAMEDIC

Standing order for the use of Valium or Ativan when versed is not available.

Dosing:

Adult

- Diazepam (Valium) SLOW IVP/IO 2-5 mg or Midazolam (Versed) 2-5 mg IV/IO/IM may repeat if seizure continues.
- LORazepam 1-2 mg IV, every 5 minutes, or 2-4 mg IM, every 10 minutes (maximum dose 8 mg).

Pediatric

- Diazepam (Valium) 0.1 mg/kg or Midazolam (Versed) 0.1 mg/kg IV/IO
- Midazolam (Versed) IM 0.2 mg/kg IM (max single dose 6 mg) Repeat once if seizure activity persists after 10 minutes. Contact MEDICAL CONTROL if seizure activity persists after repeat dose.
- Midazolam (Versed) IN 0.3 mg/kg IN (max single dose 10 mg) with maximum total dose of 0.4 mg/kg.
- LORazepam (peds 0.1 mg/kg IV/IO, max single dose 4 mg, may repeat in 5 minutes if seizure activity continues; not to exceed 0.2 mg/kg total (maximum of 8 mg)
- If seizures persist for 4 minutes repeat medication once.

Additional Medications

Zofran ODT (Ondansetron):

Indication: Nausea and vomiting.
Adult Dose: 4-8 mg orally disintegrating tablet (ODT) as needed, not to exceed 24 mg in 24 hours.
Administration: Administer orally disintegrating tablet (ODT) by placing it on the patient's tongue, where it will dissolve without water. Ensure the patient does not swallow for 30 seconds to allow absorption.

Ibuprofen:

Indication: Pain and fever.
Adult Dose: 400-800 mg orally every 6-8 hours, not to exceed 3200 mg in 24 hours.
Pediatric Dose 100 mg/5 ml (≥ 6 months old): 5-10 mg/kg orally every 6-8 hours, not to exceed 40 mg/kg in 24 hours.
Administration: Administer orally with water. Use weight-based dosing for pediatric patients.

Weight (lb)	Age (yr)	Dose (mL)
Under 24 lbs	Under 2 years	2.5
24-35	2-3 years	5 ml
36-47	4-5 years	7.5 ml
48-59	6-8 years	10 ml
60-71	9-10 years	12.5 ml
72-95	11 years	15 ml

Acetaminophen:

Indication: Pain and fever.
Adult Dose: 325-1000 mg orally every 4-6 hours, not to exceed 4000 mg in 24 hours.
Pediatric Dose (≥ 3 months old): 10-15 mg/kg orally every 4-6 hours, not to exceed 75 mg/kg in 24 hours.
Administration: Administer orally with water. Use weight-based dosing for pediatric patients.

Weight (lb)	Age (yr)	Dose (mL)
Under 24 lbs	Under 2 years	2.5
24-35	2-3 years	5 ml
36-47	4-5 years	7.5 ml
48-59	6-8 years	10 ml
60-71	9-10 years	12.5 ml
72-95	11 years	15 ml

Toradol:

Indications: Moderate to severe pain.

Adult Dose: 10-30 mg intramuscularly (IM) or intravenously (IV) every 6 hours, not to exceed 120 mg in 24 hours.

Pediatric Dose: **Not recommended** for use in pediatric patients.

Administration: Administer intramuscularly (IM) or intravenously (IV) for rapid pain relief. Use caution in patients with renal impairment or gastrointestinal bleeding.

Notes:

Paramedics are authorized to administer the above medications only to patients meeting the specified indications and within the established dosage guidelines. Prior to medication administration, paramedics must assess the patient's medical history, allergies, current medications, and vital signs to ensure safety and appropriateness of treatment.

Documentation of medication administration, including indication, dosage, route, patient response, and any adverse reactions, must be accurately recorded in the patient care report.

TXA

Intravenous TXA for hemorrhagic shock, including postpartum hemorrhage and trauma patients within 3 hours of bleeding onset.

Adult dose:

Two gram bolus in 100 mL of normal saline over 10 minutes (slow intravenous push). To avoid hypotension, administer at a rate not to exceed 100 mg per minute.

Pediatric Dose:

Weight-based, an initial dose of 20 mg/kg intravenous bolus over 10 minutes.

Contraindications

- Hypersensitivity to TXA.
- Consider if appropriate if
    - Coronary or vascular stent placed within the past 6 months.
    - DVT, PE, MI or ischemic stroke within the past 6 months
    - Subarachnoid hemorrhage.
    - Retinal vein or artery occlusion.

Cyanide Exposure Supplement Protocol for use of Cyanokit® (Hydroxocobalamin)

NOTE:

This protocol is a supplement to CBRNE Protocol for Cyanide Exposure and is intended exclusively for use by Paramedics. This protocol provides direction for use of Cyanokit® (hydroxocobalamin), when available, as an alternative antidote to sodium nitrite and sodium thiosulfate.

Indications:

The Cyanokit® is indicated for the treatment of known or suspected cyanide poisoning. If clinical suspicion of cyanide poisoning is high, Cyanokit® should be administered without delay. Patients experiencing serious symptoms from smoke inhalation, particularly when in a confined space exposure (inside a house fire,) frequently have cyanide exposure with or without carbon monoxide exposure and should be considered for the Cyanokit®.

Continue all non-pharmacologic treatment called for under the Cyanide Exposure Protocol.

The Cyanokit® is packaged in two ways:

a. A two vial kit with 2.5g of hydroxocobalamin each in powder form which must be reconstituted with 100mL of normal saline each, rotated or tipped for 30 seconds each (not shaken) and then administered through its own IV line (not used with any other medications) over 7.5 minutes each.
b. A one vial kit with 5g of hydroxocobalamin powder which must be reconstituted with 200mL of normal saline, be rotated or tipped for 60 seconds (not shaken) and administered through its own IV line (not used with any other medication) over 15 minutes.

The starting dose of hydroxocobalamin for adults is 5g (i.e., two 2.5g vials OR one 5gvial) administered as an intravenous (IV) infusion over 15 minutes. See charts below for pediatric dosing.

Two Vial Kit (2.5g/100mL):

AGE GROUP	AMOUNT	DOSAGE
Infant/Toddler (0-2 years)	¼ bottle	0.625g
Preschool (3-5 years)	½ bottle	1.25g
Grade School (6-13 years)	1 bottle	2.5g
Adult ≥ 14 years	2 bottles (entire kit)	5g

One Vial Kit (5g/200mL):

AGE GROUP	AMOUNT	DOSAGE
Infant/Toddler (0-2 years)	⅛ bottle	0.625g
Preschool (3-5 years)	¼ bottle	1.25g
Grade School (6-13 years)	½ bottle	2.5g
Adult ≥ 14 years	1 bottle (entire kit)	5g

Each vial of hydroxocobalamin for injection is to be reconstituted with diluent (not provided with Cyanokit®) using the supplied sterile transfer spike.

The recommended diluent is 0.9% Sodium Chloride injection (0.9% NaCl). Alternate solutions for dilution if NaCl not available: Lactated Ringers or 5% Dextrose injection (D5W). The line on each vial label represents the volume of diluent. Following the addition of diluent to the lyophilized powder, each vial should be repeatedly inverted or rocked, not shaken, for at least 30 seconds for the 2.5g bottles prior to infusion, 60 seconds for the 5g bottles.

Hydroxocobalamin solutions should be visually inspected for particulate matter and color prior to administration.

If the reconstituted solution is not dark red or if particulate matter is seen after the solution has been appropriately mixed, the solution should not be administered to the patient and should be discarded.

There are several drugs and blood products that are incompatible with Cyanokit®, thus Cyanokit® requires a separate intravenous line for administration.
Depending upon the severity of the poisoning and the clinical response, a second dose of 5g may be administered by IV infusion for a total dose of 10g in adults. The rate of infusion for the second dose may range from 15 minutes (for patients in extremis) to two hours, as clinically indicated. Contact medical control for second dose instructions for pediatric patients.
Contraindications: None

SPECIAL CONSIDERATION FOR SMOKE INHALATION:
Many, but not all, smoke inhalation victims will have cyanide poisoning and may present with burns, trauma, and exposure to other toxic substances making a diagnosis of cyanide poisoning particularly difficult. Prior to administration of Cyanokit®, smoke inhalation victims should be assessed for the following:

- Exposure to fire or smoke in an enclosed area
- Presence of soot around the mouth, nose or oropharynx
- Altered mental status

The Cyanokit® should be considered for all serious smoke inhalation victims (including cardiac arrest).
*NOTE: A single medical control order in a mass casualty incident may be applied to all symptomatic patients. This medication is not required to be carried on EMS vehicles and may be available through special response units.

NOREPINEPHRINE (LEVOPHED®)

Common Indications
Dyspnea – Congestive Heart Failure (Cardiogenic Shock)
Post Cardiac Arrest Treatment (Cardiogenic Shock)
Acute Coronary Syndrome (Cardiogenic Shock)
Sepsis (Septic Shock
Dialysis-Related Issues
Neurogenic Shock

For all listed situations, indication is hypotension (adult = systolic < 100 mmHg) due to cardiogenic, septic, or neurogenic shock either refractory to intravascular fluid boluses or in which intravascular fluid bolusing is contraindicated (eg. pulmonary edema)

Common Contraindications and Precautions

Hypovolemic Shock: Norepinephrine is generally contraindicated in hypotension secondary to uncorrected hypovolemia or acute hemorrhage, as profound vasoconstriction in the setting of an empty vascular bed can exacerbate tissue ischemia

Hypertension: It should not be administered to patients who are already hypertensive

Extravasation Risk: A universal precaution across all protocols is the risk of severe tissue necrosis if the medication extravasates (leaks into the surrounding tissue). Protocols mandate administration through a large, free-flowing, patent intravenous (IV) line or an intraosseous (IO) line, with frequent monitoring of the site.

Pharmacokinetics: Onset of action within 5 minutes after IV/IO infusion initiated. Rapid metabolism, requiring ongoing IV/IO infusion to maintain clinical effects. Side Effects: Few, though at higher doses, symptoms may include headache, palpitations, tachycardia, chest pain, and eventual hypertension. Bradycardia can result reflexively from an increase in blood pressure.

Dosage:
For hypotension (shock) refractory to fluids or fluids contraindicated

Start at 2-4 mcg/minute - see dosage chart - titrated to a systolic B/P ≥ 100 mmHg. Maximum infusion rate is 12 mcg/minute.

Norepinephrine Infusion Adult Dosage Chart rates reflect using a microdrip (60 drops/mL) set:

Mcg/min	2	3	4	5	6	7	8	9	10	11	12
Drops/min	15	22	30	37	45	52	60	67	75	82	90

How Supplied:

4 mg/4 mL ampule or vial.

Use only 2 mL in a 250 mL bag of D5W. (8 mcg/mL concentration) (Always check concentration and dose per container at time of patient medication administration)

Special Comments

Ensure aggressive fluid resuscitation is accomplished (unless contraindicated) prior to norepinephrine use.

Norepinephrine should be given into a large, patent vein. The vein of choice for EMS use is the antecubital vein, as this will decrease the risk of overlying skin necrosis.

Do not administer norepinephrine through an IV in the hand or leg. These veins are more likely to be affected by vasoocclusive diseases and more prone to ischemic complications.

Administration through IO in the proximal tibia or humeral head is permitted.

If local extravasation occurs, notify the receiving physician of the following FDA advisement of antidote to extravasation ischemia: "To prevent sloughing/necrosis in peripheral ischemic areas promptly use syringe w/ fine hypodermic needle to liberally infiltrate area w/ 10-15 mL saline solution containing 5-10 mg phentolamine; sympathetic blockade causes immediate conspicuous local hyperemic changes if area infiltrated within 12 hours."

Safety in pregnancy not firmly established, though when clinically indicated the benefits outweigh risks.

Safety in pediatrics not firmly established and Medical Control is to be consulted prior to pediatric usage. When possible, avoid mixing in normal saline, as NS promotes loss of potency through oxidation of norepinephrine.

Optional Medications for AGITATED / VIOLENT PATIENT

DROPERIDOL (Inapsine)

ACTION: Hypnotic, sedative, anti-emetic

- Droperidol is an antidopaminergic medication used as an anti-psychotic and anti-emetic. It also has mild alpha receptor blockade.

INDICATIONS:

- Agitated patient who may be a danger to self or others
- Anti-emetic in a patient with active or intractable nausea/vomiting.

CONTRAINDICIATIONS:

- Hypersensitivity
- Patients with known long QT syndrome (>440 males or >450 females)
- Pregnancy (category C drug)

POTENTIAL SIDE EFFECTS:

- Drowsiness and sedation
- Hypotension and tachycardia. Treat with a fluid bolus.
- Prolongation of QT interval
- Extrapyramidal side effects (e.g. dystonic reactions, akathisia or restlessness, neuroleptic malignant syndrome). Treat with Benadryl.

ADULT DOSE/ROUTE:

- Agitation
    - 5mg IM or slow IV push
    - For >65 years old, give 2.5mg
- Nausea/vomiting
    - 0.625-2.5mg IM or slow IV push

PEDIATRIC DOUSE/ROUTE:

- Not approved for less than 2 years of age.
- Nausea/vomiting
    - Age 2-12 years. 0.015-0.1mg/kg (max 5mg)

NOTES:

- Use a reduced dose in patients >65 years old
- In 2001 the FDA placed a black box warning on Droperidol for prolonged QT. However, multiple studies have not shown a true increase in adverse effects or cardiac arrythmias.

Haldol (Haloperidol)

Action: Haloperidol is a butyrophenone antipsychotic medication. Haloperidol produces a dopaminergic blockade, a mild alpha-adrenergic blockade and causes peripheral vasodilation. Its major actions are sedation and tranquilization.

Onset Onset: Within 10 minutes after IM administration. Peak effect within 30 minutes **Duration**: 2-4 hours (may be longer in some individuals)

Indications: Sedation of a severely agitated combative patient

Contraindications: Suspected myocardial infarction, Hypotension, Respiratory or CNS depression, Pregnancy, Children < 8 years old

Precautions: Butyrophenones may cause hypotension, tachycardia, and prolongation of the QT interval. Use with caution in severe cardiovascular disease. Cardiac monitor and establish an IV as soon as possible with all administrations. Some patients may experience unpleasant sensations manifested as restlessness, hyperactivity, or anxiety following Butyrophenone administration. Rare instances of neuroleptic malignant syndrome (very high fever, muscular rigidity) have been known to occur after the use of haloperidol.

Dosage and Administration Severe Agitation

Adults and Pediatrics > 13 years old

Haloperidol – 5mg IM
Adults > 65 Haloperidol – 2.5mg IM

Consider Medical Control for additional doses (consider if no effects within 10 minutes)

Special Considerations

Extra-pyramidal reactions have been noted hours to days after treatment, usually presenting as spasm of the muscles of the tongue, face, neck, and back. This may be treated with diphenhydramine. Hypotension and tachycardia secondary to Butyrophenone are usually self-limiting and should be treated with IV fluid bolus. Use reduced dose in patients age ≥ 65.

Haloperidol should be utilized if droperidol is unavailable. Droperidol is the preferred first line agent for patients with severe agitation.

Additional Procedures for Refractory VFib/Vtach

Defined as persistent V-Fib/V-Tach with no transient interruption of V-Fib/V-Tach 3 defibrillations

If ALL 3 of the below treatments have failed to convert the refractory V-Fib/V-Tach:

- 3 or more standard defibrillations have been delivered
- Correctable causes (e.g., H's & T's) have been addressed
- 450mg of AMIODARONE has been administered
- Defibrillation Pads have been reoriented anterior/posterior (may use second set of pads)

Consider **DOUBLE SEQUENTIAL DEFIBRILLATION:**

Apply an additional set of external defibrillations pads anterior/lateral OR anterior/posterior depending on where the initial pads were placed

Verify both monitors/defibrillators are attached and confirm V-Fib/V-Tach rhythm on both monitors

Charge both monitors to the maximum energy setting and ensure all team members are clear of the patient

Defibrillate by pressing both shock buttons sequentially.

Repeat every 2 minutes until termination of Refractory V-Fib/V-Tach

Atrial Fibrillation with Rapid Ventricular Response

Irregular Narrow Complex Tachycardia (Stable)

Adult patients with symptomatic atrial fibrillation or flutter with a HR ≥ 100 beats per minute and a SBP ≥ 100 mmHg

1. Initial IV diltiazem bolus over 2 minutes of 0.25 mg/kg (maximum 20 mg)
2. Second dose of 0.35 mg/kg (maximum 25 mg) if the initial response is inadequate after 15 minutes

Note: For patients older than 65 years old, recommend maximum initial dose of diltiazem 10 mg IV and a maximum second dose of 20 mg

Unstable patients: (SBP < 100 mmHg, altered mental status, ischemic chest pain, signs of shock, or acute pulmonary edema due to the rhythm) do NOT give diltiazem. Perform synchronized cardioversion, with sedation as necessary, per SOP # 107 and contact Medical Control.

Patient Refusal/Declination of Care/Transport

Standing Orders:

1. Utilize the mini-mental status exam on any patient where you have concerns regarding the decision-making capacity of the patient.
2. Confirm and document the impact of intoxicating substance or injury.
3. Confirm patient is of legal age of majority, or emancipated minor.
4. Document mechanism of injury or circumstances of illness.
5. Document pertinent past history.
6. Perform at least two sets of vital signs (one on arrival and one before leaving) and problem directed exam, document if unable to obtain vital signs.

The following are considered to lack the capacity for decision-making regarding transport:

- Patients with impaired judgment and decreased mental status (Utilize the mini-mental status exam to determine, document).
- Minors (less than 18 years of age unless they are emancipated by the courts).
- All minors must have refusal from parent or guardian, not older sibling or other relative.
- Do not release minors on the scene without parent/guardian consent.

Reasons for Non-Transport:

- Minor illness or injury and acceptable alternative transportation available.

No patient found on scene:

- Patient Care Report is to be completed in detail as to why no patient was found (i.e., no person found on scene, person located with no complaint of injury/illness and denies needing medical assistance).

The above refusal protocol was written with the intent that a paramedic on a unit cannot take a refusal from a patient that does not have decision making capacity. Always involve law enforcement/EMS supervisor. Paramedics do not have the authority to force a patient to go to the hospital. Law enforcement only has the authority to take someone against their will. The police must place the person in custody for this to happen.

Refusal Procedures: (All patients can refuse)

1. Explain all risks of refusal to patient.
2. With patient's permission, explain to family/friends on scene.
3. Contact on duty Supervisor.
4. Contact Law Enforcement and have them make the scene. Explain your concerns to the police officer of the patient's refusal. The police have the final say on the transport decision.
5. If the patient still refuses and the police do not place the patient in custody document all the following.
    a. All the findings for why the patient needs to be transported.
    b. The Supervisor was contacted about this call.
    c. The name and badge number of the police officer on the scene.
6. Have the patient sign the refusal.
7. Thoroughly document all events that occurred on the scene.

PROCEDURE
Blood Collection in Patients with Time Critical Illness

PARAMEDIC ONLY

Purpose:
To expedite the care of patients, especially those with time critical illnesses, EMS will attempt to obtain blood samples on these patients prior to arrival at the hospital. Each kit should contain tubes, labeling information, and other necessary equipment.

Indications:
Patients exhibiting signs and symptoms of time critical illnesses including Sepsis, ACS/STEMI, and stroke. Consider obtaining blood samples on any patient in which an IV is started in the field.

Procedure:

1. Obtain blood via straight stick or through IV catheter. Note that the straight stick method is preferable due to hemolysis concerns.
2. Tubes should be drawn in this order:
    a. 1 blue top tube
    b. 1 red top tube
    c. 1 light green top tube
    d. 1 dark green top tube
    e. 1 purple top tube
3. Label all tubes with the following information:
    a. Patient's last name, first name, middle initial
    b. Patient's date of birth (DOB)
    c. Date and time of stick
    d. Initials of paramedic performing blood draw
    e. EMS unit identification (i.e., U-21, etc.)
    f. ePCR #
4. Place all tubes in zip lock lab bag.
5. Ensure bag always stays with patient until delivered directly to nurse in ER when giving report.
6. Document that blood was delivered and obtain replacement blood draw kit from vending machine prior to return to service.

Order of Draw for Multiple Tube Collections

CLSI recommended Order of Draw

Blood Collection Tubes (plastic)

Tube Color	Collection Tube	Mix by Inverting
	TO FILL LINE **Must Be collected before other tubes** **PT, INR, PTT** **D-Dimer** **Fibrinogen**	**3 to 4 times**
	CMP **Enzymes** **SPT**	**8 to 10 times**
	Cardiac Enzymes **Troponin**	**8 to 10 times**
	CBC **HgB** **HcT** **Hgb A1C** **ESR**	**8 to 10 times**

Tips for Successful Venipuncture:

- **Keep angle of insertion 30 degrees or less, or as low as possible**
- **Do not leave the tourniquet on for longer than one minute prior to venipuncture to avoid altering results**
- **Instruct patient to clench and hold their fist instead of pumping it, which falsely elevates some blood levels**
- **Avoid side-to-side needle manipulation, especially in the area of the basilica vein where nerves and the brachial artery can be injured**
- **Hold pressure and observe for bleeding and hematoma formation prior to bandaging**
- **Invert each tube as indicated by above color chart**

Processing of Tubes

Why:

- Most tubes contain an additive or Clot activator that needs to be mixed with the blood sample.
- Tubes with anticoagulants such as EDTA need to be mixed to ensure the specimen does not clot.

How:

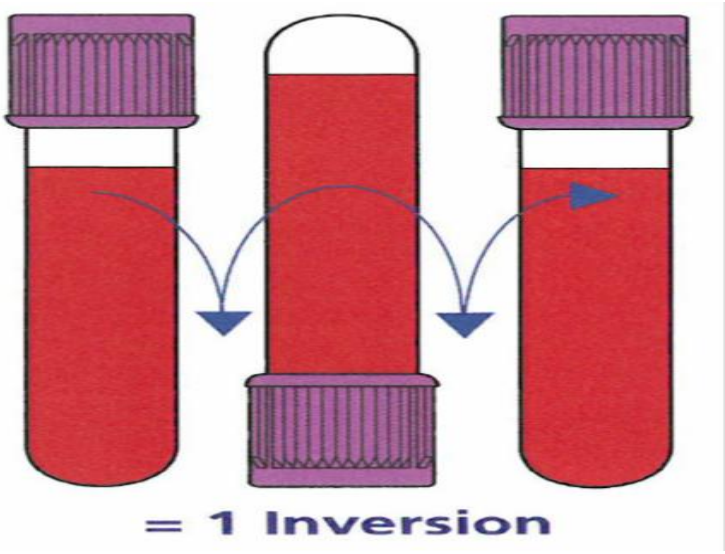
- Holding tube upright, gently invert 180° and back.
- Repeat movement as prescribed for each tube.

When:

- Immediately after drawing.

Consequences if not mixed:

- **Tubes with anticoagulants will clot.**
- **Specimen will often need to be redrawn.**



PROCEDURE
Capnography

EMT. AEMT. PARAMEDIC

Indications:

- Capnography shall be used as soon as possible in conjunction with any airway management adjunct, including endotracheal, Blind Insertion Airway Device (BIAD), or BVM.
- Capnography is recommended to be used on all patients treated with CPAP, Magnesium, and/or EPINEPHrine for respiratory distress.

Procedure:

1. Attach Capnography sensor to the BIAD, endotracheal tube, or oxygen delivery device.
2. Note CO_2 level and wave form changes. These will be documented on each respiratory failure, cardiac arrest, or respiratory distress patient.
3. Capnography shall remain in place with the airway and be monitored throughout the prehospital care and transport.
4. **Any loss of CO_2 detection or waveform indicates an airway problem or dislodgement and should be documented once assessed and corrected.**
5. Capnography should be monitored as procedures are performed to verify or correct the airway problem.
6. Document the procedure and results on/with the Patient Care Report.
7. In all patients with a pulse, an $ETCO_2$ > 20 mmHg is anticipated. In the post resuscitation patient, no effort should be made to lower $ETCO_2$ by modification of the ventilator rate. Further, in post-resuscitation patients without evidence of ongoing severe bronchospasm, ventilator rate should never be < 6 breaths per minute.
8. In the pulseless patient, an $ETCO_2$ waveform with an $ETCO_2$ value > 10 mmHg may be utilized to confirm the adequacy of an airway to include BVM and advanced devices when SpO_2 will not register.

PROCEDURE
Chest Decompression

PARAMEDIC ONLY

1. Cleanse skin on affected side using aseptic technique.
2. Using a 14 or 16 gauge 3.5” angiocath, insert between the 2nd/3rd mid-clavicular or 4th/5th mid-axillary spaces.
3. Advance needle until “pop” is felt while the needle is entering the pleural space.
4. Advance catheter until hub contacts skin.
5. Cover catheter hub with Chest Seal (ensure one-way valve effect).
6. Reassess patient for breath-sound changes.
7. If signs of tension reoccur check chest seal, consider repeating chest decompression per above steps.
8. Contact Medical Control.
9. Transport.

Use the same procedure for pediatric patients: Use 18- or 20- gauge angiocath.

PROCEDURE
Continuous Positive Airway Pressure (CPAP)

EMT. AEMT. PARAMEDIC

Continuous Positive Airway Pressure has been shown to rapidly improve vital signs, gas exchange, reduce the work of breathing, decrease the sense of dyspnea, and decrease the need for endotracheal intubation in patients who suffer from shortness of breath from asthma, COPD, pulmonary edema, CO poisoning, Near Drowning, CHF, and pneumonia. In patients with CHF, CPAP improves hemodynamics by reducing left ventricular preload and afterload.

Indications:

Any patient who is respiratory distress for reasons other than trauma or pneumothorax, and;

- Is awake and able to follow commands.
- Is over 12 years old and the CPAP mask fits appropriately.
- Has the ability to maintain an open airway.
- Has a systolic blood pressure above 90 mmHg.
- Uses accessory muscles during respirations.
- Shows signs and symptoms consistent with asthma, COPD, pulmonary edema, CHF or pneumonia.

AND who exhibit **two or more** of the following:

- A respiratory rate greater than 25 breaths per minute
- Pulse Oximetry of less than 94% at any time
- Use of accessory muscles during respirations

Contraindications:

- Patient is in respiratory arrest/apneic.
- Patient is suspected of having a pneumothorax or has suffered trauma to the chest.
- Patient has a tracheostomy.
- Patient is actively vomiting or has upper GI bleeding.
- Patient has decreased cardiac output, obtundation and questionable ability to protect airway (e.g., Stroke, etc.), penetrating chest trauma, gastric distention, severe facial injury, uncontrolled vomiting, and hypotension secondary to hypovolemia.

Precautions:

Use care if patient:

- Has impaired mental status and is not able to cooperate with the procedure.
- Has failed at non-invasive ventilation.
- Has active upper GI bleeding or history.
- Complains of nausea or vomiting.
- Has inadequate respiratory effort.
- Has excessive secretions.
- Has a facial deformity that prevents the use of CPAP.

Procedure:

Explain the procedure to the patient:

1. Connect Oxygen tubing nipple to gas source.
2. Place the face mask securely to the patient's face using head harness.
3. With nebulizer in the OFF position slowly increase gas flow to 6 or 8 LPM. Check face mask fit to patient and device connections for leaks.

4. Adjust the flow meter until desired pressure is obtained. **Maximum benefit is usually achieved at about 5.0 mm H_2O. Higher pressures result in more side effects with minimal improvements in benefits.**
5. Do not exceed 33 LPM.
6. Patient SaO_2 should be monitored using a pulse oximeter.
7. To activate nebulizer, rotate knob to the ON position.
8. If necessary, readjust flow meter to obtain desired CPAP pressure. Up to 25 LPM may be required.
9. Consider Ondansetron (Zofran) 2 – 4 mg IV ***(peds 0.15 mg/kg IV).***

Measuring Pressure:

- Pressure relief limits maximum CPAP pressure to 25 cm H_2O @ 25 LPM.
- Do not exceed pressure limit of manometer (25 cm H_2O).
- Manometer accuracy ± 3 cm H_2O up to 15 cm H_2O and ± 5 cm H_2O over 15 cm H_2O.

Specifications:

Sample guidelines for preparing Rx Dosing;

Flow meter setting L/min.	14 - 15								23 - 24							
CPAP Pressure cm H_2O	4 - 5								9 - 10							
Output	12 mL/hour								16 mL/hour							
Rx (mg/hr)	5		10		15		20		5		10		15		20	
Treatment Duration (hours)	1	2	1	2	1	2	1	2	1	1.5	1	1.5	1	1.5	1	1.5
Medication @5mg/mL (mL)	1	2	2	4	3	6	4	8	1	1.5	2	3	3	4.5	4	6
Saline (mL)	11	22	10	20	9	18	8	16	15	22	14	21	13	20	12	18

Notes:

- In the event of undesirable flow from oxygen source, simply remove the device and place on supplemental oxygen.
- Always verify delivered CPAP pressure on a manometer.
- Activation or deactivation of nebulizer may affect the delivered CPAP pressure. Always verify delivered CPAP pressure with a monometer.
- Flow meters capable of delivering up to 25 LPM may be required to operate both CPAP and Nebulizer simultaneously.
- Use of nebulizer other than the one provided may affect performance.
- Do not remove CPAP until hospital therapy is ready to be placed on the patient.
- Watch the patient for gastric distention that can result in vomiting.
- Procedure may be performed on patients with a Do Not Resuscitate order.
- Due to the changes in preload and afterload of the heart during CPAP therapy, a complete set of vital signs must be obtained every 5 minutes.

PROCEDURE
Delayed Off Loading of Patients in the ED

EMS is currently facing an increasing frequency of patient turnover being delayed in the Emergency Department due to delays in acknowledgment, assessment, and placement in the ED. These delays negatively impact the ability of EMS to maintain response capability and provide emergency responses in a timely manner. This protocol provides a method to off-load patients and return to service in a timely manner.

PARAMEDIC ONLY

Eligible patients (patients must meet **ALL** the following criteria):

- Greater than 18 years old (or with Guardian consent).
- Stable vital signs.
- IV access should be removed **or** Nurse approval to remain in place if patient will be in waiting room.

Procedure:

- Ambulance arrives in ED and notifies ED nursing staff of patient.
- If the ED Nursing Staff has not accepted report and made efforts to offload the patient from the EMS stretcher within 30 minutes of arrival, contact an EMS Supervisor.
- EMS Supervisor again requests ED Nursing staff to offload the EMS stretcher. If no progress is made within 15 minutes of the Supervisor's engagement; and the patient meets all the criteria above, perform the following:
  - Ensure the patient's condition is unchanged and is stable by obtaining another set of VS before offloading.
  - Move the patient to the triage waiting area or cot, chair, etc. in the treatment area.
  - Document all contacts with ED personnel, and record names of and times.
  - Complete an abbreviated, handwritten EMS run report to include patient demographics, complaint, vital signs and pertinent history and ensure its delivery to triage, admissions and/or nursing. EMS is responsible for ensuring the hospital is aware of patient's presence in the ED or waiting room.
  - Complete and post standard EPCR run report.
  - Return to service.

EMS Supervisory personnel have the authority to come to the ED and offload a patient in the emergency department to a cot, chair, or ED stretcher/bed as needed (regardless of meds or procedures done).

Offload may occur when EMS resources are insufficient to meet current needs.

PROCEDURE
Endotracheal Tube Introducer (Bougie)

PARAMEDIC ONLY

Indications:

- Patients meet clinical indications for oral intubation (appropriate to use with any attempt).

Contraindications:

- Introducer larger than ET tube internal diameter.

Procedure:

1. Prepare, position, and oxygenate the patient with 100% Oxygen.
2. Select proper ET tube without stylet, test cuff and prepare suction.
3. Lubricate the distal end and cuff of the endotracheal tube and the distal ½ of the endotracheal tube introducer (Bougie). Failure to lubricate the Bougie and the ET tube may result in being unable to pass the ET Tube.
4. Using laryngoscopic techniques, visualize the vocal cords if possible, using Sellick maneuver/BURP as needed.
5. Introduce the Bougie with curved tip anteriorly and visualize the tip passing the vocal cords or above the arytenoids if the cords cannot be visualized.
6. Once inserted, gently advance the Bougie until you meet resistance or “hold-up” (if you do not meet resistance, you have a probable esophageal intubation and insertion should be reattempted or the failed airway protocol implemented as indicated).
7. Withdraw the Bougie only to a depth sufficient to allow loading of the ET tube while maintaining proximal control of the Bougie.
8. Gently advance the Bougie and loaded ET tube until you have hold-up again, thereby assuring tracheal placement and minimizing the risk of accidental displacement of the Bougie.
9. While maintaining a firm grasp on the proximal Bougie, introduce the ET tube over the Bougie passing the tube to its appropriate depth.
10. If you are unable to advance the ET tube into the trachea and the Bougie and ET tube are adequately lubricated, withdraw the ET tube slightly and rotate the ET tube 90° COUNTERCLOCKWISE to turn the bevel of the ET tube posteriorly. If this technique fails to facilitate passing of the ET tube you may attempt direct laryngoscopy while advancing the ET tube (this will require and assistant to maintain the position of the Bougie and, if so desired, advance the ET tube).
11. Once the ET tube is correctly placed, hold the ET tube securely and remove the Bougie.
12. Confirm tracheal placement, inflate the cuff with 3 – 10 cc of air, auscultate for equal breath sounds and reposition accordingly.
13. When final positioning is determined, secure the ET tube, reassess breath sounds, apply Capnography, and record and monitor readings to assure continued tracheal intubation.

PROCEDURE
External Transcutaneous Cardiac Pacing

PARAMEDIC ONLY

Non-invasive cardiac pacing, also referred to as external or transcutaneous pacing, involves the temporary application of externally applied electrodes to deliver an adjustable electrical impulse directly across an intact chest wall for the purpose of rhythmically stimulating the myocardium to increase the mechanical heart rate.

Indications:

- Treatment of hemodynamically compromised patients in setting where cardiac output is compromised due either to the complete failure of cardiac rhythm or to an insufficient rate of the patient's intrinsic pacemaker.
- Bradycardia with a systolic BP < 80 mmHg with shock-like signs or symptoms.
- Patients who experience provider witnessed cardiopulmonary arrest and who present with asystole, or patients whose EKG converts to asystole while the EKG is being monitored.
- Prompt application of the transcutaneous cardiac pacemaker is appropriate prior to the administration of EPINEPHrine and Atropine when a patient converts to asystole as a primary rhythm during EKG monitoring by a paramedic.
- Pediatric patients (40 kg or less) with profound symptomatic bradycardia unresponsive to optimal airway management, oxygenation, EPINEPHrine, and Atropine.

Note: On-line Medical consultation is required for pacing pediatric patients.

Contraindications:

- Non-witnessed cardiopulmonary arrest with asystole.
- Patient not meeting blood pressure criteria.

Technique:

- Place pads in the Anterior/Posterior position.
- Start at a pacemaker heart rate of 70 beats per minute and the milliamperes (m.a.) as low as possible.
- Gradually increase m.a. until palpable pulse confirmed capture or 200 m.a.

Potential Adverse Effects/Complications:
Patients may experience mild to moderate discomfort. If patient is conscious and has adequate blood pressure, consider:

- Pain medications per pain management protocol **and/or**
- Diazepam 2.5 – 10 mg Slow IV/IO **or**
- Versed 2 – 4 mg IV/IO

Musculoskeletal twitching in upper torso may occur during pacing.

Precautions:
When properly applied, chest compressions can be performed directly over the insulated electrodes while the pacer is operating.
DO NOT USE EXTERNAL PACING ON A HYPOTHERMIC PATIENT.

PROCEDURE
Fever / Infection Control

Indications:

Age
Duration of fever
Severity of fever
Past medical history
Medications
Immunocompromised (Transplant, HIV, Diabetes, Cancer)
Environmental Exposure
Last Acetaminophen or Ibuprofen
Warm
Flushed
Sweaty
Chills/Rigors
Myalgias, Cough, Chest Pain, Headache, Dysuria, Abdominal Pain, Mental Status Changes, Rash

EMT AEMT PARAMEDIC

Procedure:

1. Use contact, droplet and airborne PPE precautions.
2. Using your IV standing order, start a normal saline bolus.
3. For a temperature greater than 100.4°F (38°C) if available administer Ibuprofen 600 mg PO ***(peds >6 months 10 mg/kg PO, max dose 600 mg)*** or Acetaminophen 1000 mg PO ***(peds >3 months 15 mg/kg PO, max dose 650 mg)*** May assist with patient medications.
4. Notify destination or contact Medical Control.

PROCEDURE
Hemorrhage Control Clamp

EMT AEMT PARAMEDIC

Indications:
Provides temporary control of severe bleeding in the scalp, extremities, axilla, and inguinal areas.

Contraindications:
Not for use where skin approximation cannot be obtained (i.e., large skin defects under high tension).

Warnings and Precautions:

- This device is intended for temporary use only; not to exceed 24 hours.
- Patients must be seen by medical personnel for device removal and surgical wound repair.
- Use device as directed to avoid needle stick injury.
- Do not use where delicate structures are within 10 mm of the skin surface (ex. Orbits of the eye).
- This device will not control hemorrhage in non-compressible sites, such as the abdominal and/or chest cavities.
- Ensure proper PPE is utilized to protect against possible splashing of blood during application.
- The device is designed for single use. Do not use if sterility seal on package has been broken or otherwise damaged.
- Dispose of the device as you would other sharps.
- For extreme extremity injuries not amenable to clamp application, consider tourniquet application per protocol.

Procedure: *(if patient is conscious, explain procedure)*

- Apply appropriate PPE.
- Open sterile package by pulling forward on outer tabs.
- Remove device from package by lifting up. Take care not to close device until it has been applied to the wound.
    - If the device has been accidentally closed, push the side buttons inward with one hand and pull the device open using the device arms.
- Locate wound edges.
- Align the device parallel to the length of the wound edge. Position the needles approx. 1-2 cm from the wound edge on either side. For very large wounds the device can be applied to one side, then pulled to the other side, or the tissue can be approximated by hand and the device applied.
- Press the arms of the device together to close the device. The device's safety seal will break with pressure.
- Ensure the entire wound is sealed and bleeding stops, using a gauze pad to wipe the area to verify no leaking of blood from the wound. More than one device may be required for large wounds.
- If bleeding continues:
    - Ensure the device is in the correct position, close the device more firmly by applying further pressure to the arms of the device.
    - If wound is too large, apply additional devices to the open section.
    - If device is applied incorrectly or not positioned properly, remove the device according to the instructions and reapply.
- Consider Pain Management protocol

Removal:
Unless you need to reposition the device, all removal should be done in a medical facility prepared to manage the wound.

- Hold the device by the gripping bars, press the device further closed to release the lock.
- While maintaining pressure on the arms, press both release buttons with your other hand.
- While pressing the release buttons, pull one of the gripping bars open and rotate the needles from the wound, one side at a time.
- Pick up the device ONLY by the buttons to prevent accidental contact with the needles.
- Dispose of the device in accordance with local guidelines for sharps.

Notes:
If desired, wound packing and/or the use of a hemostatic agent may be applied. The hemostatic agent does not need to be removed prior to application of the clamp.

PROCEDURE
Induced Hypothermia Following ROSC

The goal is to begin cooling the patient who meets criteria as soon as possible. You may initiate resuscitation with cold saline as your IVF of choice if the patient appears to be a candidate for IH. Therefore, if you have cold saline available when the first IV is started, begin cold fluids immediately. IF IV access is already established, change to cold saline when ROSC is achieved. If ROSC is not achieved, proceed as you would with any nonresponsive cardiac arrest, and document that cold saline was initiated. This will assist the medical examiner in determining times of death. Complete the remainder of the protocol.

PARAMEDIC ONLY

Criteria for Induced Hypothermia:

- Age greater than 18 years old.
- Any cardiac arrest with resuscitation efforts.
- **R**eturn **O**f **S**pontaneous **C**irculation (regardless of blood pressure) following cardiac arrest (all non-traumatic causes).
- Patient remains comatose (GCS < 8 and/or no purposeful responses to pain).
- Intubated or needs airway management (i-gel airway is acceptable) $ETCO_2$ > 20 mmHg.
- Systolic blood pressure can be maintained at 90 mmHg spontaneously or with fluids and pressors.

Patient Exclusion Criteria:

- Pregnant female with obviously gravid uterus.
- Systolic blood pressure cannot be maintained at 90 mmHg or greater spontaneously or with fluids and pressors.
- Coagulopathy or thrombocytopenia.

Procedure

1. Patient meets criteria for Induced Hypothermia?
    a. If no, proceed to post-resuscitation protocol
        i. If yes, is the ET tube placed?
        ii. If no, proceed with intubation, i-Gel Airway is acceptable
        iii. Once airway is controlled, follow remaining steps:
2. Perform neuro exam to confirm meets criteria.
3. Expose patient – apply ice packs to axilla, neck, and groin.
4. Administer cold saline bolus 30 mL/kg to max of 2 liters.
5. Consider sedation per protocol, if needed to control agitation or shivering.
6. If necessary, administer EPINEPHrine 2-10 mcg/min. for MAP 90-100.

Notes:

- If patient meets other criteria for induced hypothermia and is not intubated, then intubate according to protocol before inducing cooling. If unable to intubate, use of i-gel Airway is acceptable.
- When exposing patient for purpose of cooling, undergarments may remain in place. Be mindful of your environment and take steps to preserve the patient's modesty.
- Do not delay transport for the purpose of cooling.
- Reassess airway frequently and with every patient move.
- Patients develop respiratory alkalosis with cooling. Do not hyperventilate.
- Transport patient to hypothermia capable center.

PROCEDURE
Indwelling IV Port Access

PARAMEDIC ONLY

Indications:

- Intravenous fluid or medications **emergently** needed AND:
- Peripheral IV cannot be established AND:
- Patient exhibits one or more of the following:
    - Presence of indwelling port
    - Altered mental status (GCS of 8 or less)
    - Respiratory compromise (SaO_2 of 80% or less following appropriate oxygen therapy, and/or respiratory rate < 10 or > 40 /min.)
- Hemodynamically unstable

Contraindications:

- Infection at insertion site
- Significant edema
- Excessive tissue at insertion site
- Inability to locate landmarks

Considerations:

- Port-a-Cath access in the field should only be utilized in **EMERGENCY** situations
- Access should only be attempted under sterile conditions by those who have documented competency
- You may utilize the patient's supplies if necessary and appropriate
- **DO NOT FORCE FLUSH INDWELLING CATHETERS**

Procedure for accessing the Implanting Port:

1. Assemble supplies:
    a. 10 cc NS syringe
    b. Chloraprep
    c. Masks
    d. Sterile gloves
    e. Huber needle with attached extension tubing
    f. Transpore tape
    g. IV NS set-up
2. Cleanse hands.
3. Peel open one corner of the Huber needle package only; Extend end of extension tubing only out of the opening.
4. Attach 10 cc NS syringe to the extension tube.
5. Prime tubing and needle with NS.
6. Place Huber needle package on a secure flat surface and peel back package open. **DO NOT touch Huber needle until sterile gloves are on.**
7. Caregiver applies mask; the patient has the option of putting on a mask or turning their head away from the port area.
8. Put on sterile gloves.
9. Use repeated back and forth strokes of the applicator for approximately 30 seconds. Allow the area to air dry for 30 seconds. Do not blot or wipe away.

10. Pick up Huber needle with NS syringe attached; touch only the Huber needle as this is sterile and the syringe is not.
11. Grip Huber needle securely; remove clear protective sheath from the needle.
12. Locate and stabilize the port site with your thumb and index finger, creating a “v” shape.
13. Access the port by inserting the Huber needle at a 90° angle into the reservoir.
14. Once accessed, the needle must not be twisted; excessive twisting will cut the septum and create a drug leakage path.
15. Insert gently. Flush the port with 2 – 5 cc NS and then attempt to aspirate a blood return; this confirms proper placement; if the port is difficult to flush **DO NOT FORCE FLUSH.**
16. Slowly inject the remaining 10 cc NS; observe for resistance, swelling or discomfort; if present, assess needle placement; if still present, remove the Huber and re-access.
17. Remove empty NS syringe and attach IV solution tubing and initiate flow.
18. Hold slight pressure with 2 x 2 until bleeding, if any, stops; there should never be excessive bleeding.

Dressing the port site:

1. Assemble supplies;
    a. CVC dressing kit
    b. Flat clean work surface
2. Open the package of 2 x 2s if extra padding is needed.
3. Place one 2 x 2 under the needle to provide padding on the skin if Huber is not flush with chest.
4. Tear a piece of tape approximately 3” long; split tape lengthwise; tape over Huber needle in a “x” format.
5. Cover site with Transpore tape.
6. Secure the extra tubing with tape to prevent catching on clothes.

PROCEDURE
Intranasal Medication

EMT AEMT PARAMEDIC

Medication administration in a certain subgroup of patients can be a very difficult endeavor. For example, an actively seizing or medically restrained patient may make attempting to establish an IV almost impossible which can delay effective drug administration. Moreover, the Paramedic or other member of the medical team may be more likely to suffer a needle stick injury while caring for these patients.

In order to improve prehospital care and to reduce the risks of accidental needle stick, the use of the Mucosal Atomizer Device (MAD) is authorized in certain patients. The MAD allows certain IV medications to be administered into the nose. The device creates a medication mist which lands on the mucosal surfaces and is absorbed directly into the bloodstream.

Indications:
An emergent need for medication administration and IV access is unobtainable or presents high risk of needle stick injury due to patient condition:

- Seizures / Behavior control: Midazolam (Versed) may be given intranasally until IV access is available.
- Altered Mental Status from Suspected Narcotic Overdose: Naloxone (Narcan) may be given intranasally until IV access is available.
- Symptomatic Hypoglycemia (blood sugar less than 80 mg/dl): Glucagon may be given until IV access is available.
- Pediatric Pain Control: FentaNYL for orthopedic injuries (2 micrograms per kilogram; max single dose of 50 micrograms).

Medications administered via the IN route require a higher concentration of drug in a smaller volume of fluid than typically used in the IV route. In general, no more than 1 milliliter of volume can be administered during a single administration event.

Contraindications:

- Bleeding from the nose or excessive nasal discharge.
- Mucosal destruction.

Technique:

1. Draw proper dosage (see below).
2. Expel air from syringe.
3. Attach the MAD device via Luer Lock device.
4. *Briskly* compress the syringe plunger.

Complications:

- Gently pushing the plunger will not result in atomization.
- Fluid may escape from the nares.
- Intranasal dosing is less effective than IV dosing (slower onset, incomplete absorption).
- Current patient use of nasal vasoconstrictors (neosynephrine/cocaine) will significantly reduce the effectiveness of IN medications. Absorption is delayed, peak drug level is reduced, and time of drug onset is delayed.

Midazolam (Versed)
Precautions:
Midazolam may cause hypoventilation and potential respiratory depression/arrest. Have equipment and help readily available to manage the airway when administering this medication.
If hypotension develops after the administration of Midazolam, administer a 20 mL/kg bolus of normal saline.

Patient Age (years)	Weight (kg)	IN Midazolam volume in mL (assuming 5 mg/mL concentration) Midazolam volume dose (mg)
Neonate	3	0.18 mL - 0.9 mg
<1	6	0.36 mL - 1.8 mg
1	10	0.6 mL - 3.0 mg
2	14	0.84 mL - 4.2 mg
3	16	0.96 mL - 4.8 mg
4	18	1.12 mL - 5.4 mg
5	20	1.2 mL - 6 mg
6	22	1.3 mL - 6.6 mg
7	24	1.4 mL - 7.2 mg
8	26	1.6 mL - 7.8 mg
9	28	1.7 mL - 8.4 mg
10	30	1.8 mL - 9 mg
11	32	1.9 mL - 9.6 mg
12	34	2 mL – 10 mg
Small Teenager	40	2 mL – 10 mg
Adult or Full-grown teenager	50 or more	2 mL – 10 mg

Naloxone
Adult

- Naloxone 0.4 mg every 5 minutes until the respiratory rate improves and the patient can maintain a pulse oximetry reading of 96% OR until 2 mg has been given.
- Split dose equally between each nostril.

Pediatric:

1. ***Naloxone 0.1 mg/kg (max single dose 0.4 mg) until the respiratory rate improves and the patient can maintain a pulse oximetry reading of 96% OR until 2 mg has been given***
2. ***Split dose evenly between each nostril***

Glucagon

- Intranasal lyophilized Glucagon (Basqimi) may be given to hypoglycemic adults in the same dose as IM or IV routes.
- The dose should be split evenly between each nostril.

Fentanyl

- Dosing is 2 mcg/kg, split evenly between nostrils.

PROCEDURE
Intraosseous Access

AEMT PARAMEDIC

Indications:

1. Intravenous fluid or medications needed AND;
2. Peripheral IV cannot be established AND the patient exhibits one or more of the following:
    a. Altered mental status (GCS of 8 or less).
    b. Respiratory compromise SaO_2 of 80% or less following appropriate oxygen therapy, and/or respiratory rate < 10 or > 40/min.
    c. Hemodynamically unstable (Systolic BP < 90).
3. IV access is preferred; however, IO may be considered prior to peripheral IV attempts in the following situations:
    a. Cardiac arrest (Medical or Trauma).
    b. Profound hypovolemia with altered mental status.

Contraindications:

1. Fracture of the tibia or femur (for tibia insertion) – consider alternate tibia.
2. Fracture of the humerus (for humeral head insertion) – consider alternate humerus.
3. Previous orthopedic procedures (ex: IO within previous 24 hrs., knee replacement, shoulder replacement).
4. Infection at insertion site.
5. Significant edema.
6. Excessive tissue at insertion site.
7. Inability to locate landmarks.

Considerations:

1. Flow rates: due to the anatomy of the IO space you will note flow rates to be slower than those achieved with IV access.
    a. Ensure the administration of the 10 ml rapid bolus with syringe
    b. Use a pressure bag or pump for fluid challenge
2. Pain: Insertion of the IO device in conscious patients causes mild to moderate discomfort and is usually no more painful than a large bore IV. However, fluid infusion into the IO space is very painful and the following measures should be taken for conscious patients:
    a. Prior to IO bolus or flush on a conscious adult patient, SLOWLY administer 20 – 50 mg of 2% Lidocaine
    b. ***Prior to IO bolus or flush on a conscious pediatric patient, SLOWLY administer 0.5 mg/kg 2% Lidocaine***

Primary Insertion Site (Trauma): Tibial Plateau
If IO access is warranted the tibia shall be the insertion site of choice if possible.
Note: In the cardiac arrest patient, the Humeral Head should be the **primary insertion site.**

Primary Insertion Site (Cardiac Arrest/Medical): Humeral Head (adult patients only)
If IO access is not available via the tibia insertion site due to contraindications or inability to access the site due to patient entrapment and vascular access is imperative, the IO may be placed in the humeral head.

Notes:

- In the cardiac arrest patient, the Humeral Head should be the **primary insertion site.**
- DO NOT attempt insertion medial to the Intertubercular Groove or the Lesser Tubercle.

Pediatric Patient:

- *Defined as a patient weight 3 – 39 kg.*
- *The pediatric needle set (15 mm) shall be used for pediatric patients.*
- *Use the length-based assessment tape to determine pediatric weight.*
- *The only approved site for pediatric IO insertion is the tibia.*

Standing Order:

The Intraosseous device may be used if the indications are met and no contraindications exist.

Precautions:

- The IO is not intended for prophylactic use.
- The IO infusion system requires specific training prior to use.
- Proper identification of the insertion site is crucial.

Landmarks: Tibial Plateau

There are three important anatomical landmarks – the patella, the tibial tuberosity (if present) and the Flat aspect of the medial tibia.

- **Important: the tibial tuberosity is often difficult or impossible to palpate on very young patients!** The traditional approach for IO insertions in small patients – where the tibial tuberosity cannot be palpated – is to identify the insertion site – **"two finger widths below the patella and then medial along the flat aspect of the tibia."**
- The traditional approach to IO insertion is slightly larger patients where the tuberosity can be appreciated – generally suggests **"One finger width distal to the tibial tuberosity along the flat aspect of the medial tibia."**
- The IO should be inserted two finger widths below the patella (kneecap) and one finger medial (toward the inside) to the tibial tuberosity.
- **For the morbidly obese patient:**
    - Consider rotating the foot to the mid-line position (foot straight up and down)
    - With the knee slightly flexed, lift the foot off of the surface allowing the lower leg to "hang" dependent
    - This maneuver may improve your ability to visualize and access the tibial insertion site
    - Please use the Bariatric Needle Set in these patients

Landmarks: Humeral Head

- Place the patient in a supine position.
- Expose the shoulder and place the patient's arm against the patient's body.
- Rest the elbow on the stretcher with the forearm on the abdomen. Palpate and identify the mid shaft humerus and continue palpating toward the humeral head.
- As you near the shoulder you will note a small protrusion. This is the base of the greater tubercle insertion site.
- With the opposite hand "pinch" the anterior and inferior aspects of the humeral head confirming the identification of the greater tubercle. This will ensure you have identified the midline of the humerus itself.
- The insertion site is approximately two finger widths inferior to the coracoid process and the acromion.

Landmarks: Medial Malleolus

- The insertion site is two finger widths proximal to the Medial Malleolus and positioned midline on the medial shaft.

Procedure:

Inserting the IO;

1. Determine that the IO is indicated.
2. Ensure that no contraindications are present.
3. Locate the proper insertion site.
4. Clean the insertion site with alcohol.
5. Prepare the IO.
6. Stabilize the leg (or arm).
7. Position the IO at the insertion site with the needle at a 90° angle to the surface of the bone.
8. Activate the device to set needle through the skin. Apply firm steady pressure on the driver and power through the cortex of the bone. Stop when the needle flange touches the skin, or a sudden resistance is felt. This indicates entry into the bone marrow cavity.
9. Grasp the hub firmly with one hand and remove the driver from the needle set.
10. While continuing to hold the hub firmly, rotate the stylet counterclockwise and remove it from the needle set. Dispose of the stylet properly in a sharps container.
11. Confirm proper placement of the IO catheter tip:
    a. The catheter stands straight up at a 90° angle and is firmly seated in the tibia
    b. Blood is sometime visible at the tip of the stylet
    c. Aspiration of a small amount of marrow with a syringe
12. Attach a primed extension set to the hub and flush the IO space with 10 cc of Normal Saline. **NO FLUSH – NO FLOW**
13. If the patient is conscious, administer Lidocaine 2% 20-50 mg slowly **PRIOR** to the initial bolus ***(peds 0.5 mg/kg).***
14. Initiate the infusion per standing orders. Use of a pressure infuser or blood pressure cuff is recommended to maintain adequate flow rates.
15. Apply the wrist band and a dressing.
16. For the NIO be sure to unlock it by rotating the cap 90 degrees in either direction.
17. Place the palm of your dominant hand over the cap. Press the device against the patient's skin and maintain pressure. While pressing down on the device, pull the trigger wings upward. This action will activate the device.
18. Gently pull the NIO up in a rotational motion while holding the base of the needle stabilizer against the insertion site.
19. Continue holding the needle stabilizer in place and pull up the stylet (twisting may be necessary).

PROCEDURE
LUCAS CPR Device

EMT AEMT PARAMEDIC

Inclusion Criteria:

- The device must be present on scene within 8 minutes of the initiation of CPR
- The patient must not meet any of the exclusion criteria

Exclusion Criteria:

- Body habitus too large for the device
- Children less than 42 kg/ 90 lbs. or any individual which when fitted with the device the suction cup does not make firm contact with the chest wall
- Down time suspected to be greater than or equal to 15 minutes without CPR
- Confirmed down time without CPR > 10 minutes

If the above inclusion criteria are met, none of the exclusion criteria are present, and the LUCAS device is available, the following steps will be taken to implement its use:

1. CPR will be performed manually for at least 2 minutes and the patient will be ventilated with a BVM/ oral airway during this time.
2. After 2 minutes, the defibrillation/ monitor pads will be applied to the patient. At this time, the LUCAS device will also be applied to the patient.
3. Defibrillation performed if indicated.
4. CPR resumed using the LUCAS device. Avoid CPR pauses greater than 20 seconds when placing the device.
5. Obtain airway (adequate ventilation with OPA/NPA/BVM, i-gel Airway or ETT).
6. IV/IO access.
7. Initiation of ACLS medications.
8. Allow at least 90 seconds of CPR after any medications given before pausing to check rhythm.
9. If pulse confirmed, prepare for immediate transport. The LUCAS device may be turned off but must be left on the patient during the transport to the hospital.
10. If the patient goes back into cardiac arrest, immediate resumption of LUCAS CPR will be performed and ACLS will continue.
11. Detailed documentation with times of all initiation and termination of use of LUCAS device must be kept for statistical and feedback purposes.
12. Consider ketamine if sedation is necessary during high quality CPR.

NOTE: Placement and initiation of the device cannot exceed 20 seconds. Longer pauses result in a significant decrease in likelihood of a successful resuscitation.

NOTE: The LUCAS device may be used in cases of Traumatic Cardiac arrest.

PROCEDURE
EMS Standardized Adult Medical Radio Report

EMT AEMT PARAMEDIC

To Receiving Facility:

- This is EMS unit X, need to give report on a Priority X Medical patient.

EMS Report:

1. This is EMS unit X with a X minute ETA to your facility with a Priority X Medical.
2. We have a XX year-old male/female s/p {brief explanation of events leading to 911 call}.
3. Patient is currently GCS ___. Physical exam findings include….
4. Current vital signs are
    - Pulse
    - BP
    - Respiratory rate
    - SpO_2
    - Blood glucose level
5. Interventions performed include
    - IV
    - Medications given
    - 12 Lead Interpretation
    - Oxygen given
    - Etc.
6. Repeat ETA is ____, Do you require any further?

PRIORITY 1

- Unstable vital signs for age (hypotension, extreme tachycardia, tachypnea, etc.)
- GCS <13 with normal or unknown baseline
- Chest Pain with STEMI on 12-Lead (Go to PCI-capable facility)
- Cardiac Arrest (go to PCI capable facility)
- Acute SVT or other potentially unstable arrhythmias (v-tach with a pulse, a-fib with RVR, etc.)
- Respiratory failure or intubated patient (requiring CPAP/BVM/ventilator to support respirations)
- Septic appearing patient with poor perfusion (ETC_2 < 25, altered mental status, hypotension)
- Stroke-like symptoms w/ last known normal of >5 hours (go to comprehensive stroke center)
- Stroke-like symptoms with last known normal of < 5 hours (go to primary stroke center)

PRIORITY 2

- Abdominal pain with signs of compromise (guarding, rebound, distention, tachycardia)
- Acute chest pain with risk factors, or EKG changes (WITHOUT STEMI)
- Respiratory distress requiring oxygen via simple mask or non-rebreather to maintain SpO_2 >94%
- Acute GI bleed
- Pregnancy complications > 20 weeks (active labor, uncontrolled bleeding, preeclampsia)
- Combative patient

PRIORITY 3

- Medical patients not meeting above criteria

PROCEDURE
EMS Standardized Pediatric Trauma Radio Report

EMT AEMT PARAMEDIC

To Receiving Facility:

- This is EMS unit X, need to give report to Hospital on a Priority X Trauma patient.

EMS Report:

1. This is EMS unit X with a X minute ETA to your facility with a Priority X Trauma.
2. We have a XX year-old (< 2 years, please give months) male/female s/p {brief mechanism explanation}. Broselow color is XXX.
3. Patient is currently GCS ___. Injuries found include….
4. Current vital signs are
    - Pulse
    - BP
    - Respiratory rate
    - SpO_2
5. Interventions performed include
    - IV
    - Tourniquet
    - Needle Decompression
    - Splinting
    - Spinal Motion restriction
    - Etc.
6. Repeat ETA is ____, Do you require any further?

PEDIATRIC PRIORITY 1 TRAUMA:

- Trauma stat
- GCS <8
- Trauma arrest/CPR
- Ventilation/airway compromise/intubated
- Shock requiring ongoing fluid resuscitation (age specific hypotension)
- Hemodynamic instability assoc. WITH neurological injury
- Open Fx with Hemodynamic instability
- Open book pelvic fracture with hemodynamically instability
- Pulseless extremity
- Proximal amputation
- Spinal cord injury with paralysis
- 2nd and 3rd degree burns >30%BSA
- Penetrating head, neck, chest, torso injury

PEDIATRIC PRIORITY 2 TRAUMA:

- GCS 9-12
- Shock that stabilized with IVF
- Open or multiple fractures of extremities
- Pneumothorax
- Suspected intra-abdominal injury with hemodynamic effects
- Complex pelvic fractures
- Suspected cervical or thoracic spine injuries
- 2nd and/or 3rd degree burns 15-30% TBSA
- Penetrating wound to 2 or more extremities
- Ejection from vehicle
- Death of another occupant in same vehicle
- Struck, dragged or runover by a vehicle
- MVC with high speed impact or rollover
- Fall >20ft
- ATV/Motocross/Dirt bike with injury association

PEDIATRIC PRIORITY 3 TRAUMA:

- Trauma patients not meeting above criteria

PROCEDURE
EMS Standardized Adult Trauma Radio Report

EMT AEMT PARAMEDIC

To Receiving Facility:

- This is EMS unit X, need to give report on a Priority X Trauma patient.

EMS Report:

7. This is EMS unit X with a X minute ETA to your facility with a Priority X Trauma.
8. We have a XX year-old male/female s/p {brief mechanism explanation}.
9. Patient is currently GCS ___. Injuries found include….
10. Current vital signs are
    - Pulse
    - BP
    - Respiratory rate
    - SpO_2
11. Interventions performed include
    - IV
    - Tourniquet
    - Needle Decompression
    - Splinting
    - Spinal Motion restriction
    - Etc.
12. Repeat ETA is ____, Do you require any further?

PRIORITY 1

- Traumatic arrest
- HR > 130 (>110 for age >= 65)
- HR < 50
- Systolic < 90 (<100 for age>=65)
- GCS < 13
- Impending airway failure/intubation
- Penetrating trauma to torso, abdomen, head, or neck
- Amputation proximal to knee or elbow
- Use of a tourniquet or loss of distal pulses
- Receiving blood
- Spinal cord disruption/neurological findings
- Hip dislocation

PRIORITY 2

- Open fracture or multiple fractures
- Pregnancy > 20 weeks
- Femur fracture
- Trauma while on known anticoagulation

PRIORITY 3

- Trauma patients not meeting above criteria

PROCEDURE
RESQPOD Circulatory Enhancer

EMT AEMT PARAMEDIC

ResQPOD impedance threshold device prevents unnecessary air from entering the chest during the decompression phase of CPR. When air is slowed while flowing into the lungs as the chest wall recoils, the vacuum (negative pressure) in the thorax pulls more blood back to the heart, resulting in:

- Doubling of blood flow to the heart.
- 50% increase in blood flow to the brain.
- Doubling of systolic blood pressure.

The device should be used for all patients receiving CPR whenever ET, Blind Airway Insertion Device (i-Gel), or BVM is used.

Indications:
Cardiopulmonary arrest ages 8 and up.

Contraindications:
Patients with spontaneous respirations.
Cardiopulmonary arrest associated with trauma.

Procedure
Confirm the absence of pulse and begin CPR immediately. Assure that the chest wall recoils completely after each compression. Endotracheal intubation is the preferred method of managing the airway when using ResQPOD.

1. Using ResQPOD on a facemask:
    a. Connect ResQPOD to the facemask.
    b. Connect ventilation source (BVM) to the top of the ResQPOD. If utilizing a mask without a bag, connect to mouthpiece.
    c. Establish and maintain a tight face seal with mask throughout chest compressions.
    d. Do not use the ResQPOD's timing lights utilizing a facemask for ventilation.
    e. Perform ACLS interventions as appropriate.
    f. Prepare for endotracheal intubation.
2. Using ResQPOD on an Endotracheal Tube or I-Gel:
    a. Place Endotracheal Tube or i-Gel and confirm placement, secure the tube.
    b. Move the ResQPOD from the facemask to the advanced airway and turn on the timing lights by removing the clear tab, ventilate asynchronously over 1 second when the light flashes (10/min).
    c. Continue CPR with minimal interruptions.
    d. Perform ACLS interventions as appropriate.
    e. If a pulse is obtained, remove the ResQPOD and assist ventilations as needed.

Notes:

- Always place waveform Capnography between ResQPOD and ventilation sources.
- Do not interrupt CPR unless absolutely necessary.
- If pulse returns, discontinue CPR and ResQPOD. If patient rearrests, resume CPR with ResQPOD.
- Do not delay compressions if ResQPOD is not readily available.

PROCEDURE
Tourniquet

EMT AEMT PARAMEDIC

Indications:

- Life threatening arterial hemorrhage.
- **Serious or life-threatening** extremity hemorrhage and tactical considerations prevent the use of standard hemorrhage control techniques.

Contraindications:
Non-extremity hemorrhage.
Proximal extremity location where tourniquet application is not practical.

Procedure:

1. Place tourniquet proximal to wound.
2. Tighten per manufacturer instructions until hemorrhage stops and/or distal pulses in the affected extremity disappear.
3. Secure tourniquet per manufacturer instructions.
4. Note time of tourniquet application and communicate this to receiving care providers.
5. Dress wounds per standard wound care protocol.
6. If delayed or prolonged transport and tourniquet application time greater than 2 hours, contact Medical Control.
7. Include tourniquet use in your report to the trauma center as soon as practical.
8. Consider the need for pain management.

PROCEDURE
Vascular Access

Any patient having a condition that requires an IV or INT may receive it if the AEMT or Paramedic deems it necessary. Weigh the transport time against the time it would take to start and IV and make a good decision.

Trauma: Minimize on scene time. IVs are to be started while enroute to the hospital unless the patient is pinned in vehicle or a prolonged scene time is unavoidable. IV Lactated Ringers is for trauma patients. The rate is based on patient condition and shall be to maintain the patient's systolic blood pressure 80 – 100 mmHg.
Medical: INT or IV Normal Saline for chest pain, cardiac arrest or other medical conditions requiring possible IV access. If IV access is all that is needed, the INT is preferred

Intravenous Fluid Administration

Treatment Pathway

1. The preferred site for an IV is the hand followed by the forearm and antecubital fossa and is dependent on the patient's condition and treatment modality.

AEMT STOP HERE

2. In the event that an IV cannot be established, and the IV is considered critical for the care of the patient, other peripheral sites may be used (i.e.: external jugular, feet, legs).
3. External jugular veins should never be the first line attempted unless the patient has no limbs for the initial attempts. INTs **SHOULD NOT** be used in external jugular access.
4. The intraosseous site may be used in patients in whom IV access cannot be established within 2 attempts or 90 seconds **when IV access is critical** (refer to the EZ-IO or NIO procedure).

PARAMEDIC STOP

.

REFERENCE
Consent Issues

EMT AEMT PARAMEDIC

Tennessee Law, under a legal doctrine known as “implied consent”, allows EMS Personnel to treat and transport minors when a parent or legal guardian is not available to provide consent IF a medical emergency exists. Simply stated, a court will imply that reasonable parents will want someone to help their child in their absence if the child develops an emergent medical condition. However, implied consent only becomes legally effective after a reasonable effort is made under the circumstances to contact a parent or legal guardian to obtain their consent to treat the minor.

In non-emergent situations, “mature” minors are generally presumed to be legally competent to give consent. Whether or not a minor is “mature” depends upon multiple factors articulated by the Tennessee Supreme Court. Since it would be difficult, if not impossible, for an EMT/AEMT/Paramedic to adequately assess the factors in the field, it is highly recommended that you obtain the consent of a parent or legal guardian before treating or transporting a non-emergent minor.

Obtaining the consent of a parent or legal guardian before treating or transporting a minor with either an emergent or non-emergent condition is not necessary when the minor is married or legally emancipated. Emancipated and legally married minors are generally deemed to be legally competent.

REFERENCE
Civilians Riding in Emergency Unit During Transports

EMT AEMT PARAMEDIC

Under normal conditions and non-emergency transport one civilian that is a relative or friend of the patient will be allowed to ride to the hospital with the patient. This designated person will ride in the cab with the Unit Operator. They will be assisted by Fire Department personnel into the cab and placed in a safety belt; this may be delayed until the Operator is positioned in their seat and ready to transport.

The Paramedic in charge of the Emergency Unit has the right to deny a civilian the ability to ride during transport for the following reasons:

- The person has been drinking and/or has the smell of alcohol on their person and is deemed to be a risk to the safety of the personnel.
- The person is rude, belligerent, and/or uncooperative.
- There is a safety issue with this person riding during transport.
- The person is under the age of 18.
- Public Health concerns/ orders.

During emergency transports it will be up to the Paramedic to determine what is in the best interest of the patient and safety of the civilian as well as the personnel. In the event of an elderly person that has no transportation to the hospital or the denial of a civilian riding to the hospital, contact your EMS Supervisor for notification and assistance.

Notes:

- Be customer service responsible **at all times.** We are public servants.
- When dealing with pediatric patients every effort should be made to allow the child's parent/guardian or family member that is of age to ride during transport. This person may be allowed to ride in the patient compartment if the Paramedic/EMT deems it is in the best interest of the patient.

REFERENCE
Hospital Capabilities

EMT AEMT PARAMEDIC

	Baptist Memphis	Baptist Women's	Baptist Collierville	Methodist University	Methodist North	Methodist South	Methodist Germantown	Methodist Germantown Women's	St. Francis Memphis
Cardiac Therapeutic PCI	Yes	No	No	Yes	Yes	7a-5p M-F only	Yes	No	Yes
Suspected Stroke C-STAT score >= 2	Yes	No	No	Yes	No	No	No	No	Yes
Suspected Stroke C-STAT score < 2	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Yes
NeuroTrauma GCS <11	Yes	No	No	Yes	No	No	No	No	Yes
OB	No*	Yes	No*	No*	No*	Yes	No*	Yes	Yes for gestation > 32 weeks
Pediatrics	Please see Spence Children's Listing	Please see Spence Children's Listing	Limited (Basic)	Limited (Basic)	Limited (Basic)	Limited (Basic)	Please see LeBonheur Gemantown Listing	No	Limited (Basic)
Trauma	Equivalent to Level 2	Equivalent to Level 3	Equivalent to Level 3	Equivalent to Level 2	Equivalent to Level 3	Equivalent to Level 3	Equivalent to Level 3	No	Equivalent to Level 2
Induced Hypothermia	Primary	NA	Secondary	Primary	Secondary	Secondary	Secondary	NA	Primary
Behavioral Health	Yes	No	Yes	Yes	Yes	Yes	Yes	No	Ages 4 through Geriatric
Notes	OB < 20 weeks; other OB to Womens Hospital		OB < 20 weeks	OB < 20 weeks	OB < 20 weeks		OB < 20 weeks; other OB to Womens Hospital		

Updated 7/24

*All non OB hospitals are capable of handling OB cases < 20 weeks who are hemodynamically stable in the field except LeBonheur.

Induced Hypothermia: Primary Centers = In House Capability; Secondary = System Participation-will continue IH and transfer to Primary Center; Tertiary = No availability of IH

This document refers to "Hospital Capabilities" from an EMS standpoint ONLY. Services at individual hospitals may change without notice. These designations are unofficial, and not endorsed by the state.

	St. Francis Bartlett	Regional One Medicine	Elvis Presley Trauma Center	Rout Women's Hospital	VA Hospital	Spence Children's Hospital	Germantown LeBonheur	LeBonheur
Cardiac Therapeutic PCI	Yes	Yes	No	No	Yes, Daytime	No	No	No
Suspected Stroke C-STAT score >= 2	No	No	No	No	No	No	No	No
Suspected Stroke C-STAT score < 2	Yes	No	No	No	No	No	No	No
NeuroTrauma GCS <11	No	No	Yes	No	No	No	No	Yes
OB	No*	No*	No	Yes*	No	No	No	No
Pediatrics	Limited (Basic)	Limited (Basic)	Limited (Basic)	No	No	General	General*	Level 1 Peds Comprehensive*
Trauma	Equivalent to Level 3	Equivalent to Level 3	Level 1*	Equivalent to Level 3	Equivalent to Level 3	General	General	Level 1 Peds
Induced Hypothermia	Secondary	Tertiary	NA	NA	Tertiary	NA	NA	NA
Behavioral Health	No	Yes	Yes	No	Yes			Yes
Notes	*OB < 20 weeks	*OB < 20 weeks. Other OB to Rout Women's Hospital	Trauma for pts > 15 years of age only. Multiplace Hyperbarics Burn Center	High Risk OB/Neonatal			*Patients < 15 years of age	*Patients < 15 years of age

Updated 7/24

Note: All patients transported by EMS should be taken to the **most appropriate** facility according to the patient's condition. Careful consideration should always be given to patients with special needs. The capabilities form on the previous page should be used as a guide for making appropriate destination decisions.

Trauma "ROUND ROBIN"

When the LEVEL 1 Trauma facility is on diversion status, contact MEDCOM prior to leaving the scene for redirection to another facility. Please follow the directions given after ensuring they understand the needs of your patient.

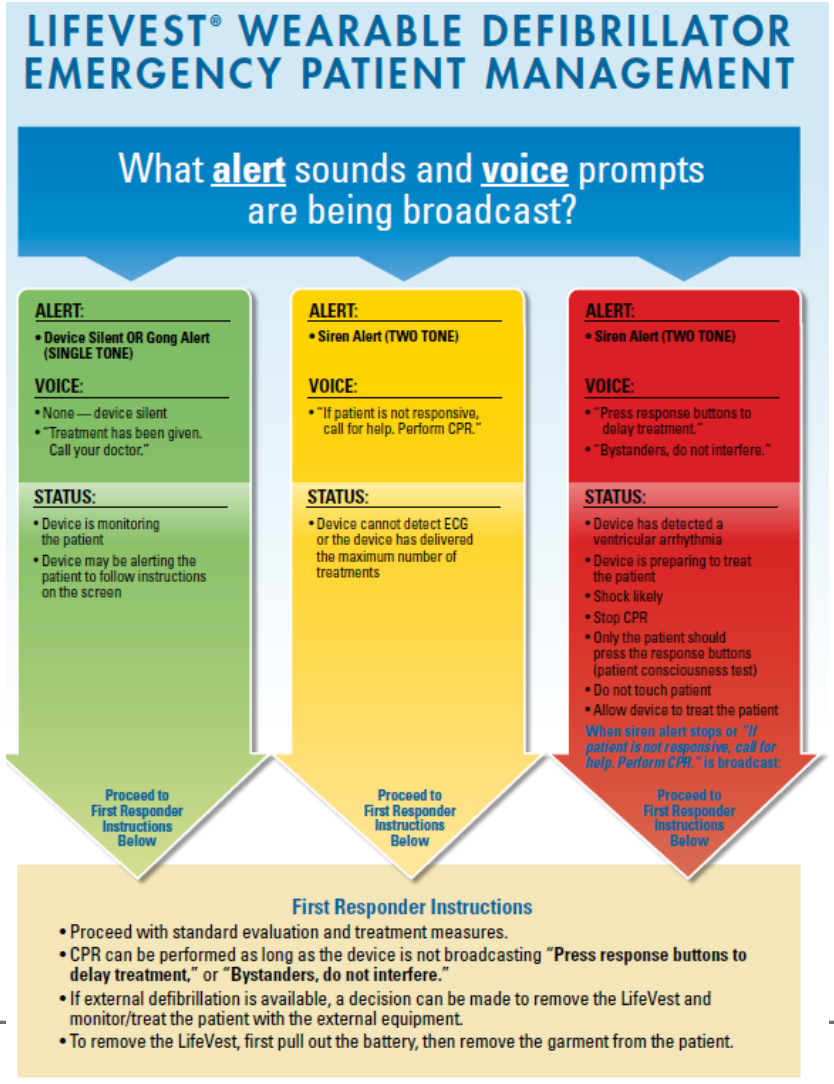
REFERENCE
Life Vest

EMT AEMT PARAMEDIC

The LifeVest wearable defibrillator is a treatment option for sudden cardiac arrest that offers patients advanced protection and monitoring as well as improved quality of life. The LifeVest is the first wearable defibrillator. Unlike an implantable cardioverter defibrillator (ICD), the LifeVest is worn outside the body rather than implanted in the chest. This device continuously monitors the patient's heart with dry, non-adhesive sensing electrodes to detect life-threatening abnormal heart rhythms. If a life-threatening rhythm is detected, the device alerts the patient prior to delivering a treatment shock, and thus allows a conscious patient to delay the treatment shock. If the patient becomes unconscious, the device releases a Blue™ gel over the therapy electrodes and delivers an electrical shock to restore normal rhythm.

The LifeVest gives off alert sounds and voice prompts. Please see the information list at the end of this reference to familiarize yourself with the LifeVest and its alert sounds and voice prompts.

If you encounter a patient with the LifeVest, contact Medical Control at the receiving hospital as soon as possible.



REFERENCE
LVAD

EMT AEMT PARAMEDIC

An LVAD is a surgically implanted mechanical pump that is attached to the heart. An LVAD is different from an artificial heart in that it replaces the failing heart completely. An LVAD works with the heart to help it pump more blood with less work. It does this by continuously taking blood from the left ventricle and moving it to the aorta, which then delivers oxygen-rich blood throughout the body.

The LVAD has both internal and external components. The actual pump sits on or next to your heart's left ventricle with a tube attached that routes the blood to your aorta. A driveline cable extends from the pump, out through the skin, and connects the pump to a controller and power sources worn outside the body.

The driveline must be connected to the controller, and the controller must be connected to power at all times to keep the pump working properly. The pump is powered by batteries or electricity. Each device has a specific carrying case.

You should be contacted by the Alarm Office prior to making the scene and/or the information will appear on the MDT. But, since an LVAD patient can be mobile, they may not be at their place of residence. So, you may not always get prior information. Should you make an LVAD patient, please contact the Cardiopulmonary Transplant Unit at Baptist Memorial Hospital Memphis 901-226-2950 or 901-226-2000 (Ask for LVAD Coordinator) if you have any questions. Please know that the patient and their family are typically very familiar with the device and will have extensive training on it.

All LVAD patients should be transported to Baptist Memphis (Hospital #11) or designated local facility.

LVAD ALARMS

Priority	System Controller Screen	Active Symbols	Alarm Means	To Resolve Alarm
ADVISORY	Connect Power :04		**One of the two power cables is disconnected.**	1. Promptly connect the disconnected power cable to power source (functioning Mobile Power Unit or two fully-charged HeartMate 14 Volt Lithium-Ion batteries). 2. If alarm persists, call your hospital contact immediately.
	Replace Power :02 + Low Battery :06		**Low battery–power input is low, with less than 15 min remaining.**	1. Promptly connect to a working or different power source (Mobile Power Unit or two fully-charged 14 Volt HeartMate Lithium-Ion batteries). 2. If alarm persists, call your hospital contact immediately.
	Call Hospital Contact Controller Fault		**System Controller Hardware Fault**	Call your hospital contact as soon as possible for diagnosis and instructions.
	Call Hospital Contact Comm Fault		**Communication Fault (Comm Fault)**	Call your hospital contact as soon as possible for diagnosis and instructions.
	Call Hospital Contact Backup Battery Fault		**System Controller Backup Battery Fault**	Call your hospital contact as soon as possible for diagnosis and instructions.
	Call Hospital Contact Backup Battery Fault +		**System Controller Backup Battery Not Installed**	Call your hospital contact as soon as possible for diagnosis and instructions.
	Call Hospital Contact Driveline Power Fault		**Driveline Power Fault**	Call your hospital contact as soon as possible for diagnosis and instructions.
	Call Hospital Contact Driveline Comm Fault		**Driveline Communication Fault (Driveline Comm Fault)**	Call your hospital contact as soon as possible for diagnosis and instructions.

Important! The Pump Running ( ) symbol is lit green when the pump is running.

Priority	System Controller Screen	Active Symbols	Alarm Means	To Resolve Alarm
HAZARD	Call Hospital Contact :07 + Low Flow :03		**Pump is off. The Pump Running symbol is black.**	1. Immediately connect to a working power source (Mobile Power Unit™ or two HeartMate® 14 Volt Lithium-Ion batteries). 2. If connecting to power does not resolve the problem, press any button on the System Controller to attempt pump start and call your hospital contact immediately.
	Call Hospital Contact :07 + Low Flow :03		**Low flow, flow is less than 2.5 lpm**	Call your hospital contact immediately for diagnosis and instructions.
	Connect Driveline :02		**Driveline is disconnected. The Pump Running symbol is black.**	1. Immediately reconnect the Driveline to the System Controller and move the Driveline safety lock on the System Controller to the locked position. Also, check that the Modular In-Line connector is secure. 2. If alarm persists after reconnecting the Driveline, press any button on the System Controller to potentially resolve. 3. If the Driveline is connected and alarm persists, replace the System Controller with a configured backup System Controller. 4. If alarm persists, call your hospital contact immediately.
	Connect Power Immediately :05 + Backup Battery :01		**Both power cables are disconnected.**	1. Immediately connect to a working power source (Mobile Power Unit or two fully-charged HeartMate 14 Volt Lithium-Ion batteries). 2. If alarm persists, call your hospital contact immediately.
	Call Hospital Contact Controller Fault		**System Controller Hardware Fault (Microcontroller Failure)**	1. No active symbols (constant audio tone). 2. Call your hospital contact as soon as possible for diagnosis and instructions.
	Low Battery :06 + Replace Power Immediately :02		**Low Battery, Power input is extremely low with less than 5 min. remaining.**	1. Immediately connect to a working power source (Mobile Power Unit or two fully-charged HeartMate 14 Volt Lithium-Ion batteries). 2. If alarm persists, call your hospital contact immediately.

REFERENCE
MCI Plan Response Levels

EMT AEMT PARAMEDIC

Below are the MCI plan response levels. Each response plan is designated to be escalating in nature. However, if any incident requires more resources are required, a higher EMS Level response may be initially requested. That response would receive the total number of resources from EMS Level 1 response to the EMS level response requested.

If you have any questions refer to EMS MCI Response Plan.

Requested by First Arriving Company:

Level 1	**Optional Responses**
5 Emergency Units	2 Additional Emergency Units
1 EMS Lieutenant	2 Additional Fire Companies
1 EMS Battalion Chief	2nd EMS Lieutenant
1 First Responder Engine Company	Mass Casualty Task Force
1 First Responder Truck Company	Additional Special Operations
1 Battalion Chief	Rescue Unit
	Aeromedical

Requested by IC after consultation with Medical Branch Director:

Level 2	**Optional Responses**
5 Additional Emergency Units	2 Additional Emergency Units
Deputy Chief of EMS	2 Additional Fire Companies
Deputy Chief of Emergency Operations	Additional Mass Casualty Unit Buses
Division Chief of EMS	Any specialized equipment
2nd EMS Lieutenant	
Additional Battalion Chief	
1 Division Chief	
Mass Casualty Task Force	
Air Mask Services	
OSHA Safety Officer	
Media Affairs	
Incident Support Team	

Requested by IC after consultation with Medical Branch Director:

Level 3	**Optional Responses**
5 Additional Emergency Units	Additional Emergency Units
Director of Fire Services	Additional Fire Companies
Deputy Director of Fire Services	Additional Mass Casualty Unit(s)
All Deputy Chiefs	Additional MATA Buses
All Division Chiefs	OEM Resources as needed
Medical Director	Additional Incident Management –
EMS Battalion Chiefs	All Teams
All EMS Lieutenants	
1 Additional ALS Fire Company	

Requested by IC on advice of Director of Fire Service or designee

Level 4

Requested by IC on advice of Director of Fire Services or designee

Level 5

REFERENCE
Non-Viable Patients on Public Scenes

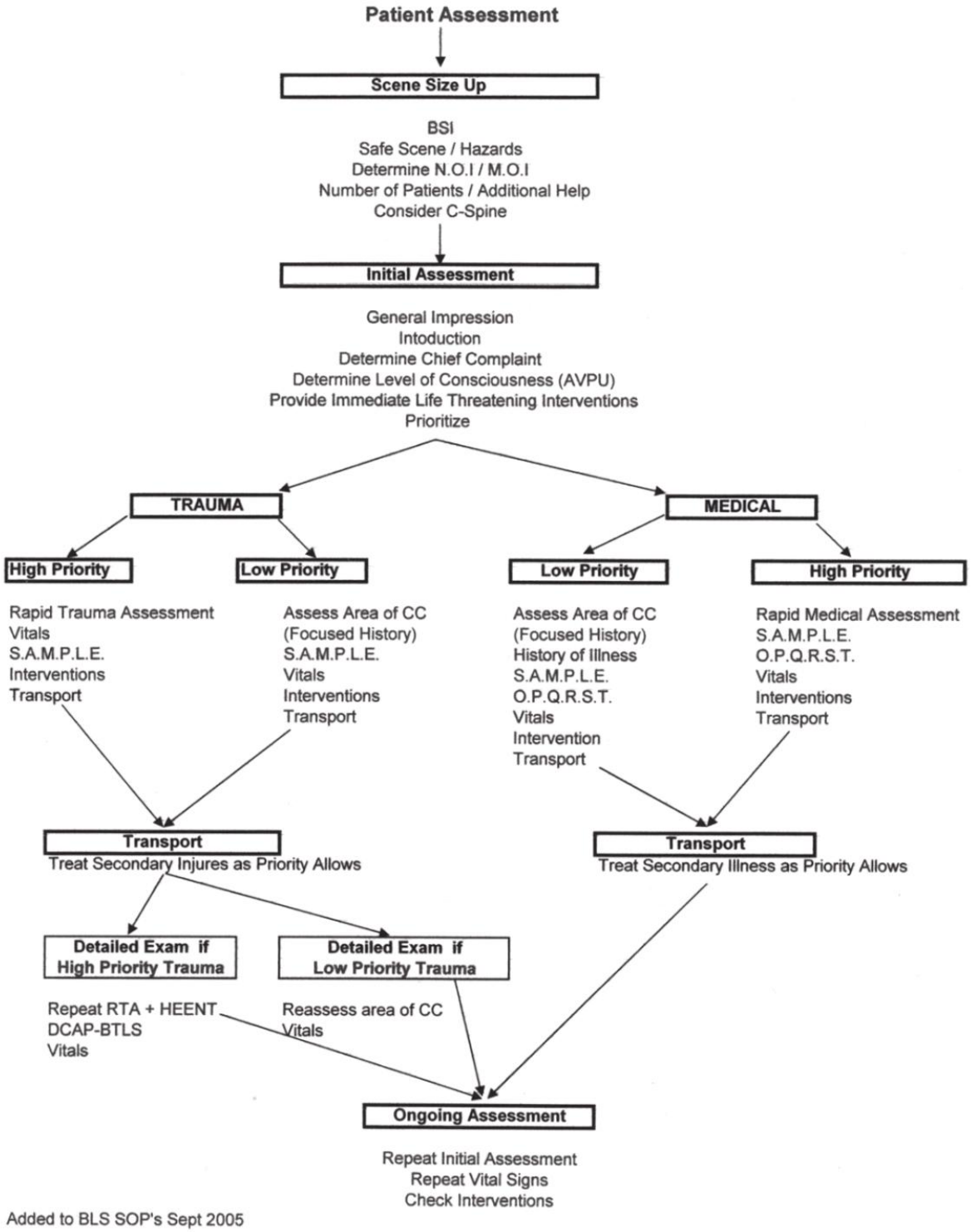
EMT AEMT PARAMEDIC

According to Tennessee state law, only a Medical Examiner (ME) can take charge of a deceased body from a crime scene or the Emergency Department. The Police Department will contact the ME, who will then make the scene (including traffic fatalities).

1. The body **WILL NOT** be touched (beyond what is necessary to determine lifelessness, if needed), no items shall be removed from the body or disturbed in any way until permission of the medical examiner's office is granted. This will generally be given by the medical legal investigator (also known as the ME investigator).
2. In the case of a traffic fatality – if there is any doubt as to whether the patient is viable, extricate the patient as per EMS protocol and follow any applicable ALS protocol(s).
3. If it is obvious and/or has been determined by a Firefighter Paramedic that an entrapped victim is non-viable, the ME's office requests that the body not be extricated until, at least, pictures of the scene have been taken by PD. At that point extrication of the victim can proceed and the body placed on the ground. After extrication, a barrier using nylon tape or rope and a drape secured to an object on scene should be used to obscure the body from public view and protect potential evidence.
4. It is imperative that EMS personnel on the scene communicate with the Incident Commander relative to patient viability in determining whether it is appropriate to extricate immediately or to wait.
5. The ME will arrange the transportation of the deceased individual(s) to the Regional Forensic Center (RFC). The Unit personnel will provide the completed hospital copy of the PCR to the RFC representative.
6. Transport products of conception or non-viable fetus with the mother.

REFERENCE
Patient Assessment Flow Chart

EMT AEMT PARAMEDIC



Added to BLS SOP's Sept 2005

REFERENCE
Physician Orders for Scope of Treatment (POST)

EMT AEMT PARAMEDIC

Directions for Health Care Professionals

Completing POST

Must be completed by a health care professional based on patient preferences, patient best interest, and medical indications.

To be valid. POST must be signed by a physician or, at discharge or transfer from a hospital or long term care facility, by a nurse practitioner (NP), clinical nurse specialist (CNS), or physician assistant (PA). Verbal orders are acceptable with follow-up signature by physician in accordance with facility/community policy.

Photocopies/faxes of signed POST forms are legal and valid.

Using POST

Any incomplete section of POST implies full treatment for that section.

No defibrillator (including AEDs) should be used on a person who has chosen "Do Not Attempt Resuscitation".

Oral fluids and nutrition must always be offered if medically feasible.

When comfort cannot be achieved in the current setting, the person, including someone with "Comfort Measures Only", should be transferred to a setting able to provide comfort (e.g., treatment of a hip fracture).

IV medication to enhance comfort may be appropriate for a person who has chosen "Comfort Measures Only".

Treatment of dehydration is a measure which prolongs life. A person who desires IV fluids should indicate "Limited Interventions" or "Full Treatment".

A person with capacity, or the Health Care Agent or Surrogate of a person without capacity, can request alternative treatment.

Reviewing POST

This POST should be reviewed if:

(1) The patient is transferred from one care setting or care level to another, or
(2) There is a substantial change in the patient's health status, or
(3) The patient's treatment preferences change.

Draw line through sections A through D and write "VOID" in large letters if POST is replaced or becomes invalid.

COPY OF FORM SHALL ACCOMPANY PATIENT WHEN TRANSFERRED OR DISCHARGED.

A COPY OF THIS FORM SHALL ACCOMPANY PATIENT WHEN TRANSFERRED OR DISCHARGED

Tennessee Physician Orders for Scope of Treatment (POST, sometime called "POLST)

This is a Physician Order Sheet based on the medical conditions and wishes of the person identified at right ("patient"). Any section not completed indicates full treatment for that section. When need occurs, first follow these orders, then contact physician.

Patient's Last Name

First Name/Middle Initial

Date of Birth

Section A *Check One Box Only*

CARDIOPULMONARY RESUSCITATION (CPR): Patient has no pulse and is not breathing.

☐ Resuscitate (CPR) ☐ Do Not Attempt Resuscitation (DNR / no CPR) (Allow Natural Death)

When not in cardiopulmonary arrest, follow orders in **B, C,** and **D.**

Section B *Check One Box Only*

MEDICAL INTERVENTIONS. Patient has pulse and/or is breathing.

☐ **Comfort Measures.** Relieve pain and suffering through the use of medication by any route, positioning, wound care and other measures. Use oxygen, suction and manual treatment of airway obstruction as needed for comfort. **Do not transfer to hospital for life-sustaining treatment. Transfer only if comfort needs cannot be met in current location. Treatment Plan: Maximize comfort through symptom management.**

☐ **Limited Additional Interventions.** In addition to care described in Comfort Measures Only above, use medical treatment, antibiotics, IV fluids and cardiac monitoring as indicated. No intubation, advanced airway interventions, or mechanical ventilation. May consider less invasive airway support (e.g. CPAP, BiPAP). **Transfer** to hospital if indicated. Generally avoid the intensive care unit. **Treatment Plan: basic medical treatment.**

☐ **Full Treatment.** In addition to care described in Comfort Measures Only and Limited Additional Interventions above, use intubation, advanced airway interventions mechanical ventilation as indicated. **Transfer** to hospital and/or intensive care unit if indicated. **Treatment Plan: Full treatment including in the intensive care unit.**

*Other Instructions:*_____

Section C *Check One*

ARTIFICIALLY ADMINISTERED NUTRITION. Oral fluids & nutrition must be offered if feasible.

☐ No artificial nutrition by tube.
☐ Defined trial period of artificial nutrition by tube.
☐ Long-term artificial nutrition by tube.

*Other Instructions:*_____

Section D *Must be Completed*

Discussed with:
☐ Patient/Resident
☐ Health care agent
☐ Court-appointed guardian
☐ Health care surrogate
☐ Parent of minor
☐ Other: _____(Specify)

The Basis for These Orders Is: (Must be completed)
☐ Patient's preferences
☐ Patient's best interest (patient lacks capacity or preferences unknown)
☐ Medical indications
☐ (Other) _____

Physician/NP/CNS/PA Name (Print)	Physician/NP/CNS/PA Signature Date NP/CNS/PA (Signature at Discharge)	MD/NP/CNS/PA Phone Number: ( )

Signature of Patient, Parent of Minor, or Guardian/Health Care Representative

Preferences have been expressed to a physician and /or health care professional. It can be reviewed and updated at any time if your preferences change. If you are unable to make your own health care decisions, the orders should reflect your preferences as best understood by your surrogate.

Name (Print)	Signature	Relationship (write "self" if patient)	
Agent/Surrogate	Relationship	Phone Number ( )	
Health Care Professional Preparing Form	Preparer Title	Phone Number ( )	Date Prepared

REFERENCE
Pulse Oximetry

EMT AEMT PARAMEDIC

Assessment:
Pulse oximetry is not without limits and must **NOT** be used to supersede other assessments.

The Firefighter EMT (or higher) shall treat the patient and **NOT** the pulse oximeter's display. The patient's other key signs and symptoms must be assessed and evaluated so that the oximeter's readings are interpreted within the context of the patient's overall condition.

The percentage of oxygen saturation measured by an oximeter only reflects the supplied pulmonary oxygenation and is not an indicator or measure of cellular oxygenation. Furthermore, it is useful both in the assessment of the patient and as an adjunct for evaluating the effectiveness of the airway management, ventilation, and oxygen enrichment provided.

Oxygen saturation pressure (SpO_2) is a different measurement than the partial pressure of oxygen (PaO_2) which is commonly measured by laboratory blood gas analysis.

Pulse oximetry should be deferred until more urgent assessment and care priorities have first been resolved. Pulse oximetry is a diagnostic tool that along with the patient's vital signs, chief complaint, mental status and other considerations, may assist us in determining the patient's respiratory status.

The pulse rate determined by the pulse oximeter is not an accurate indicator of the patient's pulse rate.

Falsely low readings may occur in the following:
- Patients with cold extremities or hypothermic patients
- Patients with hemoglobin abnormalities
- Patients without a pulse
- Hypovolemic patients
- Hypotensive patients

Falsely normal or high oxygen saturation readings may occur in the following patients:
- Anemic patients
- Carbon monoxide poisoning
- Cyanide toxicity which is being treated with the antidote
- Very bright lighting (direct sunlight or nearby strong lamp)

Other factors affecting accurate readings:
- Patient movement
- Action of vasopressor drug
- Peripheral vascular disease
- Elevated bilirubin levels
- Abnormal hemoglobin values
- IV diagnostic dye has been administered in the last 24 hours

Pulse Oximetry Values:

Normal;

- 96 – 100%
- Treatment – Non-rebreather mask (12 – 15 LPM) or nasal cannula (4 – 6 LPM) if patient cannot tolerate a mask based on patient's chief complaint

Mild Hypoxia;

- 91 – 95%
- Immediate need to increase the FiO_2
- Treatment – non-rebreather mask 12 – 15 LPM
- Consider use of CPAP if available

Moderate Hypoxia;

- 86 – 90%
- Immediate need to increase the FiO_2
- Consider possible loss of airway patency
- Treatment – non-rebreather mask 12 – 15 LPM, consider airway adjunct and bag-valve-mask @ 15 LPM, on assist
- Consider use of CPAP if available

Severe Hypoxia;

- ≤ 85%
- Treatment – assist ventilations with adjunct and bag-valve-mask @ 15 LPM. Airway management as appropriate
- Consider use of CPAP if available

REFERENCE
Quality Improvement Documentation Criteria

Documentation on all patients must include the following and any other information pertinent to patient care:

OPQRST and **SAMPLE** are the acronyms for the United States DOT EMS and Paramedic patient assessment curriculum.

O – Circumstances surrounding the **onset** of complaint.
P – What **provoked** (or provokes) the complaint?
Q – Describe the **quality** (sharp, burning, stabbing, etc.) of the complaint?
R – Where does the pain **radiate**?
S – Describe the **severity** of the pain on a 1 – 10 scale 1(minimal) – 10 (maximum).
T – **Time** of onset.

S – Signs, symptoms, physical exam findings.
A – Allergies to medications or the environment.
M – Medications, prescription or over the counter.
P – Past medical history.
L – Last oral intake.
E – Event, what happened to the patient.

All patients encountered by EMS should have at least two sets of vital signs assessed and documented. The initial set of vitals will include blood pressure (systolic/diastolic), pulse rate, respiratory rate, pulse oximetry, blood glucose (if indicated), and the time they were assessed must be recorded.

- All medications taken by the patient should be listed in the report. If medications are taken to ER, document in narrative who the medications were left with.
- When documenting the presumed presence of alcohol that is based solely upon breath odor, do so in the following manner: “Patient’s breath has the odor that is commonly associated with the consumption of alcohol.”

Abdominal Pain/ Problems:

1. Location of pain
2. Distension
3. Tenderness / radiation
4. Nausea / vomiting / diarrhea
5. Urinary complaints
6. LMP if applicable
7. Vaginal bleeding / discharge if applicable
8. Treatment / reassessments
9. Report given and signature of RN

Airway Obstruction:

1. Can patient speak / forcibly cough
2. Is patient moving air
3. Inspiratory stridor
4. What caused obstruction
5. Duration of obstruction
6. Treatment / reassessments
7. Report given and signature of RN

Alcohol Intoxication:

1. Patient's breath has odor of ETOH
2. Patient admits to drinking (type, amount, time frame)
3. Speech (normal, slurred)
4. Gait (normal, unsteady)
5. Any obvious injuries noted
6. Blood glucose level
7. Level of consciousness
8. Treatment / reassessments
9. Report given and signature of RN

Allergic Reaction:

1. Cause of reaction
2. Dyspnea
3. Facial / airway edema
4. Chest pain
5. Rash / itching
6. Urticaria / hives
7. Treatment / reassessments
8. Report given and signature of RN

Altered Mental Status:

1. OPQRST, SAMPLE as appropriate
2. ETOH / Substance abuse
3. Any obvious injuries noted
4. Blood glucose level
5. Normal mental status
6. EKG and strip attached
7. Treatment / reassessments
8. Report given and signature of RN

Animal Bite / Sting:

1. Type of animal or insect
2. Location of bite(s) / sting(s)
3. Edema at site
4. Rabies / immunization status of animal if appropriate
5. Treatment / reassessments
6. Report given and signature of RN

Assault / Fight:

1. OPQRST, SAMPLE as appropriate
2. Method of assault
3. Any obvious injuries or pain
4. Loss of consciousness, for how long
5. Treatment / reassessments
6. Report given and signature of RN

Atraumatic GI Bleed:

1. Nausea, vomiting, diarrhea, constipation
2. Active bleeding
3. Bloody emesis / stool, for how long
4. Color of emesis / stool
5. Abdominal pain – location and quality
6. Treatment / reassessments
7. Report given and signature of RN

Burn:

1. Burn source (flame, chemical, electrical)
2. Environment (enclosed, outside)
3. Entrance / exit wounds if appropriate
4. Burn surface area and thickness
5. Facial, oral, nasal area singed
6. Chest pain / dyspnea
7. Consider cyanide antidote
8. Treatment / reassessments
9. Report given and signature of RN

Cardiac Arrest:

1. Events prior to onset
2. Description / location of patient on arrival
3. Estimated down time
4. Treatment / reassessments
5. Report given and signature of RN

Chest Pain:

1. OPQRST, Sample as appropriate
2. Factors relieving or increasing pain
3. Dyspnea, cough
4. Nausea, vomiting
5. Diaphoresis
6. Aspirin within past 12 hours
7. Treatments / reassessments
8. Report given and signature of RN

CHF / Pulmonary Edema / SOB:

1. Chest pain
2. Dyspnea
3. Nausea, vomiting
4. Diaphoresis
5. JVD / lower extremity edema
6. Treatment / reassessments
7. Report given and signature of RN

Death:

1. Last time patient seen or talked to
2. Position / location of body
3. Any movement of body made by EMS
4. Any injuries noted
5. Dependent lividity / rigor mortis
6. EKG strip in two leads attached
7. Released to

Diabetic:

1. OPQRST, SAMPLE as appropriate
2. Nausea / vomiting / recent illness
3. Pre/Post treatment of blood glucose level
4. Treatment / reassessments
5. Report given and signature of RN

Hypertension:

1. Chest pain / dyspnea
2. Nausea / vomiting
3. Headache / mental status
4. Neuro assessment
5. Treatments / reassessments
6. Report and signature of RN

Hyper / Hypothermia:

1. Approximate ambient air temperature
2. Estimate exposure time
3. Type of environment (inside, outside, wet)
4. Loss of consciousness
5. Fluid intake
6. Skin turgor / condition
7. ETOH / substance abuse
8. Treatments / reassessments
9. Report given and signature of RN

Inhalation Injury (Toxic Gas/Smoke):

1. Type of gas
2. Duration of exposure
3. Area of exposure (enclosed room)
4. Heated environment
5. Burns / singing (oral, nasal, facial area)
6. Treatments / reassessments
7. Report given and signature of RN

Poisoning / Drug Ingestion:

1. Name of substance
2. Amount
3. Route of intake
4. How long ago
5. Vomiting since ingestion as appropriate
6. Intentional vs. unintentional
7. ETOH / substance use
8. Oral mucosa burns if appropriate
9. Treatments / reassessments
10. Report given and signature of RN

Pregnancy / OB Delivery:
Separate report required for mother and each delivery
Non-Delivery:

1. Abdominal pain, contractions (duration and frequency)
2. Gravida / para / abortion
3. Length of gestation / estimated due date
4. Edema (pedal) / BP / headache / visual disturbance
5. Vaginal bleeding / discharge – if yes, describe
6. Last time fetal movement
7. Treatments / reassessments

Report given and signature of R

Delivery:

1. Multiple fetuses
2. Mucous plug presented
3. Membranes ruptured – if yes, is amniotic fluid clear?
4. Crowning as appropriate

Neonate:

1. Time of birth
2. Thoroughly dried and warmed
3. Oral and nasal suctioning
4. Meconium present
5. APGAR at 1 and 5 minutes
6. General appearance
7. Treatments / reassessments
8. Report given and signature of RN

Refusals:
Documentation of:

1. Competency
2. MMSE
3. Lack of trauma
4. Situation
5. Ability to make good decisions
6. Safety of patient is assured by caretakers, family, etc.

Seizures:

1. OPQRST, SAMPLE as appropriate
2. Obvious injuries (mouth, head, tongue)
3. Duration and number of events
4. Incontinence
5. Level of consciousness (post-ictal)
6. Treatments / reassessments
7. Report given and signature of RN

Stroke / CVA / TIA:

1. OPQRST, SAMPLE as appropriate
2. Onset and duration of symptoms
3. Headache / Vision disturbances
4. Thrombolytic screening and stroke screen
5. Treatments / reassessments
6. Report given and signature of RN

Syncope / Fainting / Weakness:

1. OPQRST, SAMPLE as appropriate
2. Injuries, chest pain, dyspnea, nausea
3. Vertigo, postural, TILT changes
4. New or changed medications
5. Last meal
6. Blood glucose level
7. EKG
8. ETOH / Substance use
9. Treatments / reassessments
10. Report given and signature of RN

Trauma:

1. OPQRST, SAMPLE as appropriate
2. Description of event
3. Weapon (size, caliber, depth of penetration) if applicable
4. Description of damage, estimated speed, airbag deployment as applicable
5. Patient protection as applicable
6. Level of loss of consciousness
7. Obvious injuries and area of pain
8. Palpation / assessment of injured areas
9. Disability (PMS/SMC intact)
10. Consider use of tourniquet
11. Treatments / reassessments
12. Report given and signature of RN

REFERENCE
Sepsis Identification Tool

AEMT **PARAMEDIC**

Sepsis: Pre-Hospital Screening

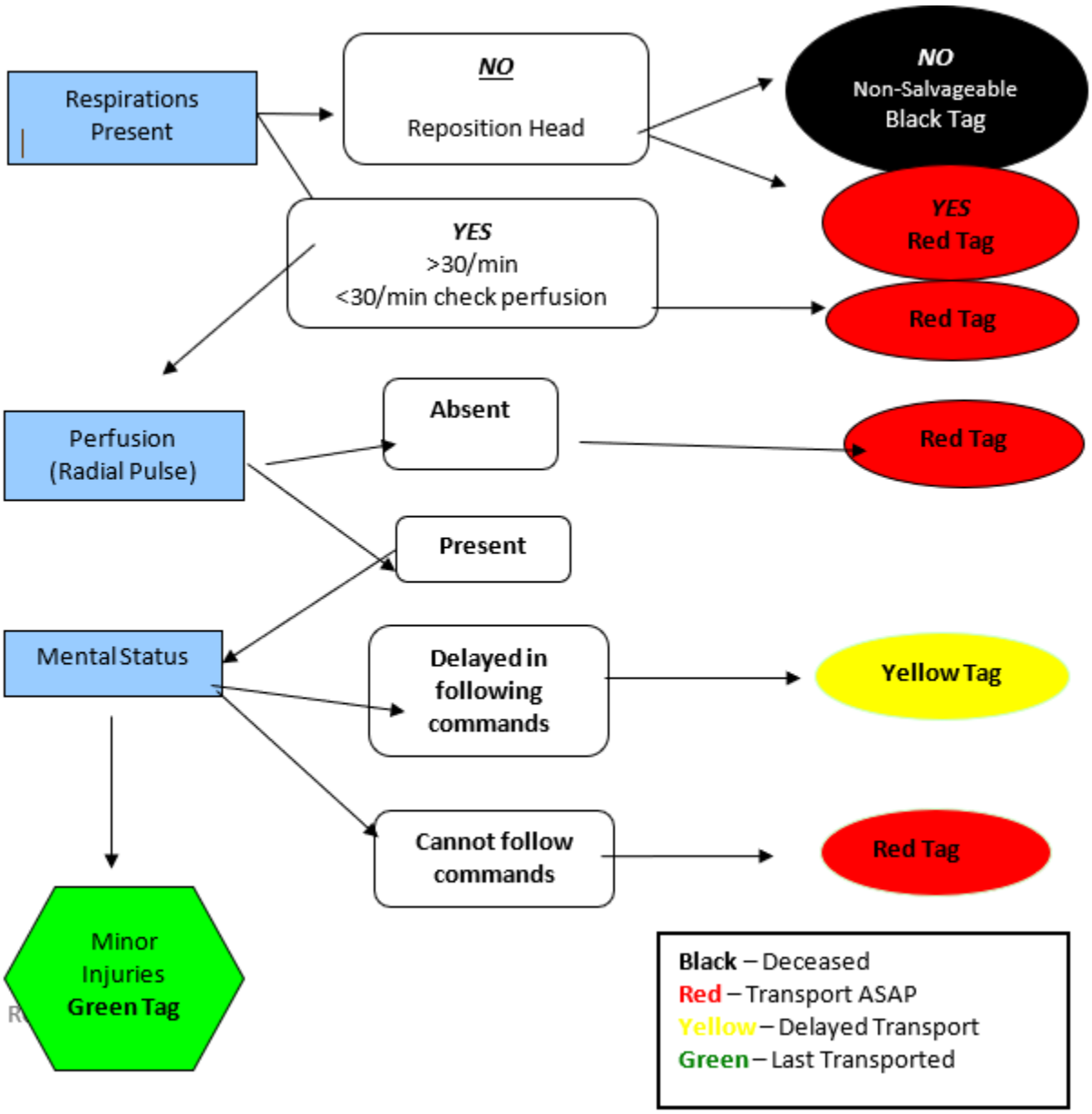
S.I.R.S (Systemic Inflammatory Response Syndrome) (2 or more)	**Infection** (Source of Infection) (1 or more)	**Severe Sepsis** (Organ Dysfunction) (1 or more)
Resp: > 20 Heart Rate: > 90 Temp: > 38 (100.5) < 36 (96.5) WBC: > 12,000 < 4,000 (If available from nursing home or other transferring facility)	Cough Painful urination Diagnosis of UTI Abscess Sign of skin infection Flu symptoms Recent chemotherapy Presence of vas cath Presence of urinary catheter Sick contacts (recent exposure)	Altered Mental Status Systolic BP<90 Oxygen sat <92% Signs of poor skin perfusion (i.e., poor cap refill, mottled skin, etc.) Lactate Level > 2 (If available from nursing home or other transferring facility.)
2 or more SIRS criteria + 1 or more sources of infection + and ETCO2 ≤ 25 **2.1 = CODE SEPSIS**		
2 or more SIRS criteria + 1 or more possible sources of infection + 1 or more organ dysfunction criteria= **2.1.1 = CODE SEVERE SEPSIS**		
In the event of a **CODE SEPSIS** OR **CODE SEVERE SEPSIS**, initiate the following Cardiac Monitor Oxygen to maintain ≥ 92% sat 2 large bore IVs draw labs Notify receiving hospital and identify patient as "**CODE SEPSIS OR CODE SEVERE SEPSIS**"		

REMEMBER…SEPSIS KILLS MORE THAN STROKE AND STEMI COMBINED!!!

The 6 hour window is closing!!!

REFERENCE
S.T.A.R.T. Triage

EMT AEMT PARAMEDIC



REFERENCE
Trauma Assessment / Destination Guidelines

EMT AEMT PARAMEDIC

- Perform primary and secondary survey
- Treat any life-threatening injuries / illness
- Obtain vital signs
- Determine mechanism of injury
- Obtain past medical history

Is transport to Trauma Center > 30 minutes

Yes	No
Initiate transport to closest appropriate facility Notify Medical Control of decision	<u>Transport to Level I Trauma Center if:</u> • GCS is < 13 and/or • Systolic BP is < 90 mmHg • Respiratory rate < 10 or > 30
Transport to trauma center may exceed 30 minutes If dictated by local Medical Control or Trauma Control	<u>Transport to Level I Trauma Center if:</u> • Penetrating injury proximal to elbow or knee • Flail chest, penetrating chest, or abdominal injury • Combination trauma with burns of > 15% BSA, or to face and/or airway • Limb paralysis • Amputation proximal to wrist or ankle • Patient ejection from vehicle • Death of passenger in the same vehicle • Extrication time > 20 minutes with above trauma
Medical Control will have final jurisdiction over destination, excluding: Any patient of legal majority (age 18 or over), the parent or legal guardian of a minor patient or an emancipated minor shall have the right to request transport to a specific facility with the county of origin.	<u>Contact Trauma Control to consider transport to Level I, II, III Trauma Center if:</u> • High speed auto accident with suspected injury • Velocity change of > 20 mph • Passenger compartment intrusion of > 12” • Auto vs. pedestrian injury with > 5 mph impact • Motorcycle accident > 20 mph or with separation of rider and motorcycle • Bicycle accident with significant impact
Transport of the patient to the requested destination shall not constitute neglect of duty imposed by law on all EMS personnel if the person making the decision has been informed that Tennessee has a trauma system, which would in their circumstance transport them to another facility.	<u>Contact Trauma Control to consider transport to Level I, II, III Trauma center if:</u> • Patient age > 55 years • Known cardiac, respiratory disease or psychosis on medication • Insulin dependent diabetic, cirrhosis, malignancy, obesity, or coagulopathy
If the patient’s condition deteriorates during transport, such that their life/health are considered in serious jeopardy if the requested/planned destination is pursued, **AND** if Medical Control deems transport to a higher level trauma center necessary, the patient may be transported to the appropriate facility	

REFERENCE

Trauma Treatment Priorities

EMT AEMT PARAMEDIC

1. If multiple patients initiate the S.T.A.R.T. and Multiple Casualty Incident System.
2. Oxygen 100% and airway maintenance appropriate for the patient's condition.
3. Treat for shock appropriate to the patient's condition.
4. Certain situations require rapid transport. Non-lifesaving procedures such as splinting and bandaging must not delay transport. Contact the responding emergency unit when any of the following exist:
    - Airway obstructions that cannot be quickly relieved by mechanical methods such as suction or jaw-thrust maneuver.
    - Traumatic cardiopulmonary arrest.
    - Large open chest wound (sucking chest wound).
    - Large flail chest.
    - Tension pneumothorax.
    - Major blunt chest trauma.
    - Shock.
    - Head injury with unconsciousness, unequal pupils, or decreasing level of consciousness.
    - Tender abdomen.
    - Unstable pelvis.
    - Bilateral femur fractures.

REFERENCE
Trauma Score

EMT AEMT PARAMEDIC

Revised Trauma Score

Respiratory Rate	10 – 24/min.	4
	24 – 35/min.	3
	> 36/min.	2
	1 – 9/min.	1
	None	0
Respiratory Expansion	Normal	1
	Retractive	0
Systolic Blood Pressure	> 90 mmHg	4
	70 – 89 mmHg	3
	50 – 69 mmHg	2
	0 – 49 mmHg	1
	No Pulse	0
Capillary Refill	Normal	2
	Delayed	1
Points to add to the RTS based on the GCS		
14 – 15		5
11 – 13		4
8 – 12		3
5 – 7		2
3 – 4		1

REFERENCE
Glasgow Coma Scale

EMT AEMT PARAMEDIC

Eye Opening	
Spontaneous	4
Opening to voice	3
Response to pain	2
None	1
Verbal	
Oriented	5
Verbal confused	4
Inappropriate words	3
Incomprehensible sounds	2
None	1
Motor	
Obeys commands	6
Localizes pain	5
Withdraws (pain)	4
Flexion	3
Extension	2
None	1

REFERENCE
Common Medical Abbreviations

1° - primary, first degree
a – before
AED – Automated External Defibrillator
AOX3 – alert and oriented to person, place, and time
Abd – abdomen
Ab – abortion
ac – antecubital
AF – atrial fibrillation
ARDS – Adult Respiratory Distress Syndrome
AT – atrial tachycardia
AV – atrioventricular
b.i.d. – twice a day
BSA – body surface area
BS – blood sugar and/or breath sounds
c – with
CC or **C/C** – chief complaint
CHF – Congestive Heart Failure
CNS – Central Nervous System
c/o – complains of
CO – carbon monoxide
CO2 – carbon dioxide
D/C – discontinue
DM – diabetes mellitus
DTs – delirium tremens
DVT – deep venous thrombosis
Dx – diagnosis
EDC – estimated date of confinement
EKG – Electrocardiogram
EJ – external jugular
ENT – ear, nose, and throat
ETOH – abbreviation of Ethanol (grain alcohol)
Fl – fluid
Fx – fracture
GB – gall bladder
g or **gm** – gram
gr. – grain
GSW – gunshot wound
gtt – drop
GU – genitourinary
GYN – gynecologic
h or **hr** – hour
H/A – headache
Hg – mercury
H&P – history and physical
Hx – history
ICP – intracranial pressure
JVD – jugular venous distension
KVO – keep vein open
LAC – laceration
LBBB – left bundle branch block
MAEW – moves all extremities well
NaCl – sodium chloride
NAD – no apparent distress/ no acute distress
NKA – no known allergies
NPO – nothing by mouth
OD – overdose
O.D. – right eye
O.S. – left eye
PERL – pupils equal and reactive to light
PID – pelvic inflammatory disease
p.o. – by mouth
PTA – prior to arrival
Pt – patient
q – every
qh – every hour
q.i.d. – four times a day
RBBB – right bundle branch block
R/O – rule out
ROM – range of motion
Rx – prescription
s – without
S/S – signs and symptoms
TIA – transient ischemic attack
t.i.d. – three times a day
Tx – treatment
V.S. – vital signs
y.o. – years old

REFERENCE
Glucose Dosage

Notes

If a glucometer reading is less than 60 mg/dL and patient is asymptomatic, start an INT and administer oral glucose. If a glucometer reading is less than 60 mg/dL and patient is symptomatic, start an IV NS and administer dextrose. Reassess patient every 5 minutes, repeat PRN.

Note: Any administration of dextrose must be given through an IV line running normal saline and **NOT VIA AN INT**. Blood glucose should be rechecked after administration of dextrose or oral glucose. Normal blood glucose values for adults are 80 – 120 mg/dL.

Blood Glucose and Stroke Screening will be performed on all patients with altered mental status. Glucose should be titrated slowly in order to restore normal levels while avoiding large changes in serum glucose levels. Be aware that elevated glucose levels are detrimental in conditions such as stroke.

Glucose (dextrose)	D50 1-2 mL/kg D25 2-4 mL/kg D10 2-4 mL/kg	>8 years 6 months - 8 years Neonate - 6 months Max Rate 2 ml/kg/min.
If D25 or D10 are not available utilize a syringe of D50. To make D25 expel 25 mL of D50 and draw up 25 mL of NS. To make D10 expel 40 mL of D50 and draw up 40 mL of NS.		
***Reminder:** Consider IO if other access unavailable and patient significantly symptomatic		

REFERENCE
Ketamine Dosage

Notes:
One of the effects of ketamine is hallucinations, and those occur within a certain dosage range. Surgical sedation dosage is 20 times higher than the dosage required to cause hallucinations, so as the drug is eliminated, patients may go through the hallucinogenic range. The dosage for pain control and hallucinations are very close together. This wearing off of ketamine is called an emergence reaction and may entail a variety of mood alteration, a floating sensation, hallucination, vivid dreams, and can be pleasant or unpleasant.

Not everyone experiences the emergence reaction, but it's more common in patients over 16, women, people who dream, those who receive large doses or rapid administration. This isn't usually an issue in the field, as the ketamine won't have worn off, and hallucinations can be managed in the ED. However, if management is required, administration of a benzodiazepine, such as midazolam, is appropriate.

Ketamine may also cause increased intraocular pressure, so use with caution in patients with glaucoma or acute globe injury. Administration of ketamine can cause nystagmus or double vision. Once patients have been sedated, they may have an eyes wide open, glazed expression.

Due to the catecholamine-induced sympathetic activity, ketamine can increase myocardial oxygen demand. Continuous monitoring of the cardiac system is required, and the drug is NOT recommended in patients with STEMI or other acute cardiovascular disease. Elevations of blood pressure, heart rate, or cardiac output that become symptomatic can be treated supportively.

Beware of laryngospasm, especially if drug administered quickly.
Ethanol also inhibits NMDA function, so use with caution in an intoxicated patient.
Continuous monitoring of the cardiac system is required.
Ketamine doesn't have a reversal agent. However, the effects that are most commonly seen in ketamine use can be managed supportively or end when the drug is eliminated from the system.

Pain Management
Indicated for an Adjunct for Musculoskeletal Pain. Not for Chest Pain, Headache, or penetrating eye trauma

Ketamine- 0.1-0.3 mg/kg IV PRN May be given 0.25-0.5mg/kg IM/IN if no vascular access
Onset: 1-3 min Peak: 3-20 min
Duration: 15-30 min
Onset 15-20 minutes

Airway Management

Ketamine is the induction agent of choice with suspected bronchospasm or asthma.

Ketamine is also the sedation / analgesia agent of choice for patient compliance and comfortability with NIPPV therapy due to its ability to maintain airway reflexes.

Ketamine 1-2 mg/kg IVP

Consider Versed 1-2 mg IVP to ensure complete sedation.

Behavioral Control/Sedation/Agitation

Ketamine- 1.5 – 2 mg/kg IV PRN
4mg/kg IM Dosing (initially preferred in acutely agitated patient)
May repeat prn
Onset: 1-3 min Peak: 3-20 min
Duration: ~15- min IV

Notes

If efforts to control seizures with Versed is unsuccessful, consider **Ketamine 1 mg/kg IV/IM.**

REFERENCE
Opiate Reference

Actions/Pharmacodynamics: Stimulates central nervous system opiate receptors, producing systemic analgesia. On a milligram weight basis, fentanyl is 50-100 times more potent than morphine. Its duration of action is shorter than morphine or hydromorphone. An IV dose of 100 mcg of fentanyl is roughly equivalent to an IV dose of 10 mg of morphine. Fentanyl has fewer emetic effects than other narcotic analgesics.

Contraindications: Hypotension
Respiratory Depression
Minor Degrees of Pain
Pain Assessed as Factitious

Side Effects: Hypotension, respiratory depression, euphoria, dizziness. Nausea and/or vomiting are rarely seen if administration is slow IVP.

Pharmacokinetics: Onset of action nearly immediate after IV administration. Peak effects occur within 3 – 5 minutes. Duration of effect is 30 - 60 minutes, with a half-life of 6 – 8 hours.

Notes

If hypotension develops after Opiate administration, give fluid bolus of 250 ml 0.9% NS or 20 ml/kg. Pediatric fluid bolus is 10 - 20 ml/kg 0.9% NS

If respiratory depression occurs after opiate administration, give Naloxone (Narcan) 0.4mg-2 mg IVP and consider intubation. **Pediatric Naloxone dosage is 0.1 mg/kg IVP**.

Fentanyl- 1 - 2 mcg/kg IV PRN. MAX Single Dose 200 mcg IV

- Onset: 1-3 min. Peak: 3-20 min.
- Duration: 15-30 min.
- Slow IV push, Repeat as needed, titrating for pain management
- Can be administered intranasally

Morphine 2 – 10 mg IV. 0.05 – 0.1 mg/kg IV PRN

- Slow IV push, Repeat as needed, titrating for pain management
- Zofran 4mg IV and may repeat as tolerated • Max total dose, 8 mg
- Onset: 1-3 min. Peak: 3-20 min.
- Duration: 15-30 min.

Doses are approximate	5-9 kg	10 kg	12 kg	15 kg	20 kg	30 kg	40 kg	50-74 kg	≥ 75 kg	Geriatric	
Fentanyl IV/IN/IO		10	12	15	20	30	40	50-75 mcg	75	25	1-2 mcg/kg
Morphine IV/IO		1 mg	1 mg	1.5 mg	2 mg	3 mg	4 mg	4 mg	4 mg	2 mg	0.05-0.1 mg/kg
Ondansetron IV/IO					3 mg	3 mg	4 mg	4 mg	4 mg	4 mg	0.15 mg/kg

If pain not controlled, Morphine and fentanyl dosing may be repeated once after ten minutes. Contraindicated in hemodynamically unstable patients.

REFERENCE
Pain Management Protocol

Notes
- Any non-intubated patient requiring pain management / sedation.
- Pain is a subjective symptom in which the patient exhibits a feeling of distress and discomfort.
- Manage pain appropriately.
- Titration over brief time periods is preferable.
- Remain alert to complications and side effects.
- Maintain adequate airway, breathing and circulation.
- Administer oxygen as indicated to maintain oxygen saturations greater than 93%.
- Monitor hemodynamics.
- Assess and document a patient's level of pain, upon initial patient contact, after any intervention that is performed to relieve pain, before transition of care, and as needed throughout care.

TREATMENT PATHWAY

1. Initiate IV Access.
2. Ensure adequate padding, splinting, etc. to reduce pain.

AEMT STOP HERE

Analgesia (Consider one or a combination of the following depending on patient needs):
Fentanyl 1 - 2 mcg/kg IV PRN. MAX Single Dose 200 mcg IV
- Onset: 1-3 min. Peak: 3-20 min.
- Duration: 15-30 min.
- Slow IV push, Repeat as needed, titrating for pain management
- Can be administered intranasally

Morphine 2 – 10 mg IV. 0.05 – 0.1 mg/kg IV PRN
- Slow IV push, Repeat as needed, titrating for pain management
- Zofran 4mg IV, May repeat as tolerated. Max total dose, 8 mg
- Onset: 1-3 min Peak: 3-20 min
- Duration: 15-30 min.

Ketamine 0.1-0.3 mg/kg IV PRN May be given 0.25-0.5mg/kg IM/IN if no vascular access
- Onset: 1-3 min. Peak: 3-20 min.
- Duration: 15-30 min.

PARAMEDIC STOP

REFERENCE
Post-Intubation Sedation

Indications:

To optimize the post-intubation treatment in terms of adequate pain management, sedation, and paralysis.

To differentiate the treatment of pain management and sedation in the intubated patient from the non-intubated patient.

Notes:

Intubated patients require aggressive pain management in the pre-hospital environment.

Consideration must be given to vital signs and non-verbal communication to adequately assess pain and anxiety in the intubated patient.

Be cautious that some agents may exacerbate hypotension where others mitigate the risk of hypotension.

TREATMENT PATHWAY

Sedation Management Therapies: Note: the following medications are not in order of succession. Use clinical judgement to guide treatment pathway.

Diazepam 0.01 – 0.1 mg/kg IV every 5 minutes
- Onset: 1-3 min. Peak: 3-20 min.
- Duration: 15-30 min.

Fentanyl 1 – 2 mcg/kg IV PRN May repeat prn
- Onset: 1-3 min. Peak: 3-20 min.
- Duration: 15-30 min.

Ketamine 1-2 mg/kg IV Peak: 3-20 min.
- Slow IV push. May be given intramuscularly if no vascular access
- Repeat as needed, titrating for pain management / sedation effect
- Consider use of Ketamine with NPPAV ventilation, to treat anxiety related to the procedure.

Versed 2 – 5 mg IV
- Slow IV push. Repeat as needed, titrating to effect

PARAMEDIC STOP

REFERENCE
Medication Dosage

EMT AEMT PARAMEDIC

Generic Name	Trade Name	Adult Dosage	***Pediatric Dosage***
Acetaminophen	Tylenol	1000 mg PO	***>3 months 15 mg/kg PO***
Adenocard	Adenosine	12 mg rapid IVP with flush	***1st dose 0.1 mg/kg max dose 6 mg 2nd dose .02 mg/kg max dose 12 mg***
Albuterol Sulfate	Proventil, Ventolin, Albuterol Sulfate	Aerosol Nebulization: 2.5 mg in 3 mL NS q 5 min. if heart rate <150	***Aerosol Nebulization: 2.5 mg in 3 mL NS q 5 min. if heart rate <200***
Amiodarone	Cordarone	300 mg then 150 mg	***5 mg/kg***
Aspirin	Aspirin	162-324 mg chewed and then swallowed	***No pediatric dosing***
Atropine Sulfate	Atropine	0.5-1 mg IVP q 3-5 min. Max dose 0.04 mg/kg	***0.02 mg/kg q 3-5 min. Max dose 0.04 mg/kg***
Calcium Chloride		500 mg IVP	***20 mg/kg***
Dextrose 50%	D_{50}, $D_{50}W$	12.5-25 gram IVP	***No pediatric dosing***
Dextrose 25%	D_{25}, $D_{25}W$		***2 mL/kg (D50 mixed 50/50 with Normal saline)***
Dextrose 10%	D10, D10W	250 cc bag of D10	**Up to 250 cc of D10**
Diazepam	Valium	2-10 mg slow IVP, titrated to effect	***00.1 mg/kg slow IVP, titrated to effect***
DOPamine		2-20 mcg/kg/min.	***2-20 mcg/kg/min.***
DiphenhydrAMINE	Benadryl	25-50 mg IM or slow IVP	***1 mg/kg***
EPINEPHrine	Adrenaline	Cardiac Arrest: 0.5-1 mg of 1:10,000 (now 0.1 mg/mL) solution IVP q 3-5 min. Anaphylaxis: 0.3-0.5 mg of EPINEPHrine 1:1,000 (now 1 mg/mL) solution IM	***Cardiac Arrest: 1:10,000 (now 0.1 mg/mL) 0.01 mg/kg IV/IO q 5 min. Anaphylaxis: EPINEPHrine 1:1,000 (now 1 mg/mL) 0.01 mg/kg IM, max dose 0.3 mg Croup: Nebulized EPINEPHrine 1:1,000 (now 1 mg/mL) diluted to 2.5-3 mL saline flush. May repeat up to 3 doses***
FentaNYL	Sublimaze	1-2 mcg/kg 50-100 mcg	***0.5-2 mcg/kg***
Glucagon	Glucagen	1-2 mg IM	***0.5 mg/dose IM/IV if <20 kg, or 1 mg/dose IM/IV if 20 kg or greater***
Ibuprofen	Motrin	600 mg PO	***peds >6 months 10 mg/kg PO***
~~Lidocaine~~	~~Xylocaine~~	IO Pain Control: 20-50 ~~mg 1-1.5 mg/kg max~~ dose 3 mg/kg	~~**IO Pain Control: 0.5 mg/kg *1.0 mg/kg***~~
Lidocaine Drip	Xylocaine	2-4 mg/min.	***20 to 50 mcg/kg***

Magnesium Sulfate		Torsades only: 1-2 gm IVP over two min. Pre-eclampsia or Eclampsia: 2-4 g slow IVP over 2 min./g Drip: 4 g in 250 mL D_5W (16 mg/mL) run at 30-60 gtts/min.	**VF/VT:** ***50 mg/kg IV/IO, max dose 2 g, over 1-2 minutes*** **Torsades only:** ***50 mg/kg IV, max 2 g***
Methylprednisolone	Solu-Medrol	62.5 or 125 mg	***Contact Medical Control***
Midazolam	Versed	2-5 mg IV or IM	***0.1 mg/kg***
Morphine	Morphine Sulfate, MS Contin, MSIR	2-4 mg IVP – see standing orders for repeat doses	***Sedation/Pain Management*** ***0.1 to 0.2 mg/kg***
Naloxone	Narcan	0.4-2 mg slow IVP	**0.1 mg/kg slow IVP**
Nitroglycerine		Oral: 0.4 mg SL or spray q 5 min. for pain Transdermal: 1" on chest wall NTG Therapy: 1 spray SL and apply 1" paste. Repeat SL spray once after 5 min. Continue therapy until pain is relieved or systolic BP <100 mmHg	
Nitrous Oxide	NitroNox	Patient self-administered gas	
Ondansetron	Zofran	2-4 mg IV; 4-8 mg ODT	***0.15 mg/kg IV***
Sodium Bicarbonate 4.2%			<1 mo, 1 mEq/kg IV/IO
Sodium Bicarbonate 8.4%		1 mEq/kg IV/IO followed by 0.5 mEq/kg q 10 min.	>1 mo, 1 mEq/kg IV/IO followed by 0.5 mEq/kg q 10 min.
Defibrillation		150 j Biphasic	***Begin at 2 j/kg***
Cardioversion		Refer to specific SOP	***0.5 j/kg to 2 j/kg***

REFERENCE
Drug Infusion Admix Dosage Guidelines

PARAMEDIC ONLY

Lidocaine:
2 grams medication/500 mL D_5W = 4 mg/mL (always use 60 gtt. Set)

1 mg/min = 15 gtt/min
2 mg/min = 30 gtt/min
3 mg/min = 45 gtt/min
4 mg/min = 60 gtt/min

Magnesium Sulfate:
4 grams in 250 cc D5W (16 mg/ml) run at 30-60 gtt/min

DOPamine:
400 mg /250 mL D5W or 800 mg/500 mL D5W = 1600 mcg/mL (always use 60 gtt. Set)

50 kg patient – 110 lbs.	*70 kg patient – 154 lbs.*	*100 kg patient – 220 lbs.*
2.5 mcg/kg/min = 5 gtt/min	2.5 mcg/kg/min = 7 gtt/min	2.5 mcg/kg/min = 10 gtt/min
5 mcg/kg/min = 12 gtt/min	5 mcg/kg/min = 13 gtt/min	5 mcg/kg/min = 19 gtt/min
10 mcg/kg/min = 19 gtt/min	10 mcg/kg/min = 27 gtt/min	10 mcg/kg/min = 38 gtt/min
20 mcg/kg/min = 38 gtt/min	20 mcg/kg/min = 53 gtt/min	20 mcg/kg/min = 75 gtt/min

Ped dose 2-20 mcg/kg/min

EPINEPHrine
2 mL (EPINEPHrine 1 mg/kg)/in 250 mL NS or D5W = 8 mcg/mL (always use 60 gtt. Set)

Ped dose – 3.6 mL (EPINEPHrine 1 mg/kg)/in 100 mL NS or D5W =32 mcg/ml (always use 60 gtt. Set)

PEDIATRIC SHOCK / TRAUMA
Pediatric Points to Remember

EMT AEMT PARAMEDIC

1. An infant is less than one year of age.
2. A child is from one to eight years of age.
3. Remember that few pediatric arrests are primary cardiac events. Most stem from respiratory (airway) problems, dehydration/metabolic, or hypothermia. Ensure that a child that arrests or is pending arrest is well oxygenated, well hydrated and warm.
4. Prognosis is extremely poor for a child that arrests.
5. Treat children aggressively before they arrest. Hypotension is a late sign.
6. **When in doubt contact Medical Control.**
7. The use of a length-based assessment tape is **required** for all pediatric patients as a guide for medications and equipment sizes.
8. Remember that with children the intraosseous drug route is quick to establish and may be easier than gaining IV access.
9. Children may be effectively ventilated using a BVM. This is the preferred method of ventilation in respiratory or cardiac arrest.

PEDIATRIC REFERENCE
Pediatric Trauma Score

EMT AEMT PARAMEDIC

(14 years of age and under)

Component	+ 2 points	+ 1 point	- 1 point
Size	Greater than 20 kg	10 – 20 kg	Less than 10 kg
Airway	Normal	Oral/Nasal airway	Unmaintainable/ intubated
Systolic BP	Greater than 90 mmHg	50 – 90 mmHg	Less than 50 mmHg
CNS	Awake	Obtunded/LOC	Coma
Open Wound	None	Minor	Major/penetrating
Skeletal	None	Closed fractures	Open/multiple fractures

Total point values from physical presentations or injury:
Trauma Score _____ (Sum of points)

PEDIATRIC REFERENCE
Triage Decision Scheme

EMT AEMT PARAMEDIC

(14 years of age and under)

Pediatric Trauma Score of 8 or less: refer to destination determinates (see pediatric shock/trauma protocol).

YES	**NO**
Transport to Level I Pediatric Trauma Center. Advise Medical Control	Assess anatomy of injury
Penetrating injury proximal to elbow and knee including head and neck	
Flail chest	
Traumatic Respiratory Arrest	
Pelvic fracture with shock	
Amputation proximal to wrist and ankle	
Combination trauma with burns of 15% BSA, or to the face or airway	
2 or more proximal long bone fractures	
Limb paralysis	
Contact Medical control for consideration of transfer to Level I or Level II pediatric trauma center. If Medical Control is unavailable, then transport to highest level trauma center.	Assess anatomy of injury
Evidence of high impact	Re-evaluate with Medical Control
Death of vehicle occupant (particularly if unrestrained)	
Fall greater than 20 feet	
Velocity change greater than 20 mph	
Passenger intrusion greater than 12 inches	
Pedestrian impact (significant) 5 – 20 + mph	
Motorcycle accident greater than 20 MPH or with separation of rider and bike	
Bicycle accident with significant impact	

PEDIATRIC REFERENCE
Age, Weight, and Vital Signs Chart

EMT AEMT PARAMEDIC

Age	Weight (kg)	Normal Diastolic BP	Normal Systolic BP	Heart Rate Per Minute	Respiratory Rate Per Minute
Birth	3.5	56 – 70	66 – 90	110 – 160	30 – 60
6 mons	7.0	56 – 70	70 – 104	100 – 140	30 – 50
1 year	10.0	56 – 76	80 – 104	100 – 140	24 – 34
2 years	13.0	56 – 76	80 – 104	90 – 110	20 – 30
3 years	15.0	56 – 76	80 – 104	90 – 110	20 – 30
4 years	17.0	56 – 76	90 – 110	80 – 110	20 – 30
5 years	19.0	56 – 76	90 – 110	80 – 110	20 – 30
6 years	23.0	56 – 76	90 – 110	70 – 100	16 – 30
7 years	25.0	56 – 76	90 – 110	70 – 100	16 – 30
8 years	28.0	60 – 76	90 – 110	70 – 100	16 – 30
9-10 years	30.0	64 – 76	90 – 114	70 – 90	10 – 20
11-12 years	37.0	64 – 80	90 – 120	70 – 90	10 – 20
13-15 years	50.0	64 – 80	110 – 124	60 – 80	10 – 20
16-18 years	65.0	64 – 90	110 – 134	60 – 80	10 – 20

Size ETT = $\frac{16 + (\text{age in years})}{4}$

PEDIATRIC REFERENCE
Age and Weight Related Pediatric Equipment Guidelines

EMT AEMT PARAMEDIC

	Premature 3 kg	Newborn 3.5 kg	6 Months 7 kg	1 – 2 years 10 – 12 kg	5 years 16 – 18 kg	8 – 10 years 25 – 36 kg
C – Collars			Small	Small	Small	Medium
Oxygen Masks	Premature or Newborn	Newborn	Pediatric	Pediatric	Pediatric	Adult
BVM	Infant	Infant	Pediatric	Pediatric	Pediatric	Pediatric or Adult
Laryngoscopes	0	1	1	1	2	2 – 3
ET Tubes	2.5 – 3.0	3.0 – 3.5	3.5 – 4.5	4.0 – 4.5	5.0 – 5.5	5.5 – 6.5
Suction Catheters	6 – 8 Fr	8 Fr	8 – 10 Fr	10 Fr	14 Fr	14 Fr
Oral Airways	Infant	Infant or Small	Small	Small	Medium	Medium or Large
IV Equipment	22 – 24 angio	22 – 24 angio	22 – 24 angio	20 – 22 angio	20 – 22 angio	20 – 22 angio
BP Cuffs	Newborn	Newborn	Infant or Child	Child	Child	Child or Adult

Authorization for Standing Orders

The Emergency Medical Services (EMS) Standing Orders and Protocols (revision project completed July, 2024) are hereby adopted. They are to be initiated by EMS personnel within their scope of practice of licensure whenever a patient presents with injury or illness covered by the protocols. Where indicated to contact Medical Control, the EMS provider should receive voice orders from Medical Control before proceeding. Other orders may be obtained from Medical Control when the situation is not covered by the protocols or as becomes necessary as deemed by the Paramedic/EMT.

Effective Date of these SOPs: 1.1.2027

"Signature on File"

Joe Holley, MD FACEP FAEMS
Medical Director

12.01.2026
Date